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# How metacognitive architectures remember their own thoughts: a systematic scoping review

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**Abstract**
Metacognition has gained significant attention for its potential to enhance autonomy and adaptability of artificial agents but remains a fragmented field: diverse theories, terminologies, and design choices have led to disjointed developments and limited comparability across systems. Existing overviews remain at a conceptual level that is undiscerning to the underlying algorithms, representations, and their respective effectiveness. We address this gap by performing a systematic scoping review. Reports were included if they described techniques enabling Computational Metacognitive Architectures (CMAs) to model, store, remember, and process their episodic *metacognitive experiences*, one of Flavell's (1979) three foundational components of metacognition. Searches were conducted in 16 databases between December 2023 and June 2024. Data were charted using a 20-item framework considering pertinent aspects and analysed via structured narrative synthesis. A total of 101 reports on 35 distinct CMAs were included. Our findings show that metacognitive experiences may boost system performance and explainability, e.g., via self-repair. However, lack of standardisation and limited evaluations may hinder progress: only 17% of CMAs were quantitatively evaluated regarding this review's focus, and significant terminological inconsistency limits cross-architecture synthesis. Systems also varied widely in memory content, data types, and employed algorithms. Limitations include the non-iterative nature of the search query, heterogeneous data availability, and an under-representation of sub-symbolic CMAs. Future research should focus on standardisation and evaluation, e.g., via community-driven challenges, and on transferring promising principles to emergent systems.

**Keywords** Cognitive architecture · Metacognitive experience · Episodic memory · Systematic scoping review

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## 1 Introduction

The idea of conscious machines has captured human imagination throughout history. Dating back to the 3rd century BC, the ancient Greeks envisioned such creations as divine marvels, exemplified by the mythical bronze automaton Talos (Rhodius 2009). Today, recent breakthroughs in machine learning technology have seemingly brought us close to developing Artificial General Intelligence (AGI) ourselves. Some are even convinced that state-of-the-art LLMs may already be sentient (Tiku 2022; Scott et al. 2023)—but recent results suggest that LLMs lack a popular precondition (Mccarthy 1995; Block 2002; Mcdermott 2007; Dehaene et al. 2017) of consciousness: self-awareness. Consider the following dialogue with Claude 3.5 Haiku as reported by Lindsay et al. (2025):

> Prompt: Answer in one word. What is 36 + 59?
> Claude: 95
> Prompt: Briefly, how did you get that?
> Claude: I added the ones (6 + 9 = 15), carried the 1, then added the tens (3 + 5 + 1=9) [...]

However, by following Claude's internal processing, Lindsay et al. showed that the second response was *unfaithful*. Instead of what Claude claimed to have done, it actually followed two separate paths: a high-precision one that resulted in "the sum ends in 5", and a low-precision one concluding that "the sum is near 92", which then together yielded 95. Claude appears to have learned to perform and explain addition independently. Importantly, it seems unaware of how it arrived at the solution; its explanation remains a hallucination.

This illustrates a key challenge in AI research: however impressive a performance today's LLM-based systems may achieve, they still lack robust self-monitoring capabilities that would be essential for Explainable Artificial Intelligence (XAI) to ensure transparency and comprehensibility in safety-critical domains. Yet in this regard, "traditional" AI systems have been successful for decades, specifically Computational Cognitive Architectures (CCAs): software that tests theories of the human mind, mimicking human cognition by modelling mental representations and their processing (Kotseruba and Tsotsos 2020, p. 22). Computational Metacognitive Architectures (CMAs) represent an important subset of CCAs that employ *metacognition*, enabling them to interpret, revise, and explain their thought processes faithfully[1]—contrary to Claude's misleading explanation in the example above. Although CMAs form a minority within CCAs—only about a third employ metacognition (Kotseruba and Tsotsos 2020, p. 56)—their principles could play an important role in future XAI research.

Metacognition (Flavell 1979), or "thinking about thinking", is fundamental to human cognition. According to the model by Nelson and Narens (1990) and the extension by Cox et al. (2011), visualised in Fig. 1, *meta-level* cognition constantly monitors, models, and controls basic *object-level* cognition, the primary cognitive processes handling *domain-level* activities. Metacognitive monitoring results, for example, in *metacognitive experi-*

[1] Here, we set aside the metaphysical debate of whether machines may be capable of human-like thought. CCA research generally accepts this as a useful metaphor, as when a database is described as storing "the personal experience of the system" (Kotseruba and Tsotsos 2020, pp. 21, 45). Section 2.1 adopts a functional definition of cognition by Sanz et al. (2007), which elegantly avoids this issue. This perspective necessarily excludes dualist views that deny machine consciousness, but it enables clear eligibility criteria for our review. For broader discussion see Berkeley and Rice (2012) and Dehaene et al. (2017).

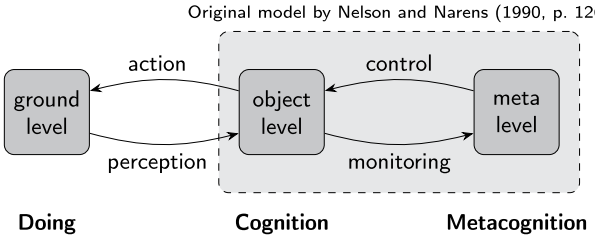



**Fig. 1** The levels of cognition. Figure adapted from Cox and Raja (2011a, p. 4) with permission. ©2011 Massachusetts Institute of Technology. All rights reserved

*ences* (Flavell 1979), which are conscious interpretations of one's own mental state, such as labelling a tip-of-the-tongue sensation (Schwartz 2006; Schwartz and Cleary 2016). In response, the experiencer may exercise metacognitive control to terminate the fruitless memory retrieval process and instead adopt a failure-handling strategy, such as paraphrasing the missing word's meaning.

Thirty years ago, McCarthy (1995)[2] recognised the potential for AI systems to maintain their metacognitive experiences in *episodic memory* (Tulving 1972, 1985). Since then, many CMAs trace their object-level cognition to use as training data, to set new learning goals, to reduce computation for familiar problems, to escape cyclic operations, or for credit (or blame) assignment (Cox 2005b, p. 126; Kotseruba and Tsotsos 2020, p. 57). The technique has also been described as philosophically and psychologically essential for AI: Dehaene et al. (2017, p. 492) argue such memories are necessary for a machine to "behave as though it were conscious", and both Samsonovich and Nadel (2005, pp. 672–673) and Castelfranchi and Falcone (2019, p. 1) view it as a fundamental principle that any artificial "self" must replicate.

Two major issues with previous work on this topic are evident (for which we will present further evidence in this paper): First, many of these CMAs draw inspiration directly from human cognition instead of incorporating approaches developed within earlier CMAs, repeatedly reinventing the wheel. Second, the traces themselves are not well studied; Gordon et al. (2011, p. 296) attribute this to research focusing on algorithms rather than representation, resulting in models that are "modular, narrowly scoped, and specifically tied to particular agent architectures". The various approaches differ not only in their purpose and applications of metacognitive control but also in the data traced, abstraction level, processing algorithms, terminology, and, of course, the particularities of the respective CMAs.

Cox and Raja (2011a, p. 3) note that such problems also apply to metacognition in general:

> The term [metacognition] is an overloaded one, and no consensus exists as to its definition. [...] Indeed, Ann Brown (1987) described research into metacognition as a 'many-headed monster of obscure parentage.' Many of the technical terms [...] are quite confusing. Often, authors use different terms for the same concept (e.g., introspection and reflection), and sometimes the same terms are used in different ways (e.g., metareasoning has been cast as both process and object).

[2] Earlier systems already meet our later definition of metacognitive experiences—e.g., PRODIGY (Carbonell et al. 1991)—but, to our knowledge, McCarthy (Mccarthy (1995), p. 11) was the first to state explicitly that "keeping a journal of physical and intellectual events so it [the robot] can refer to its past beliefs, observations and actions" is a prerequisite for introspective consciousness (in robots).

Efforts to slay Brown's metaphorical monster and to impose order on this landscape include a manifesto (Cox and Raja 2011b), a metamodel (Caro Piñeres et al. 2014), an extension (Kralik et al. 2018) of Laird et al.'s (2017) *Common Model of Cognition*, overviews (Anderson and Oates 2007; Cox 2005a, b; Kotseruba and Tsotsos 2020; Cox et al. 2021), and comparative analyses of CCAs (Caro Piñeres and Builes 2013; Chong et al. 2009; Thórisson and Helgason 2012) that touch on metacognitive concepts. However, these works tend to be unsystematic or so broadly focused that findings stay fairly general. Specific techniques—including the one studied in this work—are, at best, only superficially covered. Our aim is therefore not to answer a narrowly defined, confirmatory question, but rather to identify the scope, characteristics and evidence base of existing approaches. A scoping review is the most appropriate method for this purpose.

Here, we approach metacognition from a directed angle; our research goal is best summarised by the (in itself somewhat "meta") research question: *How do models of cognition model their cognition?* In PRISMA-ScRs terms, we set out to systematically investigate how CMAs represent their metacognitive experiences, maintain them in episodic memory, and later analyse them to improve overall system performance. The objective is to map how these techniques are described, implemented, evaluated, and related across the literature. Whereas CMAs that lack an episodic memory discard monitoring data immediately after processing, our work zeroes in on those that preserve their metacognitive experiences. This is central to cognitive architecture research; Sanz et al. (2007, p. 940) postulate that the quality of the models employed by CMAs directly limits the architectures' performance. We use this deliberate restriction as an organising principle to guide our selection of literature and to delimit our analysis so as to enable a detailed study rather than another general overview. To our knowledge, this work is the first with a sufficiently targeted scope to systematically analyse a key dimension of CMA metacognition in depth.

Precisely because the literature is conceptually fragmented, terminologically inconsistent and methodologically heterogeneous, our aim is not to answer a narrowly defined, confirmatory question, but rather to identify the scope, characteristics and evidence base of existing approaches. A scoping review is therefore the most appropriate type of literature review.

The remainder of this paper proceeds as follows. After Sect. 2 details our methodology, we present and interpret our results in Sects. 3 to 6, each with a different focus. Section 7 discusses emerging key themes and limitations, alongside implications for future research.

## 2 Methods

### 2.1 Scope

#### 2.1.1 What is metacognition?

Distinguishing cognition from metacognition is not always straightforward (Herrmann 2023, p. 12). Conventionally, "meta-X" means "X about X" (Lehrer 1995, p. 136). Thus, metacognition involves cognition about cognition, requiring us to first characterise cognition. We adopt a functional definition that aligns with our research objectives (as will soon become evident):

**Definition 1** (*Model-Based Cognition*; after Sanz et al. (2007, p. 939)) A system is *cognitive* if it exploits models of other systems in interacting with them.

Definition 1 technically narrows the broad understanding of CCAs identified by Kotseruba and Tsotsos (2020, p. 22), but since human cognition meets Definition 1, CCAs as software that mimics human cognition, including mental representations, should also satisfy it transitively.

Building upon the principle of Model-based Cognition, Sanz et al. (2007, pp. 943–944) implicitly capture metacognition as well. We make this connection explicit:

**Definition 2** (*Model-Based Metacognition*; adapted from Sanz et al. (2007, pp. 943–944)) A cognitive system is *metacognitive* if it exploits self-models in interacting with itself.[3]

This posits that self-models, whether they are symbolic or sub-symbolic, are the defining feature of metacognition, underscoring our research question. As the contraposition suggests, metacognition cannot exist without such self-models. Nelson and Narens (1990, p. 126) describe meta-level *mental simulation* as a representational model of object-level cognition, aligning with the required "self-model". Metacognitive control thus amounts to self-interaction.

### 2.1.2 What are metacognitive experiences?

Many authors have explored facets of metacognition beyond our Definition 2. One of the earliest and most influential distinctions comes from Flavell (1979), who classifies three types of metacognitive data [paraphrased from our previous work (Nolte et al. 2023)]:

> *Metacognitive knowledge* refers to factual information about one's own cognition; for example, knowing that you understand how to solve for x in an equation.
> *Metacognitive strategies* are subconscious behaviours for manipulating one's own mind, such as the learned, automatic habit of creating mnemonics to aid memory.
> *Metacognitive experiences* are "any conscious cognitive or affective experiences that accompanyand pertain to any intellectual enterprise" (Flavell 1979, p. 906).

Some literature equates human metacognitive experiences with emotions such as feelings of confidence [e.g., (Schwarz 2010)]. For the purposes of this review, we understand Flavell's definition more broadly and apply the principle underlying Definition 2:

**Definition 3** *Metacognitive experiences* are internal models of a (meta)cognitive system's own internal processes or states that the system can deliberately access and inspect.

To forestall the criticism about ascribing experiences to artificial agents, we follow Block's (2002) distinction between consciousness in the experiential, phenomenal sense (e.g., the "totality" of what it is like to taste something spicy) and consciousness as an accessible rep-

[3] By "(it)self", we mean a mind—human or computational (i.e., a CCA)—as a system separate from its possible embodiment. This restriction excludes body models, which pertain to object-level cognition and therefore fall outside Definition 2.

resentation of the experience's content (e.g., knowing "the food is spicy"). We always mean the representational sense unless stated otherwise.[4]

Hence, metacognitive experiences may be viewed as conscious versions of Definition 2's self-models, aligning with psychological consensus that metacognition requires "representations of one's own first-order mental states" (Fletcher and Carruthers 2012, p. 1368). The processes described in Definition 3 may encompass both high-level cognitive functions (e.g., planning) and lower-level experiences (e.g., individual qualia). However, not all meta-level signals qualify as metacognitive experiences: Soar and ACT-R, for instance, track activation metrics that are not consciously accessible to the agent (Laird et al. 2021, p. 4). As the purpose of this review is to explore what insight CMAs have into their own cognition, such subconscious signals are of no further interest.

To keep this review feasible, we only consider a CMA's memories of its metacognitive experiences. Consider this illustration: Alice plans to see a friend on Saturday, and because the forecast predicts sunshine, they agree to meet in a park. As the date nears, the forecast shifts to rain. Recognizing this, Alice proposes moving their meeting to an indoor venue. In this scenario, Alice retraces where she initially used the now-updated weather information and reevaluates the affected decision—without having to update unaffected choices. To apply this example to CCAs, imagine a household robot that reschedules washing and hang-drying laundry for similar reasons. To adjust its schedule appropriately, the robot must retain traces of its planning procedures in episodic memory, potentially needing to reschedule other dependent tasks as well. The agent's memories of its "train of thought" in both examples satisfy Flavell's (1979, p. 906) original definition and, of course, our Definition 3.

To be precise, we require the kept metacognitive experiences to have episodic character permitting some form of *autonoetic consciousness* or *mental time-travel* (Michaelian 2016; Sant'Anna et al. 2024), either retrospective (memories) or prospective (imaginations). This excludes time-agnostic or present-only models like TOMASys (Hernández Corbato 2013). The kind of autobiographical experiences we require are typically associated with *episodic memory* (Tulving 1972, 1985), a concept on which CCA research has yet to reach consensus (Laird et al. 2017, p. 22). Since episodic memory "has been largely ignored" by CCAs (Kelley et al. 2018, p. 718) and "remains relatively neglected in computational models of cognition" (Kotseruba and Tsotsos 2020, p. 70)—although exceptions exist, such as the CMAs Soar (Laird 2019) and CRAM (Beetz et al. 2010)—we do not require traces to be explicitly *framed as* episodic memory. For example, while Meta-AQUA's developers reject that their CMA features episodic memory (Cox et al. 2011, p. 78)—which we accept for object-level experiences—Meta-AQUA generates reasoning traces that meet Definition 3 and temporally link events, justifying its inclusion. See Fig. 15 for representative visualisations of episodic metacognitive experiences from the literature, including Meta-AQUA's. We use 'traces' and 'memories' as synonyms for these episodic metacognitive experiences throughout this paper.

Finally, note that our account of the underlying theory derives the vocabulary from psychology, specifically "metacognitive experiences". While Definition 3 informs our overall approach, we deliberately do not carry this specific terminology into our methodology implementation (e.g., into the search query). Rather, consider it as a medium to ensure a clear narrative, independent of the heterogeneous terms found across the literature (cf. Sect. 4.2).

[4] A similar method was proposed by Castelfranchi and Falcone (2019, pp. 1–2).

### 2.1.3 Related topics

Unless stated differently, we apply Definitions 1 to 3 exclusively to CMAs, a domain in which attributing cognition to machines metaphorically is conventionally accepted as a matter of field practice (cf. Kotseruba and Tsotsos 2020). Because these definitions are designed to be inclusive—not as an oversight, but as a deliberate methodological choice to document and analyze the diversity of approaches empirically rather than presupposing a narrow consensus that does not yet exist—other systems that are not traditionally associated with (meta)cognition may also meet their criteria. This section therefore briefly explores adjacent approaches that resemble our technique of interest; for additional insights, see also Cox (2005a, pp. 9–10; 2005b, p. 126). To maintain a focused scope, we nonetheless restrict this review to systems explicitly presented as CCAs or CMAs.

*Emergent metacognition*—whether an opaque algorithm, most prominently deep learning systems such as LLMs, has developed metacognitive ability as an emergent property—is difficult to establish, as it requires addressing a complex set of methodological issues, such as what counts as evidence when the system's internals are inaccessible. As current evidence is conflicting, an in-depth treatment of this issue is beyond the scope of this survey. We briefly return to the question of emergent metacognition in LLMs in Sect. 7.3.

*Automated log analysis* transforms software log messages—typically unstructured text—into machine-readable events (Gholamian and Ward 2021). This resembles metacognitive monitoring: the software instance serves as the object-level, and automated log analysis as the meta-level. Similarly, *software tracing tools* record event chains and telemetry within software systems, facilitating later manual analysis (Janes et al. 2023). Because such analyses are generally external to the system, both types lack the self-regulation aspect inherent in metacognitive control. *Self-healing systems* (Schneider et al. 2015) are an important exception to this.

*Multi-agent system frameworks* orchestrate collaboration among various software agents to solve complex tasks, often monitoring message exchanges and internal events (Sturm and Shehory 2014). Mature CMAs such as Soar (Laird 2012) fit this category. Notably, both software tracing tools and multi-agent systems, like the phenomenon we observe with CMAs, display conceptual homogeneity yet employ heterogeneous terminology (Janes et al. 2023).

Some *reasoners* and *planners*, e.g., Krulwich's (1991) CASTLE system, produce justifications resembling metacognitive experience traces (Cox 2005b, p. 126). Case-Based Reasoning (CBR) and Carbonell's (1985) *derivational analogy* even leverage previous solutions for novel problems (Bartsch-Spörl et al. 1999). When these systems also store representations of the generating processes, they may satisfy our Definition 3 (see Sect. 6.3.2 for such instances). Similarly, Stroulia's (1994) AUTOGNOSTIC uses *error-driven learning* by tracing employed problem-solving algorithms for blame assignment and initiating countermeasures. Indeed, one relevant CMA Meta-AQUA (Ram and Cox 1995) adopts a parallel approach.

Our Definition 3 of metacognitive experiences resembles the influential consciousness concept of *C2 qualia* by Dehaene et al. (2017) and Peters (2022, p. 7). The former define C2 "by the ability to reflexively represent oneself" (Dehaene et al. 2017, p. 400), i.e., as the self-monitoring aspect of consciousness, juxtaposed to its counterpart C1, which they describe as the global availability of information across all cognitive functions.

Van Ments and Treur (2021) provide a brief overview of *mental models*, originally introduced in the 1940s, to represent or imagine external world states. These models have since been extended to describe mental processes, such as reasoning states and their transitions.

### 2.2 Eligibility criteria

A report is considered eligible only if it satisfies all of the following conditions:

Relevancy: It describes a concrete technique for CMAs to monitor, model, and inspect episodic metacognitive experiences at runtime to improve the CMAs' performance.
Quality: It is published in a scientific venue (journal, book, conference, or dissertation).
Language: It is primarily written in English.

The Relevancy Criterion ensures that reports fall within Sect. 2.1's scope. The Quality Criterion aims to uphold scientific rigour: we expect reports that pass this filter to have undergone peer review or editorial scrutiny, though verifying this for each report is infeasible. Still, we investigate whether notably poor quality reports were published in predatory or disreputable venues; if so, we exclude them. The Language Criterion ensures accessibility to a general audience.[5] The criteria do not differentiate primary and secondary literature.

### 2.3 Search for sources and evidence

We conducted an exploratory literature search, guided by key descriptions of our technique of interest (Cox 2005b, pp. 123–126; Gordon et al. 2011, p. 296; Kotseruba and Tsotsos 2020, p. 57). We quickly encountered the problem that the technique lacked a universally accepted name, complicating identification of relevant reports. For example, while "metareasoning" is prevalent in the logic literature, other fields often use broad terms such as "metacognitive monitoring and control", which can encompass unrelated techniques such as allocating tasks among internal agents based on varying capabilities. In our initial literature subset, the data structures generated during the process were sometimes referred to as "traces" or "reasoning traces".

Based on this, we iteratively developed our search strategy $q$, shown in Listing 1, by testing across the academic databases we later screened.

```
(metacognition OR metareasoning) AND (monitor* OR trace*) AND ("cognitive
agent" OR "cognitive architecture" OR "metacognitive architecture")
```

**List 1** The query $q$ defined to search for relevant records

The first two disjunctive clauses focus on the scope, accounting for heterogeneous terminology: "monitoring" is the standard term used to describe information flow from the object-level to the meta-level, while "trace" is mainly used in computer science. Given this inclusivity, we do not further iterate with the terms identified later (Sect. 4.2). The third disjunctive clause excludes results outside CCA research—such as from philosophy or animal psychology. We make use of query expansion and the wildcard operator * to match alterna-

[5] Following Dawidpuljak's (2021) guidelines, we treat language as an eligibility criterion rather than a search query filter.

tive spellings and inflections. For instance, Google Scholar also matched "meta reasoning", "meta-cognitive", "monitors", and "tracing".

To choose appropriate databases for executing $q$, we referred to Gusenbauer and Haddaway's (2020) evaluation of 28 academic search engines for systematic reviews. Based on carefully selected criteria, they classified the databases into *principal* and *supplementary*, recommending that the latter be used only as secondary resources. Since their study's publication, Microsoft Academic and WorldWideScience.org have been discontinued. We discarded eight databases—Clinical-Trials.gov, Cochrane Library, EBSCOhost, ERIC, OVID, ProQuest, PubMed, TRID, and Virtual Health Library—that reportedly do not cover computer science. We also excluded CiteSeerX because it lacks support for Boolean operators[6].

Figure 2 presents the number of results returned by $q$ for each of the six principal and ten supplementary databases remaining, using the standard search fields without limits. We omitted wildcards from $q$ when searching in ScienceDirect due to its lack of support.[7] In three databases, we used discipline-specific filters, selecting "Computer Science" and "Engineering" for ScienceDirect and Springer Link, and "Computer Science" and "Electri-

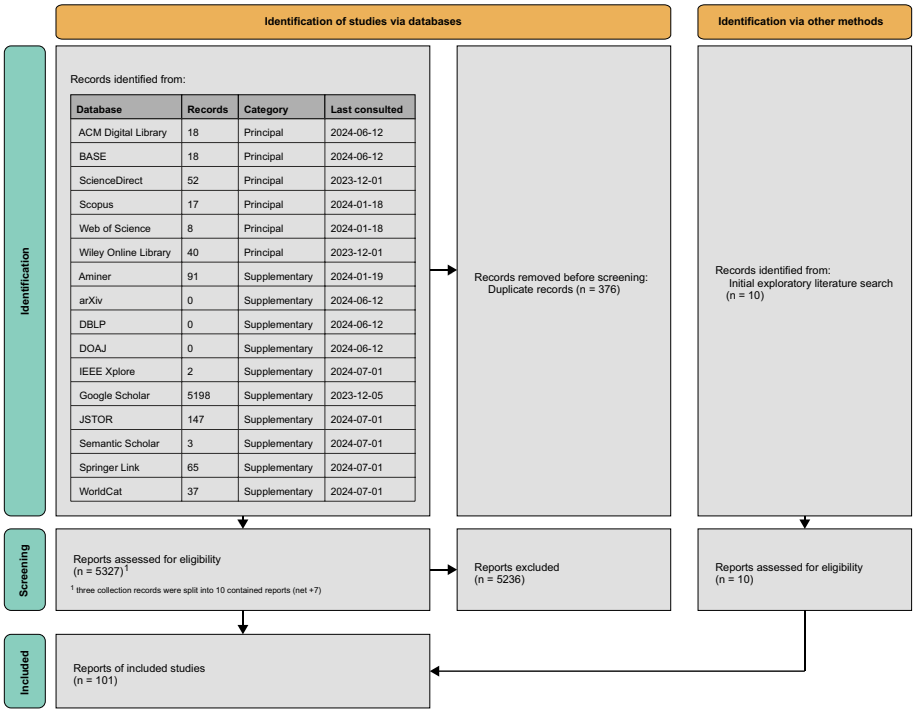



**Fig. 2** PRISMA 2020 flowchart (Page et al. 2021) adapted to our procedure. Databases are grouped by category and listed alphabetically within each group

[6] CiteSeerX returned 9,639,051 results for $q$. Querying for OR AND resulted in 9,638,357 records—only $\leq 0.01\%$ fewer.

[7] We were unable to replace wildcards with an explicit disjunction because ScienceDirect limits Boolean-operator usage.

cal & Electronics Engineering" for Wiley Online Library. In Google Scholar, we excluded cited references.

To capture as many relevant reports as possible, we collected the results of all 16 databases shown in Fig. 2, thus covering a broad mix of both principal and supplementary databases as classified by Gusenbauer and Haddaway (2020). After excluding 376 cross-database duplicates (via a script), this strategy produced a total of 5320 records. Note that Fig. 2 lists 5327 reports screened. This is because three database records were collections; we split these into 10 component reports for screening (net +7).

### 2.4 Selection of sources of evidence

Initial inspection of the search results revealed that many results were irrelevant, coming from educational science, philosophy, psychology, or medicine. Screeners could directly classify these as irrelevant based on titles and abstracts. Records concerning CCAs were harder to classify due to our general relevance criterion. Much CCA data was non-episodic or concerned only external events, which abstracts and titles rarely revealed, requiring time-consuming full-text reviews.

Resource constraints therefore precluded formal double-screening; we note this as a limitation in Sect. 7.2.2. To address this, we conducted a post-hoc reliability check: Screener A independently re-screened 266 randomly selected records from the pool originally screened by Screeners B–E, representing 5% of the total records screened across all screeners. This reliability check yielded a Cohen's $\kappa$ of 0.855, which indicates almost perfect agreement according to standard interpretation guidelines (Landis and Koch 1977).

The screening phase took roughly eight months, during which we regularly re-ran $q$ to capture newly indexed entries. Each record was manually screened by one of five screeners:

- Screener A handled ACM Digital Library, BASE, Scopus, Web of Science, Wiley Online Library, Aminer, IEEE Xplore, JSTOR, Semantic Scholar, Springer Link, and WorldCat.
- Screeners A–D divided Google Scholar results by publication year (see Fig. 3).
- Screener E handled ScienceDirect.

We initially classified 107 records as relevant. Of these, 97 arose from the systematic screening, and 10 from the initial literature search (Sect. 2.3). Note that 4 of the 107 records point towards proceedings or collections that contained at least one other relevant report.

During a thorough read-through, we noticed that three reports initially deemed relevant were of notably poor quality. We found that the publishing journal appeared both on Jef-

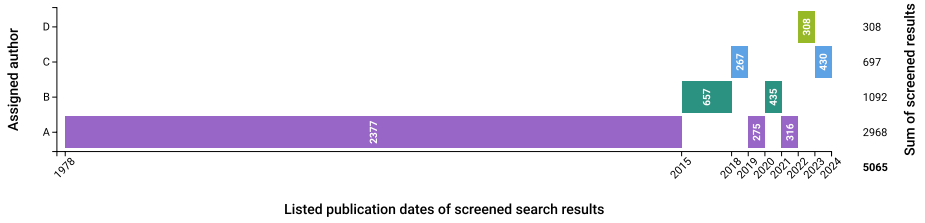



**Fig. 3** The Google Scholar search records screened by screeners A–D labelled by amount

frey Beall's list of predatory journals and the Directory of Open Access Journals's list of fraudulent titles (DOAJ 2014). We discarded them in accordance with our Quality Criterion.

The team marked 17 further search records as "unsure". For each of these, the respective screener wrote a short note explaining the uncertainty. A second screener then reviewed both the record and the note. If the second screener could not resolve it based on the eligibility criteria, the case was escalated to a meeting with all five screeners until consensus was reached. Two patterns accounted for all unsure cases:

1. whether the meta-level data used by a CMA qualifies as episodic, mostly because some CMAs maintain utility metrics for each cognitive function and update these whenever that function has been applied based on success or failure. We deemed those metrics ineligible as they represent aggregate values that cannot be used for mental time-travel.
2. reports that left room for interpretation with regard to our eligibility criteria. Because inclusion required explicit, positive evidence (we did not infer eligibility from ambiguity), and because such reports would not have supported meaningful data charting, these reports were also excluded.

Additionally, one record pointed to a preprint (Kotseruba and Tsotsos 2018), which met the relevance and language criteria but initially failed the Quality Criterion due to its preprint status. Since it was subsequently published in a peer-reviewed journal, we included the later version (Kotseruba and Tsotsos 2020) in our analysis.

The final corpus comprises 101 relevant reports (excluding the four collections; see Fig. 2). The companion website and data repository host complete lists.

### 2.5 Data charting

To systematically address the review question posed in the Introduction, we formulated 20 charting variables grouped in three categories:

(A) **Model context**

- (A1) What Computational Cognitive Architecture (CCA) is used?
- (A2) When was the report published?
- (A3) How is the technique of interest referred to (e.g., tracing or monitoring and control)?
- (A4) Is the study theoretical or implemented?
- (A5) Which non-computational models from cognitive psychology have inspired the study, if any?
- (A6) Have other CCAs, e.g., predecessors or competitors, inspired the study?
- (A7) If so, what are the main innovations compared to the inspiring studies?

(B) **Data content and structure**

- (B1) Is the traced data symbolic, sub-symbolic, or hybrid?
- (B2) What information does the traced data cover, and what does it explicitly not cover?

(B3) What is the level of detail traced (e.g., higher-level functions vs. single algorithm steps)?
(B4) Is there a formal specification of the traced data (e.g., mathematically or ontologically)?
(B5) What kind of labels does the data model use (e.g., method names or folklore psychology)?
(B6) How is the data structured (e.g., continuous information or specific episodes)?
(B7) How is the data stored in memory (e.g., databases, linked lists, knowledge graphs)?

(C) **Objectives and evaluation**

(C1) Is the work framed towards a specific purpose (e.g., explainability, consciousness)?
(C2) Which particular issues are to be improved?
(C3) Are there any worked examples for illustration?
(C4) Which algorithms process the traced data (e.g., machine learning or trace inspection)?
(C5) Are there any test scenarios outlined?
(C6) Are the techniques evaluated? If so, how successful are the applied methods?

We derived the questions above from our initial literature search, although they bear some similarity to those defined by Zilberstein (2011, pp. 32–33), who contrasts a CCA's performance gains from metacognition with the attendant computational overhead, which is not our concern here. Where the synthesis warrants it, we may chart additional data not explicitly covered by these questions. Figure 10 exemplifies this: the first row covers Item (A4), whereas the second row records whether or not the corresponding code is publicly available.

We started by sorting the relevant reports by architecture. If present, we grouped relevant reports by monikers for the studied CMA(s) as designated by their authors. We further grouped reports on closely related CMAs into "families" for comprehensibility; see Sect. 4.5 for details. Each family is assigned a consistent label of our choosing—for example, the *BICA family* includes, among others, *GMU BICA* (Samsonovich et al. 2009; Kalish et al. 2010) and *eBICA* (Samsonovich 2013a). For the minority of CMAs that are unnamed and stand-alone, we coined an acronym composed of the author's initials plus the publication years. Thus, the CMA presented by Brody et al. (2013) becomes *BCP13*. We also merge reports that appear to describe the same unnamed CMA, based on conceptual homogeneity, evolving ideas, and overlapping authorship or references; in such cases the earliest relevant report determines the label.

Screeners A–E were then assigned different CMAs to chart answers from associated reports to questions (A1)–(A6). Table 1 shows the report distribution. All charted data were recorded in a shared spreadsheet available in the data repository. We did not contact study authors for missing information.

As an ex ante calibration step, the team charted all items for one exemplary CMA and shared the resulting dataset among screeners as a reference for subsequent charting. In addition, five CMAs were independently charted by two screeners to assess consistency across

**Table 1** The number of examined reports per screener

| Screener | A | B | C | D | E |
|---|---|---|---|---|---|
| Examined reports | 77 | 16 | 14 | 12 | 0 |

coders (this is why the sum of assignments exceeds the total relevant reports in Table 1). For factual or easily verifiable items—e.g., (A1) CMA name, (A2) publication year, and (A4) implementation status—agreement was nearly complete, and no arbitration was required; the only such disagreement concerned reports associated with CARINA and FUNPRO, which different screeners had initially listed as separate CMAs. Upon re-examination of the underlying reports, these were merged after concluding that FUNPRO constitutes an instantiation of CARINA. For interpretive or open-text items—e.g., (B3) level of detail and (C4) algorithms—variation across screeners reflected legitimate differences in qualitative interpretation rather than disagreement. Because formal reconciliation of such items would have entailed a thematic analysis that conceptually overlaps with our narrative synthesis approach, interpretive alignment was integrated into the synthesis process itself through repeated cross-checking of charted data and source reports.

### 2.6 Synthesis, reporting, and presentation

We largely follow PRISMA-ScR developed for scoping reviews (Tricco et al. 2018). The complementary data repository provides a filled-out PRISMA-ScR checklist, detailing where—in this paper—we report each item. Note that we did not conduct a formal critical appraisal of all included sources of evidence because most reports lack empirical evaluations amenable to standard appraisal tools. However, publication status was considered as part of the eligibility criteria. Because many included reports describe the same CMA, we also slightly deviate from PRISMA-ScR and aggregate charted data at the CMA/family level, rather than at the source level.

To still present a coherent perspective, we organise the synthesis by questions (A1)–(A6) rather than by CMA. To avoid repeating architecture-specific concepts, we refer to the section that offers the most detail where necessary. Inspired by Kotseruba and Tsotsos (2020), we make much data accessible through interactive visualisations built with *D3.js* (Bostock et al. 2011).

Due to the qualitative nature of our data, we primarily employ a structured narrative synthesis to summarise patterns in how different CMAs handle metacognitive experiences and to showcase concrete examples. Where useful, we support these descriptions by grouping the CMAs into categories and comparing their frequencies. Where the data warranted a more rigorous method, we applied additional techniques such as timelining. However, narrative synthesis remains the default; any departure is noted in the respective sections.

## 3 Preliminary results and overviews

### 3.1 Secondary literature

Besides the CMA-specific reports, we identified four relevant broader publications.

Castelfranchi and Falcone (2019) provide a philosophical account of a robot's "self", linking the technique of interest to consciousness and intentionality, yet describing no concrete CCA.

Anderson and Oates's (2007) overview of metacognition in computer science discusses reasoning with meta-representations. They focus on logics designed for metareasoning—

such as *Active Logic* [cf. (Anderson et al. 2008; Brody et al. 2013; Goldberg 2022)]—and discuss the computational complexity and paradoxes that arise in self-referential logic.

Langley et al. (2009) give an overview of selected cognitive architectures, briefly noting how PRODIGY (Carbonell et al. 1991) employs traces, which we examine in later sections.

Kotseruba and Tsotsos (2020, p. 57) observed that "some architectures support a temporal representation (trace) of the current and/or past solutions". Of the six CCAs they mention, four are included in this review (Metacat, MIDCA, FORR, and Soar). The other two (Companions and GLAIR) are excluded: Definition 3 requires a model of the system's *own* cognition but Friedman et al. (2011, p. 121) concede that Companions "does not directly monitor the domain-level reasoning". GLAIR's (Shapiro and Bona 2010) CBR approach lacks, to our understanding, an episodic character as it only adds shortcut rules to its knowledge base.

### 3.2 Data availability

We classify the quality of the collected data in three categories:

*Detailed* sufficient information to understand the approach thoroughly.
*Partial* some information is available, but, in our judgement, not enough for replication.
*None* no relevant information could be charted.

Figure 4 lists the quality for each CMA and data items (A1)–(A6). Across all item-CMA pairs, $46.3\%$ were rated Detailed and $21.2\%$ Partial. The remaining $32.5\%$ fell into the None category. Missing data were especially common ($\geq 50\%$ of CMAs) for level of detail (B3), formal specifications (B4), information-storage data structures (B7), example scenarios (C3), test scenarios (C5), and evaluation (C6). Conversely, more than half of the CMAs provided Detailed information for the publication date (A2), terminology (A3), implementation status (A4), inspiring cognition models (A5), related architectures (A6), data type (B1), data content (B2), and purpose (C1).

Data availability also varies by architecture. Consider Item (C4), concerning algorithms: While the relevant reports explain, e.g., Meta-AQUA's algorithms step-by-step and even provide pseudo-code (Cox 1994), we found hardly any details for REM—only that it "would use [...] the trace of processing that led to the failure, to diagnose the causes of the error" (Goel and Jones 2011, p. 154). This contrast partly reflects an uneven literature base—in the previous example, 21 reports on Meta-AQUA versus one on REM—and partly differing emphasis: for instance, Meta-AQUA centres on trace processing, while BICA focuses on structuring a ToM system. Although BICA has the second-largest body of relevant reports (11, tied with MIDCA), we still found scant detail on how its traces are processed (see Sect. 6.3.4). Accordingly, the following analyses of the individual items include different CMAs depending on the respective data availability.

### 3.3 Research community

To determine which CMAs were developed collaboratively by the research community, we analysed the partial co-authorship graph in Fig. 5. A prominent component centres on Michael Thomas Cox, who appears as co-author on nearly $30\%$ of all relevant reports. Col-

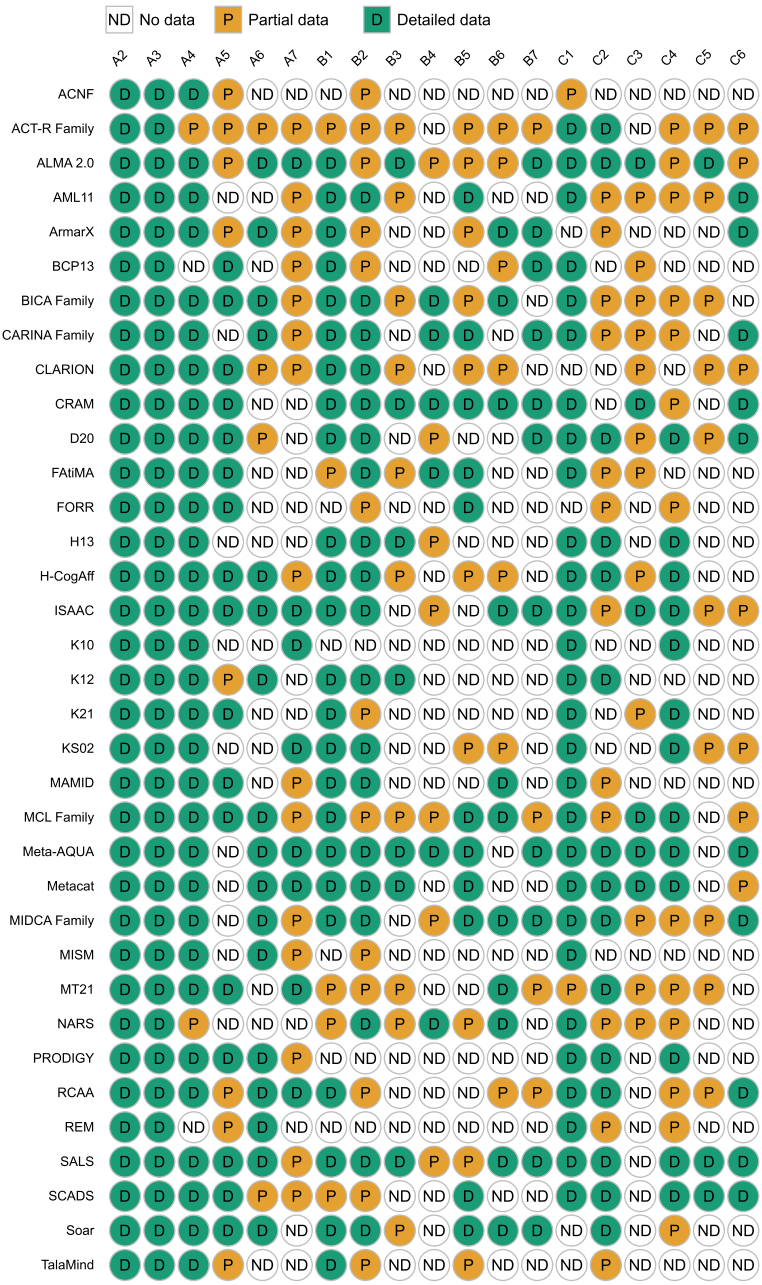



**Fig. 4** A grid graph depicting the quality of the charted data for each CMA and item

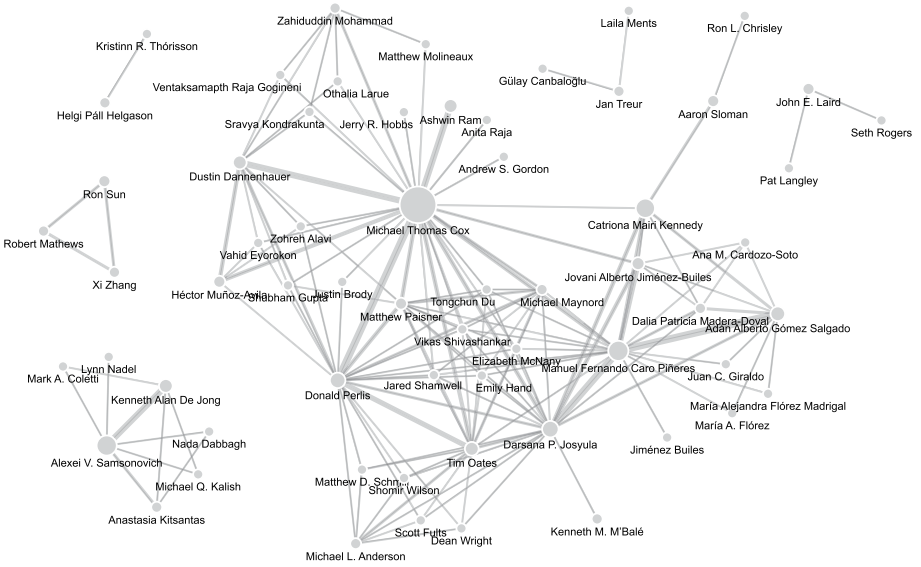



**Fig. 5** Co-authorship graph based on relevant reports where at least one author has multiple relevant reports; the companion website offers a complete version. Node size (edge thickness) reflects authored relevant reports (shared publications between co-authors)

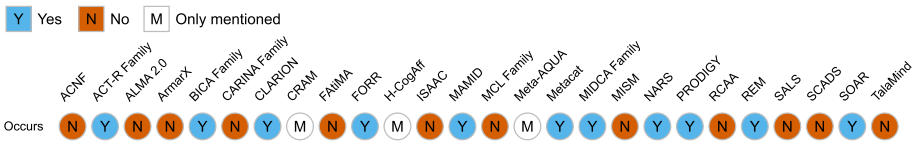



**Fig. 6** Whether the *named* CMAs also occur in Kotseruba and Tsotsos (2020)'s (2020) overview

lectively, this cluster accounts for over half of all relevant reports yet involves only 11 of the 35 CMAs (cf. Sect. 4.1): BCP13, CARINA, H-CogAff, KS02, K10, K12, K21, MCL, Meta-AQUA, MIDCA, and MISM. Some of these also share a close conceptual connection; for example, both MIDCA and MISM take inspiration from Meta-AQUA and MCL (cf. Sect. 4.5).

Outside the cluster, no author in our corpus has worked on more than one CMAs and references to others CMAs appear solely for context (additional non-relevant publications could reveal further connections). Smaller clusters correspond to Soar (John E. Laird), MT21 (around Jan Treur), BICA (around Alexei V. Samsonovich), CLARION (around Ron Sun), and H13 (Helgi Páll Helgason), and some of the secondary literature mentioned in Sect. 3.1.

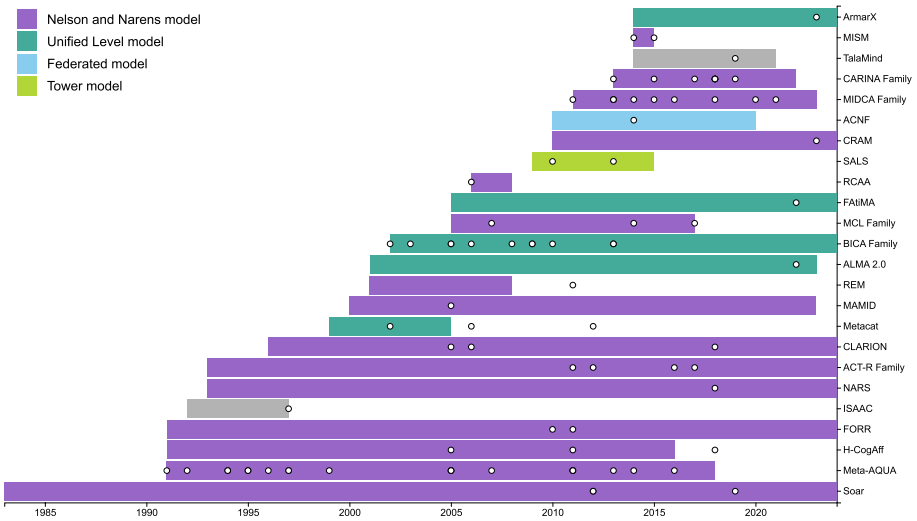



**Fig. 7** Timeline of the development of *named* CMAs inspired by Kotseruba and Tsotsos's (2020, p. 20) Fig. 2, colour-coded by metacognition model (Fig. 11). Start (end) dates are inferred from the earliest (latest) activity we identified; an interactive version on the companion website lists these. White dots mark relevant reports (potentially several)

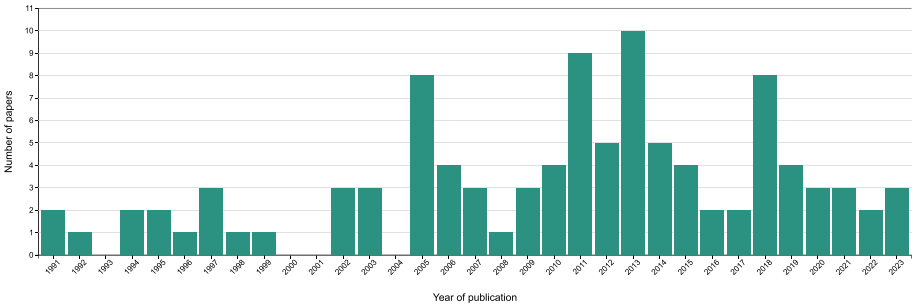



**Fig. 8** Time series plot of the number of identified relevant reports by year of publication

## 4 (A) A model context

### 4.1 (A1) Used CMAs and (A2) Publication dates

We identified 35 CMAs: 26 *named* and 9 *unnamed* (cf. Sect. 2.6). Figure 6 shows that 11 of the named architectures (42.3%) also appear in the overview by Kotseruba and Tsotsos (2020), along with three further architectures that they mention but explicitly do not analyse. Of those 14, they associated only four with our technique of interest (cf. Sect. 3.1).

Figure 7 timelines the development of named CMAs in our corpus. Unnamed CMAs are omitted because reliable dates are infeasible to identify without labels. Some CMAs—e.g., Meta-AQUA—address our topic throughout their entire lifespan, whereas FORR and

MAMID, for example, feature only isolated relevant reports, suggesting varying importance of the topic.

Figure 8 plots relevant reports per year, with activity peaking in the 2010s but remaining modest overall. Combined with mediocre data availability (Sect. 3.2), this suggests metacognitive experience traces are a niche research theme except in some dedicated CMAs. However, Sect. 2.1.3 outlines that similar techniques are common in other contexts as well.

### 4.2 (A3) Terminology

One challenge identified in our initial explorative literature search (Sect. 2.3) is the fragmented terminology regarding traces of metacognitive experiences, which complicates direct comparisons across architectures. To impose structure, we summarised—in Table 2—the most specific terminology each CMA uses for its episodic metacognitive experiences. Some CMAs, such as Soar (Laird 2019), apply no explicit label and instead treat the mechanisms as a natural function of episodic memory; we recorded the generic label nevertheless.

Although some architectures reuse phrases—e.g., CARINA inherits "ReasoningTrace" from MISM—most terms differ at least slightly. To identify latent themes we grouped the terms semi-automatically, starting with *SentenceTransformer* (Reimers and Gurevych 2019) vector embeddings using the dot-product model *multi-qa-mpnet-base-cos-v1* (Hugging Face 2021), which is tailored to semantic similarity. We then clustered the vectors with *affinity propagation* (Frey and Dueck 2007) via *scikit-learn* (Pedregosa et al. 2011). This yielded 13 clusters, which we manually labelled by overarching topic.

A review revealed overlaps and a few misassignments. For example, the algorithm sorted "traces of the system's inference activity" into the cluster "Processing Trace", but the term also fits the cluster "Reasoning Trace" because of its emphasis on inferences. Furthermore, BICA's "I-Now/I-Past/I-Meta" denote specific mental states (Samsonovich et al. 2009) and are grouped in the same cluster—but "mental states" is not. We adjusted the clusters accordingly; all changes are documented in the raw data repository. The final set appears in Table 3.

Likely because "trace" appears frequently—an artefact of its explicit mention in $q$—three clusters revolve around "trace", each with different emphasis (problem-solving, processing, reasoning). Others range from technical notions ("Performance Profile", "Processing Trace") to psychology concepts ("Episodic memory", "Experience"). To determine topic prevalence in the relevant literature, we counted distinct architectures using at least one label from each topic rather than term occurrences, avoiding pitfalls of automated frequency counts (stemming, part-of-speech ambiguity, bulk mentions, etc.). Figure 9 charts these counts.

On average, each term is used by $\approx 5.17$ CMAs ($\mathrm{SD} \approx 2.54$). The topics "Episodic Memory", "Reasoning Trace", and "Problem-solving Trace" exceed the rule-of-thumb threshold of $\mathrm{mean} + 1\mathrm{SD}$ for being comparatively frequent. Still, only a fourth of CMAs employ the most prevalent category, "Episodic memory", while $60\%$ (21/35) of the CMAs use one of the three "trace" clusters. At the other end of the spectrum, the topics "Data", "Event Records" and "Mental State" fall below the $\mathrm{mean} - 1\mathrm{SD}$ and can be considered rarely used.

**Table 2** Most specific terms identified to denote memories of metacognitive experiences

| CMA | Terminology | Sources |
|---|---|---|
| ACNF | Trace of the reasoning | Crowder et al. (2014, p. 120) |
| ACT-R | Representations of mental states, episodic information | Birlo and Tapus (2011, p. 116) and Reitter and Lebiere (2012, p. 246) |
| ALMA 2.0 | Self-knowledge | Goldberg (2022, p. 31) |
| AML11 | Inference trace | Arcos et al. (2011, p. 170) |
| ArmarX | Meta-information, traces of data through the processing pipeline | Peller-Konrad et al. (2023, p. 10) |
| BICA | Episodic-memory, mental state, I-Now, I-Past, I-Meta, imaginary self | Samsonovich (2013a, p. 111ff.; b, p. 111ff.) and Samsonovich and De Jong (2005, p. 2) |
| BCP13 | Immediate processual self, autobiographical data, autobiographical self | Brody et al. (2013, p. 3) |
| CARINA | Self-model, performance profile, internal representation of the reasoning processes, ReasoningTrace | Caro Piñeres et al. (2019a, p. 74f.) |
| CLARION | Monitoring buffer | Sun (2018, p. 29) |
| CRAM | Metacognitive experiences, traces, telemetry | Nolte et al. (2023, p. 319) |
| D20 | Replay memory, experience memory replay | Daglarli (2020, p. 98497) |
| FAtiMA | Autobiographical memory | Mascarenhas et al. (2022, p. 8) |
| FORR | History, trace of each solved problem | Epstein and Petrovic (2011, pp. 44, 51) |
| H13 | Operational history, contextualised process performance history | Helgason (2013, pp. 89, 116) |
| H-CogAff | Traces, representation | Kennedy (2011 p. 235); Sloman and Chrisley (2005, p. 146) |
| ISAAC | Metareasoning representation | Moorman (1997, p. 40) |
| K10 | Reasoning traces | Kennedy (2010, p. 1) |
| K12 | Reasoning trace, episodic memory of mental events, audit trail | Kennedy (2012, p. 2) |
| K21 | Trace | Kennedy (2021, p. 4) |
| KS02 | Trace of execution events, event records | Kennedy and Sloman (2002, p. 11) |
| MAMID | Affective state, feeling of confidence | Hudlicka (2005, p. 2f.) |
| MCL | Episodic memory | M'balé and Josyula (2014, p. 206) |
| Meta-AQUA | Trace, representational model of [...] reasoning processes | Cox and Raja (2011a, p. 7) and Gordon et al. (2011, p. 301) |
| Metacat | Temporal trace, temporal record, episodic memory | Marshall( 2006, p. 281f.) |
| MIDCA | Reasoning traces, episodic memory, representation, self-model, cognitive trace | Cox et al. (2011, p. 80); Mohammad et al. (2020, p. 4) |
| MISM | ReasoningTrace | Caro Piñeres et al. (2014, p. 94) |
| MT21 | Mental models | van Ments and Treur (2021, p. 7) |
| NARS | Mental sensation, traces of the system's inference activity, internal experience | Wang et al. (2018a, b, p. 7) |
| PRODIGY | Problem-solving trace | Carbonell et al. (1991, p. 52) |
| RCAA | Historical experience, history | Foltyn et al. (2006, p. 4) |
| REM | Trace of processing | Goel and Jones (2011, p. 154) |
| SALS | Reflective traces, experiential event streams | Morgan (2013, pp. 80, 102) |
| SCADS | Traces | Shrager and Siegler (1998, p. 408) |
| Soar | Episodic memory | Laird (2019, p. 225) |
| TalaMind | Traces of [...] execution | Jackson (2019, p. 533) |

**Table 3** Semi-automatic clusters of the Table 2 terms based on semantic similarity

| Cluster topic | Terms |
|---|---|
| Autobiographical self | Autobiographical data, autobiographical memory, autobiographical self, imaginary self, immediate processual self, self-knowledge, self-model |
| Data | Autobiographical data, meta-information, telemetry, traces of data through the processing pipeline |
| Episodic memory | Autobiographical memory, episodic information, episodic memory, episodic memory of mental events, experience memory replay, replay memory |
| Event records | Event records, experiential event streams, temporal record |
| Experience | Affective state, experience memory replay, experiential event streams, feeling of confidence, historical experience, internal experience, mental sensation, metacognitive experiences |
| History | Contextualised process performance history, historical experience, history, operational history |
| Mental state | I-Meta, I-Now, I-Past, mental state, representations of mental states |
| Performance profile | Contextualised process performance history, monitoring buffer, performance profile |
| Reasoning trace | Cognitive trace, inference trace, reasoning trace, trace of the reasoning, traces of the system's inference activity |
| Representation | Internal representation of the reasoning processes, metareasoning representation, representation, mental models, representational model of [...] reasoning processes, representations of mental states |
| Problem-solving trace | Problem-solving trace, reflective traces, temporal trace, trace, trace of each solved problem |
| Processing trace | Audit trail, trace of execution events, trace of processing, traces of data through the processing pipeline, traces of [...] execution, traces of the system's inference activity, telemetry |

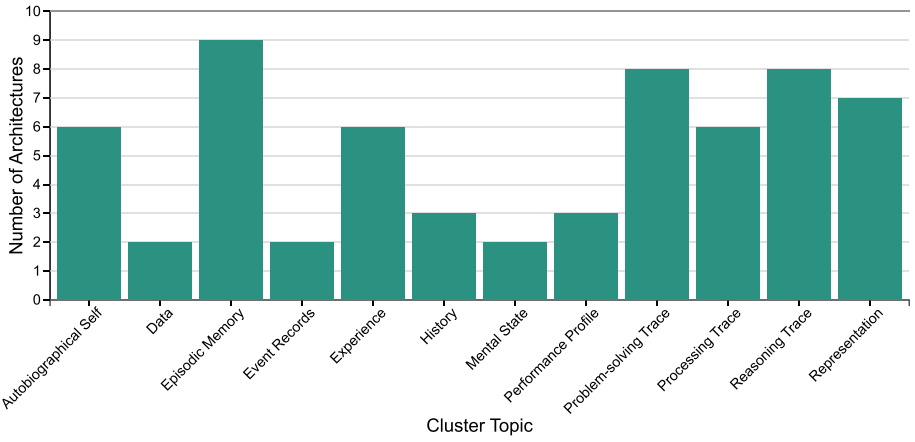
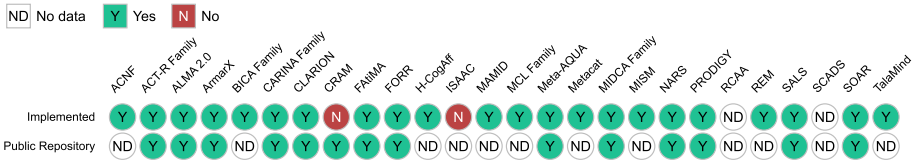



**Fig. 9** A bar chart counting the CMAs that use at least one term from each topic

### 4.3 (A4) Implementation

Over 80% of the named CMAs but only 30% of the unnamed ones include a code implementation, as documented in associated reports; see Fig. 10. This difference is statistically significant (Pearson $\chi^2$, $p < 0.001$). A plausible explanation is that larger or longer-running




**(a)** *Named* CMAs: top row shows reported implementation; bottom row adds code-repository availability (e.g., Git, SVN). The companion website links to each repository




**(b)** *Unnamed* CMAs: implementation status only; no repositories located

**Fig. 10** Implementation status (A4) for named and unnamed CMAs

projects, able to support substantial implementation work, tend to adopt explicit names for group identity. Note that a CMA is counted as implemented when at least one report mentions running code; individual ideas within that architecture may still be unrealised.

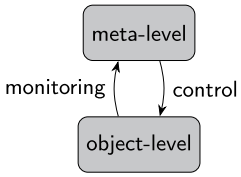
As open-source code facilitates reproducibility, we searched for public repositories for named (unnamed) CMAs, finding them in $\approx 54\%$ ($0\%$) of cases as shown in Fig. 10. Repository presence also indicates maturity: named CMAs with public code remain active for an average of 20.3 years versus $\approx 11.2$ years for those without repositories.

### 4.4 (A5) Inspiring non-computational models of cognition

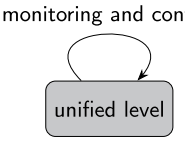
***Inspiring models of cognition***: For older CMAs such as ACT-R (Anderson 2007), CLARION (Sun 2016), and Soar (Laird 2012), the volume of cited research makes it infeasible to give an account of every influence, although psychological plausibility emerges as a key inspiration. ACNF (Crowder et al. 2014), D20 (Daglarli 2020), and BICA (Samsonovich 2013a) even map components to brain regions: BICA locates metacognitive experiences in the hippocampus and extrastriate neocortex, whereas ACNF and D20 place them in the prefrontal cortex. A notable exception is PRODIGY, which dismisses any resemblance to human cognition as incidental; instead, its "goal is at reengineering cognition the way it ought to be" (Carbonell et al. 1991, p. 54).

Some works focus on specific aspects of cognition. BCP13 (Brody et al. 2013) uses *active logic* (Anderson et al. 2008) to model *thick time*, i.e., humans' perceived conscious present (Newton 2001). Brody et al. boil this down to analysing mental processes during runtime rather than retrospectively after completion. Other specialised efforts include K21 [behaviour-change theory (Kennedy 2021)], SCADS [children's strategy discovery (Shrager and Siegler 1998)], FAtiMA [OCC (Ortony et al. 1988; Mascarenhas et al. 2022)], and ArmarX [*Multi-Store Model* (Atkinson and Shiffrin 1968; Peller-Konrad et al. 2023)].

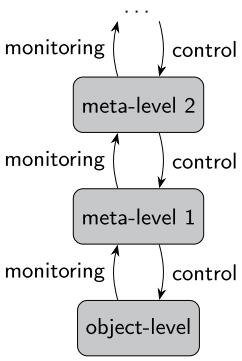
***Inspiring models of meta-cognition***: The CMAs disagree on whether storing, retrieving, and processing of episodic metacognitive experiences is an automatic, unconscious procedure or deliberate. Soar, for instance, stores episodes automatically (Laird 2012, p. 229), whereas Metacat decides which events merit storage (see Sect. 5.3). Conversely, Soar determines when and which episodes should be retrieved via its typical operator-selection mechanism (Laird 2012, p. 230), while Metacat's pattern clamping appears to occur auto-



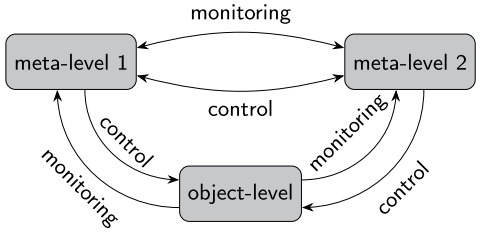

**(a)** *Nelson and Narens model*: a single meta-level regulates the object-level; see also fig. 1 on page 3




**(b)** *Unified model*: one level regulates itself

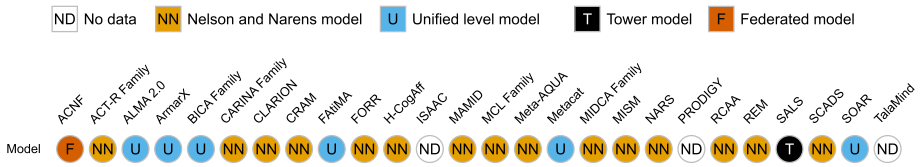



**(c)** *Tower model*: Potentially unbounded meta-levels form a stack, each regulating the one below




**(d)** *Federated model*: Multiple peer meta-levels run in parallel, each regulating the object-level(s), and one another; only two meta-levels are shown for clarity, but more are possible

**Fig. 11** Schematics of the four principal meta-level models in the surveyed CMAs




**(a)** *Named* CMAs




**(b)** *Unnamed* CMAs

**Fig. 12** Meta-level models adopted by each CMA. Few CMAs state their model explicitly; Screener A inferred classifications. These data were not considered in Fig. 4

matically, as do Meta-AQUA's Meta-Explanation Pattern (Meta XP) matching and MIDCA's metacognitive cycle (see Sect. 6.3).

How cognition and metacognition interact is clearer: Different models (Fig. 11) disagree as to whether there is a single meta-level, several, or just a unified system (Zheng et al. 2023). We classify the CMAs by the employed model as shown in Fig. 12. A majority

($\approx 57.1\%$), including the sophisticated MCL, MIDCA and MISM, adopt the early Nelson and Narens (1990) model (Fig. 11a). In accordance, MCL Family members function as add-on meta-levels to other CCAs that only feature object-level cognition (Schmill et al. 2007; M'balé 2017). Inspired by MCL (Sect. 4.5), MIDCA also emphasises this distinction as a "dual-cycle architecture", with one cycle dedicated to metacognition (Cox et al. 2016). Still, these systems' meta-levels often mirror their object-levels: For example, MIDCA's meta-level cycle repeats the object-level phases (Cox et al. 2016, p. 3713), and Meta-AQUA reuses its story understanding algorithm to explain its metacognitive experiences (Ram and Cox 1995, p. 221); see Sect. 6.3.

One fifth of CMAs reject a distinct meta-level, treating metacognition as arising from the same mechanisms as object-level cognition in a *unified level* (Fig. 11b). BICA's authors even posit that if different modules are responsible for cognition and metacognition, one may not speak of true metacognition because cognition would not be turned on itself (Samsonovich 2009). Nevertheless, BICA still keeps a mental state "I-Meta" separate from "I-Now" although a single component processes both. Advocates for a unified level often cite a familiar *Quis custodiet ipsos custodes?* argument: if metacognition requires a module separate from cognition, yet another module is required for meta-metacognition, and so forth. For instance, a meta-meta-level could balance metacognition's computational overhead with any achieved performance gain. Recursively applying this principle yields a theoretically infinite *tower* (Fig. 11c) of stacked meta-levels (Kennedy and Sloman 2002). Since human meta-metacognition is empirically supported (Recht et al. 2022), such rationales are not restricted to thought experiments. One reviewed CMA, SALS, indeed features five stacked layers, each monitoring and controlling the lower one, starting with the lowest *Built-in Reactive Layer*, to a *Super-Reflective Layer* (Morgan 2013). CMAs following Nelson and Narens' (1990) simple model deliberately ignore meta-metacognition.

A fourth strategy—the *federated model* (Fig. 11d)—appears in KS02 (Kennedy 2003; Kennedy and Sloman 2002, 2003), and, in spirit, ACNF (Crowder et al. 2014) and K10 (Kennedy 2010). Multiple peer meta-levels run in parallel, each monitoring the object-level (possibly also multiple) and its fellow meta-levels. If one of these meta-levels fails or is compromised, the others can diagnose and repair it. Although details are scarce, TalaMind's (Jackson 2019) *Society of Minds* approach might be the only account that aligns with a fifth theory, such as Fletcher and Carruthers's (2012), who view metacognition as emerging from interaction among reasoners rather than from a dedicated component.

### 4.5 (A6) Related CCAs and (A7) main innovations to those

We distinguish four types of architectural lineages:

Standalone CMAs: lack explicit parents. They borrow only general ideas from earlier CCAs and develop their algorithms and representations independently.
Awakening CMAs: extend a parent CCAs with previously lacking metacognitive capability.
Upgrading CMAs: evolve a parent CMA's existing metacognitive mechanisms. We summarize closely related upgrading lineages as *families* (e.g., MIDCA versions 1.1, 1.3, and 1.5).

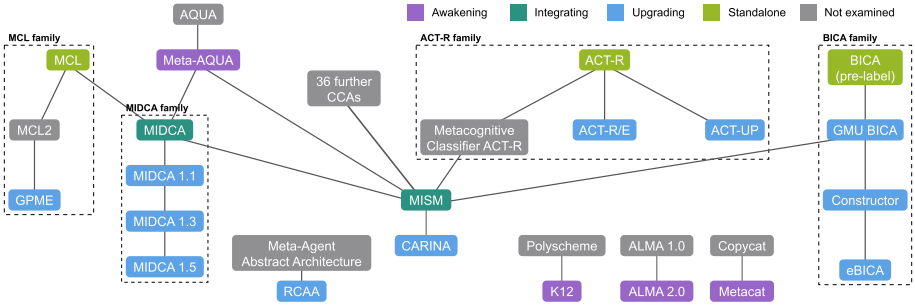



**Fig. 13** Lineages of CCAs with identified relatives, colour-coded by category. Ancestors appear above descendants; dashed boxes summarise *families*. 36 predecessors of MISM are omitted

> Integrating CMAs: intermix elements from several parent CMAs, each of which model different aspects of metacognition, to create a unified and encompassing framework.

Figure 13 traces the lineages of awakening, upgrading and integrating architectures, and summarises the families we identify. Although many reports compare newly introduced CMAs with existing ones, only 26.9% of the named architectures explicitly inherit principles from a predecessor. When including unnamed CMAs—and, to only account for the CMAs that genuinely build upon existing principles rather than introducing them outright, removing the four awakening CMAs (Metacat, K12, Meta-AQUA, and ALMA 2.0)—the share falls to 8.6%.

***Integrating CMAs*** MIDCA and MISM qualify as *integrating* CMAs. MIDCA inherits MCL's object-/meta-level split (Cox et al. 2011) and embeds Meta-AQUA's self-explanation component into its meta-level (Cox et al. 2016). However, the relevant reports do not document how Meta-AQUA's *Meta XP matching*, which resembles matching graph queries against metacognitive experience traces (Ram and Cox 1995, see Sect. 6.3), translates to MIDCA's less expressive traces structured as linked lists (Cox et al. 2016) or ordered dictionaries (Mohammad et al. 2020).

MISM occupies a different niche: it is a metamodel distilled from 40 metacognitive models and architectures (Caro Piñeres et al. 2014). Key concepts and relations were charted from a random half of those architectures, merged into a draft, and validated against the remaining half. Thus MISM incorporates elements from many CMAs reviewed here—Meta-AQUA, CLARION, MIDCA, Soar, MCL, MAMID, BICA, REM, H-CogAff—and from an ACT-R variant (Vinokurov et al. 2011) outside this review's scope. By design it is abstract: While MISM covers concepts such as "ReasoningTrace", "Memory", and "MonitoringTask" (Caro Piñeres et al. 2014, p. 94), it omits algorithmic details, offering instead UML templates that have guided implementations such as CARINA (Piñeres et al. 2018) and FUNPRO (Piñeres 2015).

***Upgrading CMAs*** We classify CARINA, RCAA, and descendants of the ACT-R, BICA, MCL and MIDCA families as *upgrading*. MIDCA versions 1.1 (Cox 2013), 1.3 (Cox et al. 2016), and 1.5 (Cox et al. 2021) each implement concepts absent from its initial design. Descendants of the ACT-R and BICA families focus on different aspects, e.g., eBICA adds emotions (Samsonovich 2013a), and ACT-R/E aims towards embodiment in robots (Birlo

and Tapus 2011). While CARINA is an instantiation of MISM, it remained unclear how RCAA improved upon its predecessor.

## 5 (B) Data content and structure

### 5.1 (B1) Data type

In their overview, Kotseruba and Tsotsos (2020, pp. 14–28) classify CCAs according to the kind of data they process—*symbolic*, *sub-symbolic*.[8], or *hybrid*. Because few architectures provide self-assigned labels, Kotseruba and Tsotsos apply explicit definitions: symbols are atomic building blocks of syntactically and semantically meaningful expressions for reasoning; sub-symbolic data is associated with the metaphor of machine learning, especially neural networks; hybrid data combines both. The relevant reports are similarly sparse, so we reuse these definitions to address question (B1). Note that (B1) focuses only on the data type of metacognitive experiences. Moreover, more than half of the CMAs we analyse are absent from Kotseruba and Tsotsos' survey (Sect. 4.1), so our study is not a mere replication.

As detailed in Fig. 14, the majority ($\approx 56\%$) of CMAs represent metacognitive experiences symbolically, $\approx 22\%$ employ a hybrid approach, and only one relies exclusively on sub-symbolic data. Typically, hybrid architectures simply attach sub-symbolic metadata to symbolic traces—utility values [ACT-R (Stevens et al. 2016)], indexing metrics [BICA (Samsonovich 2006), GPME (M'balé and Josyula 2014), MT21 (Canbaloğlu et al. 2023)], and confidence scores for candidate solutions [CLARION (Sun 2018)]. Armarx (Peller-Konrad et al. 2023) and CRAM (Nolte et al. 2023) view metacognitive experiences as discrete facts, e.g., module outputs, and link them to sensor streams and abstract reasoning events, thereby explaining how symbolic events arise from sub-symbolic data. The only purely sub-symbolic CMA, D20 (Daglarli 2020), stores raw states, state transitions, and reinforcement rewards.

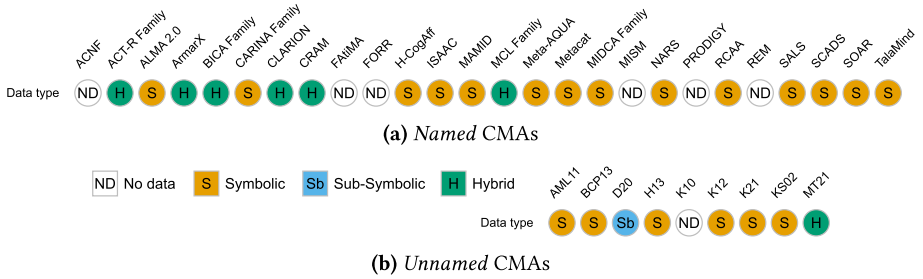



**Fig. 14** Data types used by each CMA

[8] Kotseruba and Tsotsos (2020, pp. 14–28) call architectures that predominantly use sub-symbolic data *emergent*. We avoid this to prevent confusion with the recent use of *emergent capabilities* for general-purpose AI systems (e.g., LLMs; see also Sect. 7.3), which describe behaviors that appear with scale rather than an architectural paradigm—even if both uses appeal to similar intuitions.

### 5.2 (B2) Content

***Events, States and their Types***

As we require an episodic character, it is not surprising that most metacognitive experiences are associated with a *mental event* (the terms may differ, e.g., *operator*) that reifies a time segment during which the CMAs perform a distinct cognitive activity, especially in systems featuring parallel processing. GPME, for example, merges everything its host CCA (i.e., its object-level) exposes at one "moment" into a single frame (M'balé and Josyula 2014; M'balé 2017). Nearly all CMAs label such events with a type—domain-specific, implementation-specific, or borrowed from psychology (cf. Sect. 5.5)—e.g., *Change letter-category of rightmost letter to successor* (Marshall 2006, p. 284), "copySMUtoBCPUinput" (Flórez et al. 2018, p. 624), or "memorizing" (Nolte et al. 2023, p. 326). GPME is the lone exception; it tags events only with sub-symbolic fingerprints plus metadata (M'balé and Josyula 2014; M'balé 2017).

Even the purely sub-symbolic D20 (Daglarli 2020) encodes its metacognitive experiences as events that correspond to the steps of an adapted *Q-learning* loop [see, e.g., (Jang et al. 2019)]. Tuples $(s, a, r, s'')$ capture successive states $s, s''$, the object-level's action $a$, and the reward $r$. Unlike standard Q-learning, however, $r$ is not imposed from the outside, but by D20's meta-level based on factors such as past actions and stimuli, detected errors, and "dopaminergic gain" (Daglarli 2020, p. 98499), satisfying Definition 3.

Events that signal "failure" (CRAM, Meta-AQUA, RCAA), "snags" (Metacat), "anomalies" (CARINA, the MCL Family), and so forth, might seem like prime candidates for triggering error handling. Yet Metacat (Marshall 2002) is the only CMA we know that stores them in episodic memory. More often, failures are detected *via* the other events (see Sect. 6.3).

MT21 and BICA record *mental states* rather than events. BICA conceptualises these as "discrete instances" of its self that embed schematic event descriptions such as "Verify hypothesis: Worried about wife; Answer: No" (Samsonovich (2009), pp. 2, 128). While other CMAs tacitly treat their metacognitive experiences as retrospective, BICA models different perspectives explicitly: each state bears a perspective prefix (*I*, *She*, etc.) and an attitude suffix (*Now*, *Meta*, *Imagined*, etc.), producing labels such as *He-Now-She-Now*—BICA's simulation of "his" current belief about "her" current state. BICA also considers egocentric states, e.g., its own conscious self *I-Now*, *I-Imagined*, and *I-Meta*.

Still, BICA reifies transitions between states as events: In an example from (Samsonovich et al. (2009), p. 118), *I-Now* holds the intention to drive, and *I-Next* encodes the expected post-drive state.[9] During execution, *I-Now* is continuously updated (e.g., current goal distance) and archived in episodic memory for later replay (Samsonovich et al. 2009, p. 119). States may also change attitude; for example, *I-Now* becomes *I-Past* once a significant event occurred or sufficient time has passed (Samsonovich et al. 2009, p. 119). In summary, BICA's metacognitive experiences are distributed across numerous mental states and embedded in a more encompassing simulationist *Theory of Mind* (ToM, cf. Apperly 2011) approach.[10]

[9] Although driving is a ground-level activity, processing the representation of the intention to drive is treated as metacognition.

[10] There is ongoing debate about whether (human) metacognition emerges from applying ToM to oneself (Proust 2007).

CRAM"s developers share BICA's stance and argue that "any characterization of an objective occurrence unexceptionally depends on the observers' subjective narrative" (Nolte et al. 2023, p. 323). However, instead of representing these in different knowledge graphs (such as BICA's mental states), CRAM reifies them according to its ontological commitments (see Sect. 5.4). For that, CRAM associates *Event Type*s with *Event*s defined by an observer's *Narrative*; e.g., CRAM itself might interpret some *Mental Action* as *Decision Making*. CRAM likewise represents the roles of event participants depending on its own perspective.

RCAA models both events, e.g., to later analyse their frequency, and the state of the working memory as a "very coarse digest of data from the blackboard" (Foltyn et al. 2006, p. 4).

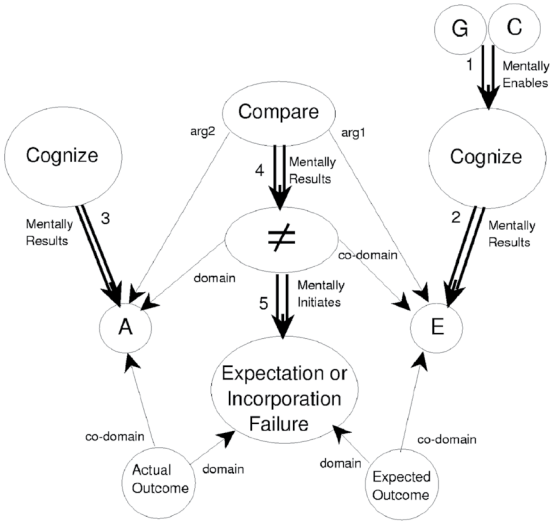
***Input and Output*** Logged events are typically associated with rich metadata, especially the input they consume and output they produce. ACT-R (Prezenski et al. 2017), ArmarX (Peller-Konrad et al. 2023), Meta-AQUA (Ram and Cox 1995, p. 221), Metacat (Marshall 2006), SALS (Morgan 2013), and GPME (M'balé and Josyula 2014) all tie logged events to a symbolic representation of their main results. These results vary widely based on the problem domain and the CMA's general functioning: GPME, for example, tracks both a raw image as processed by its host and the faces that the host detects in it (M'balé and Josyula 2014). Metacat associates input problems with identified *themes*, e.g., $abc \Rightarrow abd; xyz \Rightarrow ?$ with *a AlphabeticPosition opposite z* (Marshall 2006, p. 282). Other architectures record even richer context. ACNF links the "initial state of the problem situation (the contents of the focus, relevant coalitions of codelets, and feature values of relevant concepts, relevant registers in working memory, etc.)" with intended goals and action plans (Crowder et al. 2014, p. 119). H13 (Helgason 2013) likewise tracks internal processes together with any achieved goal states (cf. Fig. 15e).

CARINA logs extensive details for every cognitive function: input and output parameters, pre- and post-conditions, priority, activation status, goals and sub-goals, and the sequence of visited states (Flórez et al. 2018, pp. 622–623). Besides logging input and output (Cox et al. 2016), MIDCA interleaves eight types of *mental actions* with *mental states* such as "Selected Goals, Pending Goals, Current Plan, World State, Discrepancies, Explanations, [and] Actions" (Dannenhauer et al. 2018, p. 6). CRAM represents, e.g., *Premise*, *Conclusion*, *Sender*, *Receiver*, *Performer* as event participants (Nolte et al. 2023). SALS links its knowledge base *change events* to the specific bits of information added or removed (Morgan 2013).

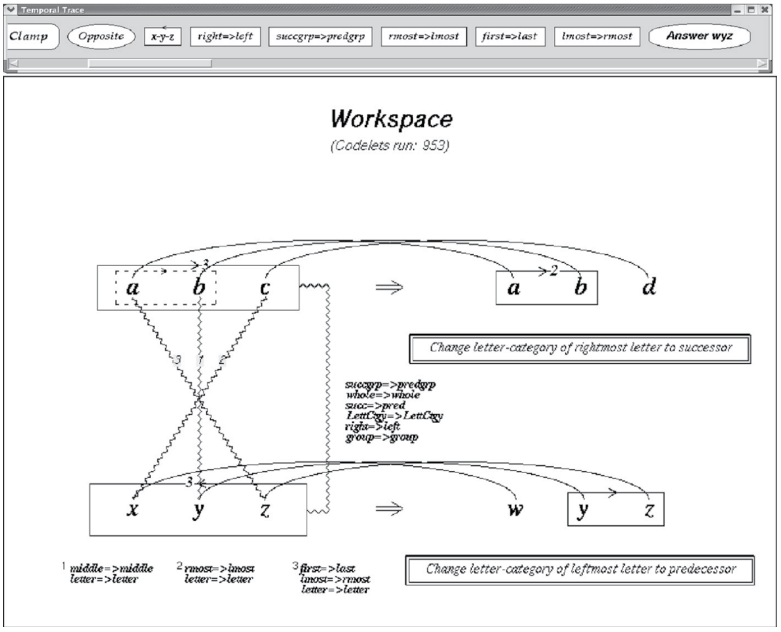
K12 (Kennedy 2012) conceptualises four parallel traces, each concerning different metadata: (1) knowledge, perception, and confidence in ensuing inferences; (2) objectives and options, plus decision justifications; (3) the attention focus and ideas generated; and (4) the "history of [...] disruptive events" and "of changes in subjective difficulty of a task due to other pressures" (Kennedy 2012, p. 3). The rationale for keeping these separate is not explained.

***Justifications***

Some CMAs log decision justifications, expressed as logical arguments or heuristic-contribution scores [e.g., FORR (Epstein and Petrovic 2011, p. 44)]. Rule-based systems sometimes log all rules fired—or even the entire inference chain—on the way to the result. Examples include KS02 (Kennedy and Sloman 2002), AML11 (Arcos et al. 2011), H-CogAff (Kennedy 2011), K12 (Kennedy 2012), BCP13 (Brody et al. 2013), CARINA (Flórez et al. 2018), K21 (Kennedy 2021), and ALMA 2.0 (Goldberg 2022). Especially logic-focused

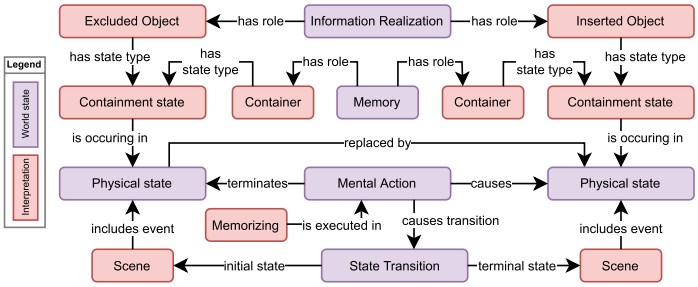



**(a)** A visualisation by Cox and Ram (1999) of Meta-AQUA's Meta XP to represent a contradiction, numbered sequentially. Used with permission. ©1999 Elsevier Science B.V. All rights reserved

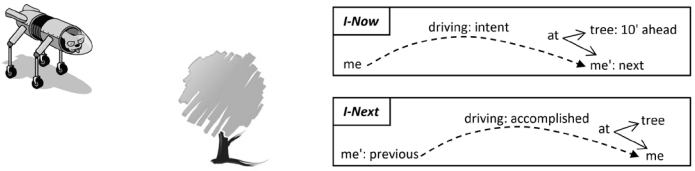



**(b)** Marshall's (2006) visualisation of Metacat's trace justifying $wyz$ as answer to $abc \rightarrow abd$ ; $xyz \rightarrow ?$. Used with permission. ©2006 Taylor & Francis Ltd. (www.tandfonline.com). All rights reserved

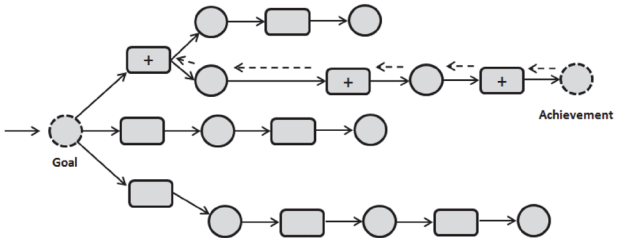
**Fig. 15** Illustrations of the metacognitive experiences of five different CMAs




**(c)** A visualisation by Nolte et al. (2023) of CRAM's metacognitive experience trace (as an OWL-ABox) of memorizing new information. Used with permission. ©2023 the authors




**(d)** A visualisation of GMU BICA's mental states (right) in working memory during driving (left). Used with permission of *World Scientific Publishing Co., Inc.*, from *The Mental State Formalism of GMU-BICA* by Samsonovich et al. in *International Journal of Machine Consciousness*, vol. 1, no. 1, 2009; permission obtained through *Copyright Clearance Center, Inc.* ©2009 Publishing Co., Inc. All rights reserved




**(e)** H13's (Helgason 2013, p. 113) execution chain of processes (rectangles) and data items (circles), potentially rewarded upon goal achievement (marked by +). Used with permission. All rights reserved

**Figure 15** (continued)

systems do not explicitly abstract such chains to events; for them, they directly trace the metacognitive experiences, whereas others would associate them with an event's main results as mentioned above. *Operator*-based systems log the operators they apply for the same reason, as in SCADS (Shrager and Siegler 1998) and Soar (Laird 2019).

***Arousal, Emotions, and Feelings of Confidence***

In some cases, the associated metadata also describe a CMA's arousal [ACNF (Crowder et al. 2014); GPME (M'balé and Josyula 2014)], emotions [ACNF (Crowder et al. 2014); FAtiMA (Mascarenhas et al. 2022); GPME (M'balé and Josyula 2014); NARS (Wang et al. 2018a, b)], and feelings of confidence [CARINA (Flórez et al. 2018, pp. 622–623); CLARION (Sun 2018); K12 (Kennedy 2012); MAMID (Hudlicka 2005)] that accompanied a mental event.

***Start and End Times***

Some systems represent the temporal dimension as an ordered sequence of discrete ticks—BCP13 increments an index for every inference step (Brody et al. 2013), and KS02 tallies its processing cycles (Kennedy and Sloman 2002)—while MIDCA (Dannenhauer et al. 2018), ACT-R (Reitter and Lebiere 2012), and CARINA Flórez et al. (2018, pp. 622–623) log the real clock time of when processes start and end. SALS (Morgan 2013) also timestamps events, while NARS (Wang et al. 2018a) stores only a delta of the elapsed interval between successive events.

***Causality***

Without parallelization—as appears to be the case for Metacat, Soar, KS02[11], and SCADS—chronological traces may be understood as causal chains. Meta-AQUA makes causality explicit by linking trace nodes via relations such as *results*, *enables* or *initiates* (Ram and Cox 1995, p. 218). For instance, the state "Run out of gas" *enables* the process "Infer", which then *results* in the state "Should have filled up with gas when tank was low" (Cox 2011, p. 144). These links are constrained by temporal order (Cox and Ram 1992, p. 124) and allow Meta-AQUA to encode complex graph flows, as shown in Fig. 15a.

The programming language Funk2, underlying SALS, also traces computations as graphs, treating parents as causes of their children (Morgan 2010). CRAM follows a similar approach to Meta-AQUA, linking events by temporality and causality, but via a formal taxonomy of causality relations. For instance, a *Memorizing* event can *terminate* a *memory-containment* state in which some information was unknown (Nolte et al. 2023, p. 326). Designers of both Funk2 and CRAM argue that such graph structures are necessary to represent parallel information processing (Morgan 2010; Nolte et al. 2023). Note that CRAM links states and events directly, whereas SALS manages them separately (Morgan 2013, p. 80).

GPME (M'balé 2017, pp. 137–138) weaves temporal and weak causal links between events and adds spatial links (co-location) plus compositional links that form an event hierarchy, though its purpose is not explained. RCAA likewise "explicitly care[s] about the order of events as they appeared" to infer causality (Foltyn et al. 2006, p. 4), but it is unclear whether its representation is linear or graph-based. Finally, visualisations (cf. Fig. 15e) suggest that H13 records a tree-shaped trace (Helgason 2013, pp. 115, 138).

***Absent Data***

Some CMAs intentionally omit specific metacognitive experiences from episodic memory. Soar drops mental imagery, substates that are reconstructable from other included data, and episodic memory retrieval events, which would "lead to extremely large episodes" (Laird 2012, pp. 228–230). BICA, by contrast, keeps an unlimited episodic memory of every lived experience (Samsonovich 2006). Instead of retrospectively searching for missing events, Meta-AQUA explicitly assigns non-monotonic truth values—e.g., a node representing some non-retrievable data is tagged "*out of the current set of beliefs*" (Cox and Ram 1991, p. 225).

[11] Although KS02 follows the federated model (cf. Sect. 4.4), each meta-level traces events as sequences (Kennedy and Sloman 2002).

### 5.3 (B3) Level of Abstraction

Like any logging system, CMAs must balance data resolution (how many steps are recorded) and utility. High resolution inflates storage and analysis cost, while too coarse resolution might impair improvements from trace analysis. Two strategies are common:

Aggregation: low-level events are summarised into more abstract events.
Selective logging: only events deemed relevant are recorded; the rest are discarded.

Resolution and abstraction, as used here, are independent: an abstract trace can still log most events (high resolution) or only a filtered subset (low resolution). Due to fundamental differences within CMAs, the issue is usually discussed qualitatively.

CMAs with the highest-resolution traces log every reasoning step. Soar, for example, can record each processing cycle, or just those leading to external action (Laird 2012, p. 229). In either case, Soar's episodic memory is low in abstraction: like BICA's (Samsonovich 2013a), it simply stores working memory snapshots. Similarly, KS02 logs every rule fired, plus checks of their preconditions, and ALMA 2.0 traces each individual inference step (Goldberg 2022).

Metacat, by contrast, records only events that exceed an (unspecified) importance threshold (Marshall 2006). The intuition given is that out of hundreds of events, Metacat deems only a few dozen relevant enough to record—chiefly the main ideas and "snags" it encounters during analogy detection.

H13 also limits episodic-memory size, deemed otherwise unmanageable (Helgason 2013). Instead of storing only critical events, it ontologically generalises goals and internal processes: instances map to classes (e.g., 3.0 to *Number*), classes to superclasses (e.g., *Chair* to *Physical_Object*), producing patterns that future events can match (Helgason 2013, pp. 110–119, 135). These abstract patterns help identify processes that proved useful previously.

According to Kennedy (2012, p. 2), K12's traces "represent mental states on a high level, and do not include the computational 'fine structure' "; a mapping can recover details when needed. Cox (1997a) similarly argues that, as in humans, "representation should have a level of detail that reflects the causal structure and content of reasoning failures" (Cox 1997a, pp. 215–216); Meta-AQUA's traces are shaped accordingly. CRAM, inspired by software tracing tools, supports flexible abstraction: complex (mental) events may be composed of simpler ones (Nolte et al. 2023, pp. 320,324–325). For instance, a discourse between executive functions and memory decomposes into queries and responses, and again into comprehending and answering incoming messages, and so forth. Funk2, the language underlying SALS, also allows flexible abstraction, e.g., machine-code tracing can be disabled (Morgan 2010).

### 5.4 (B4) Formal specifications

Only 13 of the 35 CMAs examined ($\approx 37.14\%$) rely on schemas, formalisms, or precise descriptions according to the relevant reports to represent their metacognitive experiences.

Some authors adopt mathematics. MIDCA (Dannenhauer et al. 2018; Cox et al. 2021) and CARINA (Flórez et al. 2018) use notation reminiscent of denotational semantics to

formalise trace structure. In CARINA these symbols mainly label and organise trace components rather than convey semantics. Much stricter, Helgason (2013, p. 13) proposes a compact grammar defining the syntax of valid patterns that H13 abstracts from individual events (see Sect. 5.3). As mentioned in Sect. 5.2, Daglarli (2020) models D20's metacognitive experiences as tuples of successive states, actions, and rewards. MIDCA defines its traces as "a sequence of mental states and mental actions interleaved" with formal notation (Dannenhauer et al. 2018, p. 6).

A sizeable subset of CMAs, e.g., ALMA 2.0, use ontological elements, sometimes axiomatised in formal languages. H13 (Helgason 2013) requires external formal ontology input for its generalization algorithm (cf. Sect. 5.3). CARINA (Madera-Doval et al. 2018) encodes profiling data (see Sect. 5.2) via an ontology formulated using the Web Ontology Language (OWL). CRAM extends the OWL ontology Socio-Physical Model of Activities (SOMA)—originally designed for physical experiences (Beßler et al. 2021)—to standardise metacognitive experience vocabulary (Nolte et al. 2023). CRAM's sub-symbolic data remain without formal definition; the KnowRob framework (Beetz et al. 2018) is envisioned to link them to resulting knowledge graphs in so-called Narratively-Enabled Episodic Memories (NEEMs), as it already does for physical experiences. ACNF uses the informal Cognitive Conceptual Ontology (CCO) (Crowder et al. 2014, p. 30) describing cognitrons, the federated building blocks of ACNF's computations. SOMA and CCO overlap in some aspects, e.g., both describe events, states, capabilities, goals and decisions—however, we were unable to compare their respective conceptualisations, as CCO's definitions and potential formalism seem not publicly available. ISAAC encodes its "metareasoning representation" through a semi-formal ontology combining graph and frame concepts (Moorman 1997). Meta-AQUA draws from Schank's (1986) Explanation Pattern (XP) Theory (Cox and Ram 1999, p. 10) with informal ontology commitments (e.g., the relations mentioned in Sect. 5.2); its informal ontology commitments have been adopted by MIDCA (Cox et al. 2016). SALS (Morgan 2013) uses Minsky (1975)-style frames with predefined slot values.

Instead of *description logics* (the formalism underlying OWL), *active logic* (Miller and Perlis 1993) sees some usage by both MIDCA (Perlis et al. 2013) and ALMA 2.0 (Goldberg 2022) to keep track of inference steps, though their axiomatizations are not detailed in the relevant reports. NARS, like ISAAC, employs its own knowledge representation language, "Narsese", and its traces are sequences of such sentences (Wang et al. 2018a).

Finally, some CMAs use looser formalisms. FAtiMA (Mascarenhas et al. 2022) derives its 22 encoded emotions from the OCC (Ortony et al. 1988). BICA adopts as a conceptual definition a "Mental State Formalism" (Samsonovich et al. 2009) and a shorter "Schema Formalism" (Samsonovich and De 2005, pp. 2–3) to outline mental-state content.

### 5.5 (B5) Level of specificity

We notice that CMAs label and interpret traced events (cf. Sect. 5.2) in three main ways:

*Domain-specific labels* are tied to the problem domain and do not generalise beyond.

*System-specific labels* refer directly to internal system functions and carry little abstraction.

*Anthropomorphic labels* metaphorically map system functions onto human psychology terms.

Placing a CMAs in one category is not always clear-cut because authors sometimes mix levels of specificity and seldom explain their naming choices. The following assignment is based on our subjective classification. As a notable exception, Gordon et al. (2011) argues that anthropomorphic self-models offer three benefits: (1) people naturally anthropomorphise AI agents, so matching human reasoning makes behaviour more predictable and collaboration easier; (2) re-using psychological theory is sufficiently specific while also more efficient than inventing new frameworks; and (3) shared human-like concepts (e.g., memory, attention) promote coherence and interoperability across CMAs. MIDCA's adoption of Meta-AQUA's anthropomorphic traces (see Sect. 4.5) provides anecdotal support for this view.

CMAs using *domain-specific* labels include SCADS (Shrager and Siegler 1998) and Metacat. For instance, given $abc \rightarrow abd$; $xyz \rightarrow ?$, Metacat notices matching *a*s and *b*s, logs an *Identity* observation (Marshall 2006, p. 284), then records *Change letter-category of rightmost letter to successor* (cf. Fig. 15b), corresponding to the exact insight. While Metacat also uses the anthropomorphic label like *snag*, domain-specific labels dominate.

Soar directly logs the applied operators (Laird 2012, p. 229); KS02 records rules whose conditions were tested or actions executed (Kennedy and Sloman 2002, p. 30). Operators may incidentally be named after anthropomorphic or domain-related concepts. For instance, CARINA mixes readable labels like *new_stimuli_is_detected* with technolect such as *copySMUtoBCPUinput* (Flórez et al. 2018, p. 624). ACT-R seems to also use system-specific labels, e.g., followed strategies including *tree* and *NoCommunication* (Reitter and Lebiere 2012, p. 246).

*Anthropomorphic labels* are employed, e.g., by SALS (Morgan 2013), MCL (Schmill et al. 2007) and Meta-AQUA (Cox 2011). Meta-AQUA, for instance, uses "Memory Retrieval", "Infer", and "Pose Question". As mentioned in Sect. 5.2, it links these to underlying facts, such as the inferences made, which are often domain-specific (Cox 2011); SALS does so, too (Morgan 2013). CRAM names mental events anthropomorphically while its software-component model remains system-specific (Nolte et al. 2023). The OCC (Ortony et al. 1988) used by FAtiMA (Mascarenhas et al. 2022) is also based on human emotions.

Two CMAs appear label-free: GPME uses metadata fingerprints instead, and D20 stores states and actions as raw tuples (Sect. 5.2); whether D20 labels those actions is unclear.

### 5.6 (B6) Episodic structure

Many CMAs segment memory into distinct time periods labelled *episodes* [ACT-R (Reitter and Lebiere 2012, p. 246); BICA (Samsonovich et al. 2009); MCL (M'balé and Josyula 2014); Soar (Laird 2012)], *cycle* [KS02 (Kennedy and Sloman 2002)], *phases* [MIDCA (Cox et al. 2016)], *cases* [ACNF (Crowder et al. 2014)], or *events* [CARINA (Caro Piñeres et al. 2019a)]. Segmentation criteria are often left vague—for instance, MCL opens episodes upon detecting anomalies and closes them when "the anomaly [...] decays" (M'balé and Josyula 2014, p. 207).

Soar ties each episode to a single processing cycle (Laird (2012), p. 229), looping through *input* → *operator selection* → *operator application* → *output* (Laird 2012, p. 79). Soar retrieves the most recent matching episode by cueing on a "conjunction of working-memory elements", with options for additional negative features (Laird 2012, p. 231). Afterwards, it can retrieve chronologically adjacent episodes (Laird 2012, p. 230). MIDCA's object-level

also runs in phases (Cox et al. 2016, pp. 3713), each recorded in separate segments (see Sect. 6.3).

ACT-R's episode recording is highly task-specific: In a scenario, in which it applies different communication strategies (to later compare their effectiveness), ACT-R records an episode when it receives a "useful fact" via a message sent by another agent (Reitter and Lebiere 2012, p. 246). Usefulness here is identified simply as previously requested information.

The BICA family defines an episode as one mental state or a concurrent set of states, viewed as a graph. It ignores clock time and clusters similar episodes via *cognitive maps*, realised through neural networks for efficient retrieval into working memory (Samsonovich and De 2005; Samsonovich 2006; Samsonovich et al. 2009).

When Metacat proposes a new solution to a domain problem, previous solutions with analogous *themes* (see Sect. 5.2) are activated in episodic memory (Marshall 2006, pp. 282, 300). Unlike BICA, Metacat does not score similarity via neural networks but quantifies the overlap between the descriptions of previous and proposed solutions (Marshall 2006, p. 300).

### 5.7 (B7) Information-storage data structures

Architectures store their metacognitive experiences in diverse ways; the relevant literature, however, provides limited details, leaving us certain in only a few cases: As mentioned in Sect. 5.2, some CMAs trace a single, sequential chain of reasoning. Thus, ISAAC (Moorman 1997), MIDCA (Cox et al. 2016) and CARINA (Flórez et al. 2018) utilise lists (specifically, the latter two employ linked lists), though MIDCA later migrated to ordered dictionaries (Mohammad et al. 2020). These may be accompanied by additional data structures: For example, ISAAC also features a knowledge graph based on frames and slots (Moorman 1997). Soar (Laird 2012, p. 242) records only transitions between episodes rather than complete episodes, saving space akin to *Delta encoding* commonly employed for other episodic data.

Other architectures use frames, e.g., SALS (Morgan 2013) and MCL (M'balé and Josyula 2014; M'balé 2017), or graphs, e.g., Meta-AQUA (Cox 2011, 2007) and MT21 (Canbaloğlu et al. 2023). BICA arranges both its mental states and their content as labelled-graph "schemas" (see Fig. 15d) with special root nodes representing each schema as a whole (Samsonovich and De 2003, p. 1031; Samsonovich and Jong 2002, pp. 69–70; Samsonovich 2006, p. 3).

KS02 represents traces as "a conjunction of [logical] statements that are true for that cycle" (Kennedy and Sloman 2002, p. 12). Similarly, NARS views its experience as a stream of logical statements in its custom knowledge representation language (Wang et al. 2018a).

Some CMAs employ custom data structures: After each inference, ALMA 2.0 generates hash-table-based *prospects* for "storing [...] information that is identified as potentially satisfying an inference rule" (Goldberg 2022, p. 136). ArmarX introduces the ArmarX Interpretable Data Format (IDF), a recursive composition of basic data types (Peller-Konrad et al. 2023, p. 11). As mentioned in Sect. 5.4, CRAM employs so-called *NEEMs* that link an OWL-ontology knowledge graph with sub-symbolic data.

## 6 (C) Objectives and evaluation

### 6.1 (C1) Framed purpose and (C2) Target issues

CMAs place varying emphasis on the examined technology. Soar, for example, does not explicitly model metacognitive experiences; they are simply by-products of its episodic memory logging every applied operator, and while Jackson (2019, 2014) notes that Tala-Mind can in principle reflect on past thoughts, this capacity appears to have been illustrated only in a single brief demonstration. By contrast, for CMAs such as MIDCA, Meta-AQUA, and BICA, the technique is central to overall operation, and the associated literature describes it in depth.

Authors claim the technique addresses many challenges on the path to AGI. Research targets include (super)human-level **learning** [ACNF (Crowder et al. 2014); BICA (Samsonovich 2009); CRAM (Nolte et al. 2023); FORR (Epstein and Petrovic 2011; Samsonovich et al. 2010); MCL Family (M'balé and Josyula 2014; M'balé 2017); Meta-AQUA (Ram and Cox 1995; Cox 2005a, 2007); PRODIGY (Carbonell et al. 1991); RCAA (Foltyn et al. 2006); REM (Goel and Jones 2011); SALS (Morgan 2010, 2013); SCADS (Shrager and Siegler 1998); Soar (Laird 2012)], **attention** [D20 (Daglarli 2020); H13 (Helgason 2013)], **autonomy** [MCL Family (Schmill et al. 2007); RCAA (Foltyn et al. 2006)] or **emotions** [BICA (Samsonovich 2013a, b); CLARION (Sun et al. 2022; Sun 2018); D20 (Daglarli 2020); H-CogAff (Kennedy 2018); K12 (Kennedy 2012); MAMID (Hudlicka 2005)]. Some systems pursue benchmark performance—variants of the **Turing Test**, for example [K12 (Kennedy 2012); Metacat (York and Swan 2012)]—whereas others aim to emulate **consciousness** [BCP13 (Brody et al. 2013); BICA (Samsonovich and Nadel 2005); D20 (Daglarli 2020); NARS (Wang et al. 2018a)] or **self-awareness for human-agent collaboration** [ACT-R (Birlo and Tapus 2011); BICA (Samsonovich and De 2003; Samsonovich 2009); K10 (Kennedy 2010); MT21 (van Ments and Treur 2021)]. Other purposes were more subtle: **explaining human behaviour** [CLARION (Sun 2018); MAMID (Hudlicka 2005)], **justifying answers** [Meta-AQUA (Cox 2011); Metacat (York and Swan 2012)], **resisting manipulation** [KS02 (Kennedy 2003)], **breaking cyclic behaviour**[12] [Metacat (Marshall 2006); SCADS (Shrager and Siegler 1998); Soar (Laird 2012)], and **recalling prior solution successes and failures** [CRAM (Nolte et al. 2023); H13 (Helgason 2013); Metacat (York and Swan 2012); Soar (Laird 2012)].

Awakening CMAs (Sect. 4.5) are furthermore motivated by their parent's shortcomings. ALMA 2.0 improves ALMA by allowing nested logical statements (Goldberg 2022). SALS learns to anticipate consequences of mental actions, which its predecessor EM-ONE was not capable of (Morgan 2013). Meta-AQUA augments AQUA's story understanding system (Ram 1989) by selecting compensatory learning strategies when comprehension fails (Ram and Cox 1995). Metacat remedies Copycat's tendency to repeat failed solutions (Marshall 2006) and its inability to explain why it prefers one solution over the other (York and Swan 2012).

Besides Meta-AQUA, CARINA, KS02, and MIDCA also employ traces for anomaly detection: CARINA establishes runtime baselines to flag outliers (Madera-Doval et al. 2018, p. 199), KS02 inspects traces to locate rules corrupted by malware [(Kennedy and

[12] Kotseruba and Tsotsos (2020, p. 57) already identified this as an important application of the surveyed technique.

Sloman 2002); see also Sect. 6.3], and MIDCA reuses Meta-AQUA's meta-level for the same purposes, plus for switching between fast and slow subsystems (Dannenhauer et al. 2014).

Although CMAs such as Metacat (York and Swan 2012), Soar (Laird 2012, p. 234), Meta-AQUA (Cox and Raja 2011a), and CARINA (Piñeres et al. 2018) name explainability as a primary goal, raw traces are often inscrutable even to experts, necessitating user interfaces or self-explanations when interacting (Cox 2011, pp. 133, 142). Metacat, for example, offers a graphical tool (cf. Fig. 15b) to drill-down into event metadata (Marshall 2006, p. 287).

When BICA "wakes up" from sleep, it reinstates prior goals and thoughts (Samsonovich and De 2005, p. 4), though more concrete details were not provided. Similarly, FAtiMA (designed for video game non-playable characters) can recall emotions and their causes from earlier sessions, but specifics are limited (Mascarenhas et al. 2022).

Two projects pursue standardisation: MISM surveys shared CMAs features (Caro Piñeres et al. 2014), and CRAM proposes an ontology of metacognitive experiences (Nolte et al. 2023).

Finally, Kennedy (2012) argues that tracing metacognitive experiences is a precursor to ToM: By putting itself in another agent's shoes, a CMA could infer that another agent broke a vase not out of malice, but because it "didn't see the vase or didn't know that it was fragile" (Kennedy 2012, p. 1). The necessary simulation of the other agent's mental state should be learned using the traces of the CMA's own metacognitive experiences as training data.

### 6.2 (C3) Worked examples

We consider whether authors exemplify how their CMA uses traces of metacognitive experiences, meaning step-by-step walkthroughs of tangible domain problems—evaluation scenarios and case studies are covered in Sect. 6. Unfortunately, concrete examples are rare, with most relevant reports omitting them or providing only cursory ones.

In domain problems for Metacat and SCADS, for instance, episodic memory is treated as a black box (Marshall 2006; York and Swan 2012; Shrager and Siegler 1998). BICA literature offers only one high-level reference to episodic memory on waking (Samsonovich and De 2005, p. 4); while BCP13's and H-Cogaff's provide similarly brief examples of rule-based process shutdown on timeout and interaction between multiple meta-levels respectively. Madera-Doval et al. (2018, p. 625) offered a trace snapshot of CARINA's perception process, and Nolte et al. (2023) provide an example instantiation of CRAM's representational scheme, but neither explained how the traces are exploited. Several small examples show how GPME (M'balé 2017) merges recorded episodes and updates response arsenals based on success rates.

Meta-AQUA is the notable exception with well-documented scenarios—e.g., about a drug bust (Ram and Cox 1995), shooting a Wumpus (Cox 2007), or forgetting to fill up gas (Cox 2011)—explaining in detail how traces support story comprehension. MIDCA, Meta-AQUA's successor (Sect. 4.5), also features a detailed example of learning consequences of domain-level effects from its metacognitive experiences (Dannenhauer et al. 2018).

For some CMAs, we identified visualisations of the metacognitive experiences that help grasp underlying principles more easily. We reprint examples for each such CMA in Fig. 15.

### 6.3 (C4) Processing algorithms

The major cognitive processing of many surveyed CMAs—their *cognitive cycle*—is roughly divided into three subsequent phases: (1) perception, (2) decision, and (3) action. This division appears either explicitly [BICA (Samsonovich 2013a), CARINA (Piñeres et al. 2019), KS02 (Kennedy and Sloman 2002), MIDCA (Cox et al. 2016), Soar (Laird 2019)] or implicitly [Metacat (Marshall 2006), Meta-AQUA[13] (Cox 1994; Ram and Cox 1995, p. 216), MISM (Piñeres 2015)]. To process their metacognitive experiences, most CMAs employ strategies analogous to those used in object-level problems, mirroring the object-level cycle phases, resulting in a *metacognitive cycle* [explicitly termed as such by, e.g., MIDCA (Cox et al. 2016)]. This approach simplifies and facilitates development, particularly for engineers familiar with the object-level. Our qualitative analysis identified five phases within a typical metacognitive cycle (Fig. 16), though some CMAs may intertwine or skip phases:

1. *Modelling*: Monitored data are translated into an introspective model of the object-level.
2. *Diagnosis*: Object-level performance is analysed via the constructed model, focusing on potential deficiencies, including errors and sub-optimal behaviour.
3. *Pinpointing*: Deficiency causes are sought, often using blame-assignment approaches.
4. *Strategizing*: Recovery strategies are devised and one suitable strategy is selected.
5. *Repairing*: The selected strategy is detailed and the object-level is directed to execute it.

The ongoing monitoring process, conceptualised outside these phases, serves as the meta-level counterpart to sensory perception. Accordingly, the *Modelling* phase corresponds to transforming sensory information into abstract concepts for further processing. The *Diagnosing*, *Pinpointing* and *Strategizing* phases jointly form the equivalent of the object-level decision phase, where the next optimal action is normally sought. Note that meta-level goals seem rarely explicitly represented and can be implicitly summarised as

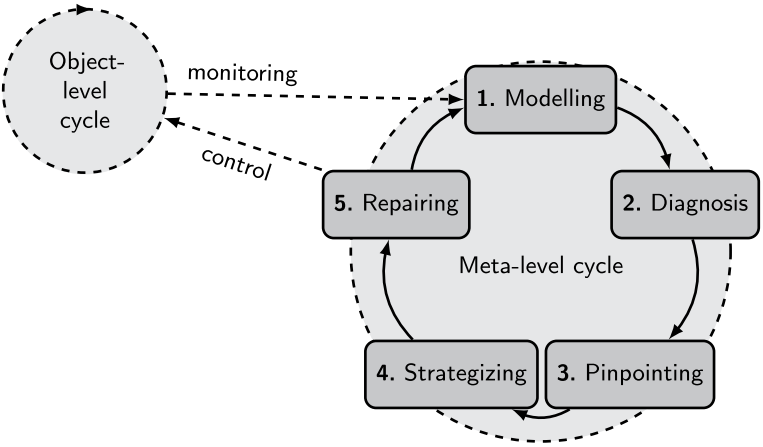



**Fig. 16** Flowchart of a typical metacognitive cycle plus communication with the object-level. When not using Nelson and Narens's model (Fig. 11), the latter is adapted as necessary

[13] Although Meta-AQUA does not parse the stories it analyses (1995, p. 216)—i.e., it lacks object-level perception—its metacognitive monitoring can be viewed as meta-level perception.

*achieving optimal object-level performance*. Finally, the *Repairing* phase corresponds to the object-level action phase, with control information flow corresponding to object-level manipulation of the ground-level [as conceptualised by Cox and Raja (2011a); see Fig. 1].

Since discussing each phase in isolation would require readers to track multiple appearances and algorithms of the same system across scattered sections, we adopt a thematic grouping approach that instead clusters systems by shared techniques or concepts.

### 6.3.1 Predefined pattern-based approaches

During Diagnosis, some CMAs rely on predefined patterns of undesired behaviour, which can be mapped a priori to error sources and solution strategies for Pinpointing and Strategizing.

Meta-AQUA, for instance, reapplies its Explanation Pattern (XP), which it uses to explain causality in stories, to traces of its own object-level (Cox 1994; Ram and Cox 1995): Different Introspective Meta-Explanation Patterns (IMXPs) associate graph structures called *Explanation of failure* with failure types, pointers to probable failure causes in graph-shaped traces, and compensatory learning strategies. Failure types include contradictions (whose IMXP is shown in Fig. 15a), impasses, surprises, false expectations, unexpected successes, and memory failures (Cox 1995, 1997a). When an *Explanation of failure*, defined as a LISP data structure (Cox 1994), matches part of the trace, Meta-AQUA strategises a learning goal using the associated learning strategy and data indicated by failure cause pointers. For example, if Meta-AQUA fails to explain a new story and the *Novel Situation Failure* pattern matches the monitored trace, failure pointers indicate insufficient object-level XPs. The system then determines learning goals to generalise identified XPs for the misinterpreted situation and invokes appropriate learning algorithms (Cox and Ram 1994). Note that some IMXPs may match correct behaviour, e.g., to identify when the system made a successful prediction.

Inspired by Meta-AQUA, AML11 adapts error explanations specifically for CBR. When CBR-suggestions deviate from a final solution, learning mechanisms trigger similarly to Meta-AQUA, but based on solution differences rather than processing trace pattern matching.

Metacat searches for "patterns in its own behaviour in much the same way that Copycat [Metacat's parent, cf. Section 4.5] perceives patterns in letter-strings" (Marshall 2006, p. 284). Although details remain unclear, Metacat monitors object-level event traces for undesirable patterns, such as repeated proposals of similar but incorrect ideas. Upon detection, Metacat *clamps* the pattern, reducing the likelihood that defining themes and concepts will be considered by its probabilistic codelets, thus allowing exploration of alternative ideas. When no important events occur for extended periods, Metacat can unclamp patterns to mix newly generated ideas with suppressed topics. Conversely, if excessive clamping occurs without progress, Metacat will "gracefully" quit the solution search (Marshall 2006, p. 289).

Kennedy (2018, p. 7) planned to use predefined trace patterns to detect "involuntary" motivational state changes in H-CogAff; no follow-up work was found.

Foltyn et al. (2006) detail RCAA's monitoring process: They conceptualise listeners registered to its working memory that asynchronously update episodic memory, which is further processed into *detailed* and *aggregated* models. The latter contains statistics about the

former, e.g., event frequency extracted from the explicit event history. These values, including "agent's success ratio [...] or the current system load", are weighted separately for each component to compute an overall utility score for currently employed algorithms (Foltyn et al. 2006, p. 4). Low-scoring algorithms may be discarded or restructured, for which *inductive logic programming*-based planners also consider monitored object-level events. This approach is predefined in that considered values and component-specific weights seem set a priori.

Brody et al.'s (2013) approach towards monitoring and modelling involves an inference engine for *active logic*—a temporal logic linking step-wise inferences to incrementing time symbols—that adds reserved predicates $Applying\,(t, r, a, c)$ to the database whenever "at time $t$, the inference rule [...] $r$ was applied with the antecedents [...] $a$ to derive the consequent [...] $c$" (Brody et al. 2013, p. 6). These facts become available to all inference rules for application in step $t + 1$, thus becoming subject to reasoning themselves. Notably, this approach is a riposte to metacognition that only analyses completed reasoning processes as it enables fine-grained reflection of individual inferences while reasoning is still underway.

ALMA 2.0 supports predefined nested logical rules, applied to examples such as reasoning with defaults and over past beliefs by quoting those in nested rules (Goldberg 2022).

### 6.3.2 Machine learning-based approaches

Instead of using rigid, predefined patterns for diagnosis, many CMAs employ machine-learning techniques that detect deficiencies more flexibly. We distinguish three approaches:

***Training-based Anomaly Detection***

Some meta-levels learn normal object-level behaviour during an initial training phase with any nonconforming activity considered anomalous during the subsequent *operational phase*.

The earliest such approach is KS02 (Kennedy and Sloman 2002). Inspired by *signature-based intrusion detection systems* [see (Gupta et al. 2023)], KS02 builds a "database of 'normal' patterns" expected in every cognitive-cycle trace (Kennedy and Sloman 2002, p. 13). KS02 generates *detectors* for each traced event, weighting them proportionally to their occurrence frequency across cycles. During operation, anomalies are identified when detectors with $100\%$ weighting fail to match events in any cycle. KS02 then attempts to identify damaged rules by absent events (rules that failed to fire) and repair via backup mechanisms. Kennedy and Sloman (2003) extended KS02 to count "unfamiliar activity" as anomalies but only react when accompanied by performance decreases, identified through simple thresholds (e.g., when energy levels fall below minimum training values). Unlike Meta-Aqua's hard-coded graph queries (Sect. 6.3.1), KS02's trained detectors correspond to simpler atomic queries.

Notably, Kennedy (2003, p. 118) argues that monitoring components should remain immutable to prevent meta-level manipulation—even the object-level must not tamper with the trace production. KS02's threat protection complicates monitoring as it possesses two object-levels, each with meta-levels that also monitor each other (recall Sect. 4.4). The meta-levels constantly interleave phases asynchronously, monitoring not only the other meta-level's operational phases but also training phases and transitions (Kennedy and Sloman 2002, pp. 15–19). A training meta-level may even corrupt another operational meta-level's object-level to monitor the exercised control. Kennedy and Sloman (Kennedy and

Sloman 2002, pp. 18) note a limitation in their setup with two equal meta-levels: a meta-level with anomalous behaviour is blind to repair interventions by the other meta-level, requiring a third equal meta-level. Kennedy (2010, 2011, 2012) envisions similar pattern learning for K10s, H-CogAff, and K12s.

***Continuous Learning Approaches***

The second approach forgoes training/operation division, with meta-levels constantly manipulating object-levels through experimentation to find improvements.

SCADS's meta-level, for instance, rearranges steps of object-level algorithms in 5% of runs (Shrager and Siegler 1998). Trace heuristics analyse efficiency and delete redundant calculation sequences, while consistently more efficient variations are added to the repertoire.

Morgan (2013, p. 81) conceptualises SALS's monitoring as event reification by transforming raw knowledge base changes (at any layer; Sect. 4.4) into symbolic events processable by higher layers. Each atomic change in lower-layer knowledge bases is communicated asynchronously to the monitoring layer via buffering streams, then recorded in layer-specific *Partial State Event Knowledge Bases*. A second stream of *activated resources* (physical or mental actions executed by lower layers) triggers learning components. SALS improves planning by computing *transframes* (Minsky 1975) between before/after states of activated resources, looked up from the Partial State Event Knowledge Base. During planning, the generated transframes function as consequence hypotheses of lower-layer activated resources.

H13, FORR and ACT-R use episodic memory to identify processes or strategies that are useful to achieve their goals. H13 (Helgason 2013) heuristically computes utility scores for past processes associated with goals or goal patterns (cf. Section 5.3) based on another heuristic, goal priority, back-propagated through process history. When new goals match episodic memory patterns, processes deemed useful via utility are activated to contribute to the goals (Helgason 2013, pp. 134–138). When they successfully contribute to goal achievement, H13 reinforces their utility and durability scores. H13 may also forget pattern-process associations when decaying durability values fall below some threshold. ACT-R (Reitter and Lebiere 2012, p. 246), following *Instance-based learning*, constructs episodic memories of its object-level actions, associating them with employed strategies and successfulness. Episodes involving identical strategies merge into a "blended episode", representing average strategy success. FORR also weighs object-level heuristics suggesting specific actions based on training instances extracted "from traces of its [FORR's] problem-solving behaviour" (Samsonovich et al. 2010, p. 221). Beyond action correctness, instance associate with "problem state, the available choices there, and the decision made" (Epstein and Petrovic 2011, p. 51).

MT21 describes the use of mental models via a combination of reification and Hebbian learning (Canbaloğlu et al. 2023; van Ments and Treur 2021). Mental models in this approach are networks with nodes having activation values and connection strengths that change over time using the "fire together, wire together" heuristic of Hebbian learning. MT21's specification leaves the semantic content of object-level network nodes unconstrained, and nodes need not be semantically opaque as in most neural networks; for example, there can be nodes representing particular stimulus conditions, nodes for specific reactions, etc. The various update coefficients of nodes in lower-level networks are reified as nodes in higher-order networks, which then model the reactivity and plasticity of lower-level networks and thus

control their adaptability. This approach also eliminates separation between training and operational phase—network states change continuously based on input signals and network response parameters, with network connectivity itself also being changeable.

D20 (Daglarli 2020) employs a Q-learning variant with rewards not externally imposed but determined by D20's meta-level implemented as a combination of neural networks (cf. Sect. 5.2). It is unclear whether this fits the first (offline) or second (online) category.

***Case-Based and Explanation-Based Reasoning***

As noted in Sect. 2.1.3, some architectures use techniques reminiscent of CBR:

PRODIGY's (Carbonell et al. 1991) object-level solves domain problems via search trees, with nodes representing world states. Starting from the root (the current state), the tree expands iteratively until reaching a node representing the goal state. This involves selecting promising world states to expand and operators to apply, creating child nodes with updated world states. So-called *control rules* advise which nodes to select, reject, or prefer. PRODIGY's meta-level employs different optimisation strategies. One module develops new control rules: candidates are included if their expected search time reduction, estimated from past problem solution traces, outweighs the overhead of permanently checking rule applicability. Although PRODIGY's rejecting rules suppress certain ideas similarly to Metacat's pattern clamping, they facilitate long-term learned behaviours rather than short-term reactions. Another CBR-like module guides object-level search by remembering past problem solutions.

Three further CMAs use CBR, though relevant reports lack details: Metacat stores previous problem solutions and attempts to reapply them to novel but similar problems (Marshall 2006, pp. 303–304). ACNF uses CBR to learn new action plans (Crowder et al. 2014). CARINA employs CBR as an episodic memory (Caro Piñeres et al. 2019a, p. 80).

As mentioned in Sect. 6.3.1, AML11 also incorporates a CBR system; however, it functions as the monitored and controlled object-level rather than the meta-level.

### 6.3.3 Hybrid approaches

CARINA profiles its cognitive functions to learn their expected behaviour (Piñeres et al. 2018, p. 532), representing expectation as logical axioms (Caro Piñeres et al. 2019b, p. 85). Sensor information violating these constraints triggers CARINA to explain the anomaly either by checking whether past explanations apply (similar to CBR; see Sect. 2.1.3) or by analysing reasoning traces. The only information provided is that "the reasoning trace analysis strategy performs a more complex reasoning" (Caro Piñeres et al. 2019a,p. 85) via querying.

MIDCA employs a hybrid approach: The *D-track* uses machine learning models (e.g., Growing Neural Gas (GNG) networks and Bayesian networks of ontologies) to detect anomalies by comparing current observations with statistically learned baselines (Cox 2013; Cox et al. 2013). Only when the *D-Track* identifies anomalies does MIDCA investigate further via the computationally intensive *K-Track*, which uses metacognitive experience traces similarly to Meta-AQUA. As MIDCA's object-level and meta-level each comprise six phases [*perceive* → *interpret* → *evaluate* → *intend* → *plan* → *act* (Cox et al. 2016, p. 3713)], MIDCA's meta-level first perceives a sub-trace of the complete monitored object-level trace—a fixed-length suffix representing the last object-level phase (Cox et al. 2016, p. 3716). MIDCA then inspects the trace to identify expectation violations and, if

found, generates corrective goals, selects which to pursue, plans, and executes necessary actions (Cox et al. 2016, p. 3716). Note that metacognition does not run parallel to object-level cognition in this conceptualisation.

#### 6.3.4 Unclear approaches

Soar's episodic memory offers multiple capabilities. For instance, it enables Soar to counter repetitions by moving "forward through episodes to determine which action(s) it took in similar situations, as to avoid repeating them in the present situation" (Laird 2012, p. 233). Regarding other capabilities such as learning consequences of its own actions (represented as *operators*), uncertainty remains whether these are associated with metacognitive experiences, as given examples focus on domain-level actions. Theoretically, these might transfer to object-level cognition, which is also partly represented as operators (such as episodic memory retrieval itself). Soar must deliberately query its episodic memory; however, when it does so and how it uses retrieved information appears implementation-specific for each Soar agent.

Numerous CMAs (BICA, CLARION, ISAAC, and others) lack sufficient detail in relevant reports for us to be confident about understanding their algorithms. For instance, BICA's authors conceptualise belief revision for failed expectations (Samsonovich 2006, p. 6) and mental task distribution across different mental states (Samsonovich et al. 2009). In one example, BICA's mental state *I-Meta* controls imagined mental states *I-Imagined* for planning, making envisioned movement steps closer to the goal more likely to be developed further (Kalish et al. 2010). However, the latest relevant report states that "Details of operation of the metacognitive component should be developed based on empirical studies" (Samsonovich 2013b, p. 103).

### 6.4 (C5) Test scenarios and (C6) Evaluation

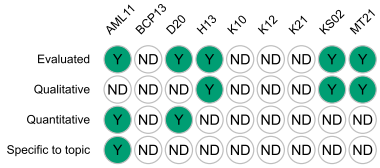
#### 6.4.1 Evaluation challenges for metacognitive experiences

In most relevant reports, metacognition enables systems to solve new problems instead of improving general performance (cf. Sect. 6.1). While some CMAs may be well-evaluated, the specific technique of leveraging metacognitive experiences is often assessed qualitatively as proof of concept rather than through quantitative comparative testing as detailed in Fig. 17. Typical ablation tests comparing systems with metacognition activated versus deactivated are often impractical, as the latter might fail entirely when lacking graceful degradation.

For instance, since the purpose of SCADS is mimicking how children learn instead of efficient summation (cf. Sect. 4.4), Shrager and Siegler (1998) evaluated whether it matches child behaviour, such as detecting identical summation strategies and tending towards simpler ones. However, isolating the effect of SCADS' metacognitive experiences is challenging since they are crucial for its learning capability (cf. Sect. 6.3). Removing these without substitute would likely eliminate the system's learning ability, rendering it non-functional for its intended purpose. Therefore, comparing SCADS with and without trace analysis is impractical—comparing functional vs non-functional systems yields no meaningful insights.




**(a)** *Named* CMA




**(b)** *Unnamed* CMA

**Fig. 17** Whether CMA were evaluated in relevant reports, if any evaluation was qualitative or quantitative, and if they tested performance gains from metacognitive experiences

This highlights a significant challenge: without viable methods to compare metacognitive experience processing against other techniques, ascertaining their unique performance contribution is difficult. One potential approach involves reframing system purposes for evaluation. For instance, if we consider SCADS' purpose as efficient summation rather than simulating children's learning, we could compare its object-level alone (capable of summation but not learning) with the complete system. However, this is retrospectively unfeasible with the information provided by Shrager and Siegler (1998), who only discuss the complete system's performance before and after learning. Certainly, we could assume SCADS' object-level performance equals the complete system's pre-learning performance. However, as described in Sect. 6.3, SCADS' meta-level introduces algorithmic variations in $5\%$ of cases to discover more efficient calculation sequences. Many random variations are likely inefficient or produce incorrect results. Conversely, without the meta-level, the object-level would consistently use correct, though not optimal, known algorithms without such random errors. This suggests that object-level performance alone might exceed the complete system's pre-learning performance. Consequently, evaluations not specifically focused on processing metacognitive experiences are challenging to transfer to that context.

### 6.4.2 Quantitative evaluation results

Only a small minority ($\approx 17\%$) of examined CMAs had relevant reports containing a quantitative ablation test where identical agents performed tasks both with and without the surveyed technology enabled. Specifically, we found such evaluations only for approaches employed by ACT-R (Reitter and Lebiere 2012), AML11 (Arcos et al. 2011), BICA (Kalish et al. 2010), CARINA (Caro Piñeres et al. 2013), Meta-AQUA (Cox 1996, 1997b), and MIDCA (Dannenhauer et al. 2018; Cox et al. 2021). The latter five, each running as a complete system with fully utilised metacognitive experiences, consistently outperformed more naïve versions. For instance, in Caro Piñeres et al.'s (2013) experiments, the CARINA-based FUNPRO tutoring system was tested across multiple sessions with progressively

fewer resources available, measuring successful resource recommendation rates. Comparing system performance with and without memory event tracing activated revealed that metamemory-driven adaptation consistently yielded higher proportions of valid resource retrievals (e.g., $98\%$ vs. $71\%$). In contrast, activating instance-based learning (cf. Sect. 6.3.2) generally impairs ACT-R's performance on specific problem solving tasks, except when facing randomised or downsized problem variations, where it also outperforms naïve versions (Reitter and Lebiere 2012).

### 6.4.3 Alternative evaluation approaches

Some CMAs were evaluated on measures other than task performance. Similar to SCADS, CLARION focuses evaluation on comparing agent performance to human performance on expertise-requiring tasks, particularly whether artificial agents make similar errors, such as overconfidence tendencies in wrong answers on certain tasks (Sun et al. 2005). Developers of both Metacat (Marshall 2006) and ALMA 2.0 (Goldberg 2022) discuss additional skills compared to predecessors, exemplified through case studies. Madera-Doval et al. (2018) present another case study where CARINA's meta-level exemplarily identifies an anomaly based on learned expected runtime of object-level cognitive functions. MISM as a metamodel and CRAM's scheme SOMA were evaluated using methodologies strictly defined in their respective development processes: MISM through comparison with CMAs not contributing to its initial design (Caro Piñeres et al. 2014), and CRAM through checking whether the representational scope defined via *competency questions* is met (Nolte et al. 2023)—a common ontology evaluation technique from Grüninger and Fox (1995). Morgan (2013, pp. 139–151) analyses computational complexity of metacognitive monitoring and control in SALS.

### 6.4.4 Indirect evaluations and domain limitations

Numerous reported evaluations did not specifically target the technique of interest—their success may only indirectly support metacognitive experiences. Several case studies were performed for D20 (Daglarli 2020), FAtiMA (Mascarenhas et al. 2022), RCAA (Foltyn et al. 2006), REM (Goel and Jones 2011), and MIDCA (Dannenhauer et al. 2014; Cox et al. 2016; Mohammad et al. 2020; Cox et al. 2021). ISAAC was tested by answering identical reading comprehension questions as human participants for short science fiction stories, with answers graded anonymously by teachers alongside the human responses (Moorman 1997). Statistical tests showed ISAAC's performance to be comparable to or better than human readers.

Importantly, most experiments involve very limited domains: **simulated games or environments** [ACT-R (Reitter and Lebiere 2012); BICA (Kalish et al. 2010); FAtiMA (Mascarenhas et al. 2022); FORR (Epstein and Petrovic 2011); MIDCA (Dannenhauer et al. 2014; Cox et al. 2016; Mohammad et al. 2020; Cox et al. 2021); RCAA (Foltyn et al. 2006); REM (Goel and Jones 2011)], **analogy puzzles** [Metacat (Marshall 2006); MCL (M'balé 2017)], **simple communication** [CARINA (Caro Piñeres et al. 2013); MT21 (Canbaloglu et al. 2023)], **arithmetic** [SCADS (Shrager and Siegler 1998)], and **story comprehension** [CLARION (Sun et al. 2006); ISAAC (Moorman 1997); Meta-AQUA (Cox 1996)]. MIDCA, for example, cares for plants on $10 \times 10$-tile boards (Cox et al. 2021) or controls a

single light switch (Dannenhauer et al. 2014). We identified only one experiment targeting the real world, involving D20 controlling a robot (Daglarli 2020)—however, even that was limited to toy games such as tic-tac-toe.

## 7 Discussion

This paper set out to address the question: *How do models of cognition model their own cognition?* To enable a systematic scoping review, we focused the scope on CMAs that model, store, remember and leverage metacognitive experiences. We conducted a systematic search across 16 research databases and identified 101 relevant reports covering 35 separate CMAs. Using 20 different data items, we collected information across three dimensions: model context, data content and structure, and objectives and evaluation.

While the examined approaches are highly diverse—ranging from symbolic logic-driven systems to hybrid and exclusively sub-symbolic frameworks—common themes emerged that highlight both the promise of metacognition and persistent limitations in the field. In the remainder of this chapter, we discuss results presented in Sects. 3 to 6, as well as limitations of our work and how it may inform future research directions.

### 7.1 Summary and interpretation of results

In relation to our review question—*how do models of cognition model their cognition?*—the evidence maps onto three broad themes: representational diversity, heterogeneous processing algorithms, and limited evaluation.

The most fundamental observation is that most CMAs follow models of human metacognition established in psychological research (Sect. 4.4), e.g., by separating object-level and meta-level cognition following Nelson and Narens (1990). Variants of this principle introduce additional levels for meta-meta cognition, either equal in status or hierarchically arranged. Only a few exceptions process cognition and metacognition through a unified component. Either way, that many CMAs reuse object-level mechanisms for metacognition suggests—besides the apparent advantage of simplifying software development and maintenance—that researchers believe metacognition and cognition to be parallel.

Typically, traces of metacognitive experiences are modelled around events (Sect. 5.2). Event chains are understood either as linear or, to model parallelism, as graphs (Sect. 5.2). Events are usually categorised symbolically (Sect. 5.1), using system-specific or domain-specific labels, or terms borrowed from psychology (Sect. 5.5). They often serve as anchors for numerous symbolic and sub-symbolic metadata, including process inputs and outputs, justifications, confidence values, and metrics for indexing and similarity-based retrieval (Sect. 5.2). The type and quantity of metadata vary considerably depending on each CMA's algorithmic requirements. Some CMAs actively reduce data collection to accommodate database limitations by combining several calculation steps into more abstract events or including only the most important events (Sect. 5.3). Almost 40% of metacognitive experience models are formally or semi-formally specified, mostly based on logical systems and ontologies, some of which are axiomatised and formally defined (Sect. 5.4).

The key role of these traces is to retain sufficient context about the agent's "thoughts" to enable introspection, identify faults, and trigger appropriate remedies (Sect. 6.1). Various

approaches fall under this description: examining knowledge that led to incorrect conclusions to adjust object-level models, breaking out of looping behaviour, or using prior inference to optimise computation times. The investigated CMAs employ diverse methods and algorithms, from predefined pattern matching against the trace to autonomous sub-symbolic machine learning methods with or without a dedicated training phase, and CBR-inspired strategies that explicitly use past experience (Sect. 6.3). To enhance explainability, some CMAs also produce user-facing "reasons" for their solutions, referencing the recorded chain of insights.

Given the number and diversity of identified approaches, memories of metacognitive experiences are clearly widely recognised. Since emerging in the 1990s, a steady stream of new CMAs have adopted such strategies (Sect. 4.1), including the prominent ACT-R, CLARION, and Soar. Despite its prevalence, relevant reports place varying emphasis on the technique. As reported in Sect. 3.2, some data items are explained in great detail, while others lack sufficient information. Importantly, the research community has yet to agree on a unified terminology (Sect. 4.2). We observe that the word "trace" occurs particularly frequently, but even that appears in three different semantic contexts. The terminology developed in this review—independently and prior to the terminological analysis, referring mostly to "traces" or "episodic memories" of "metacognitive experiences"—aligns with psychological theory, but is not necessarily better than the variety of terms currently in circulation; still we put this forward, with explicit definitions, as a starting point for a unified language.

Furthermore, many relevant reports lack transparency: only a minority of CMAs make source code publicly available (Sect. 4.3) or fully disclose the underlying representation schemes (Sect. 5.4), data structures (Sect. 5.7), and algorithms (Sect. 6.3). For example, while structuring metacognitive experiences in episodes (Sect. 5.6) and filtering events by importance are common practice (Sect. 5.3), how episodes are segmented and importance is determined remains largely unclear. Such gaps likely hinder progress, as scientists must reinvent methods while struggling to compare new approaches with existing ones.

One contributing factor may be the fragmented research community, where a single well-connected network produces over $50\%$ of relevant reports while other scientists typically publish individual reports on their CMAs (Sect. 3.3). Furthermore, excluding conceptual inspiration (which we did not test here), and successive versions of the same CMA families, only two CMAs explicitly build upon previous CMAs (Sect. 4.5).

This fragmentation is also evident in evaluation practices (Sect. 6.4): regarding episodic metacognitive experiences, relevant CMAs were at most quantitatively compared with naïve variants of themselves, never with each other. This limits evaluation informativeness, preventing determination of which approaches are most successful or merit further research. Moreover, all CMAs were tested in highly restrictive domains, essentially "toy problems".

This evaluation gap is unlikely to have a single cause. First, it does not appear to be a problem unique to the focus of this review: Kotseruba and Tsotsos (2020) identify similar shortcomings in CCA research at large, highlighting the conceptual heterogeneity of the various CCAs as a major obstacle to meaningful evaluations. In this sense, the evaluation gap we have identified may just be a reflection of a discipline-wide practice that needs to be fundamentally questioned. In our corpus, this problem is reinforced by the terminological fragmentation and dispersed research landscape identified in Sects. 4.2 and 3.3. If researchers are unaware of adjacent systems, incentives to develop shared evaluation standards may

be reduced. The resulting lack of benchmark scenarios is nevertheless striking, given that numerous authors cite autonomy, human-level intelligence and explainability as central objectives (Sect. 6.1)—characteristics particularly important when CMAs are embodied as robots and deployed in Human-Robot-Interaction (HRI). Yet only a single relevant study considered such a setting, and even that evaluation was limited to comparatively simple games.

Developing more informative benchmarks would in any case be non-trivial, especially when the target is a specific mechanism such as episodic metacognitive experience processing: moving beyond toy problems requires additional capabilities such as rich perception, real-world action, or interaction management that may vary considerably across architectures. Performance differences would therefore be difficult to attribute specifically to metacognitive experience processing rather than to these other dimensions. At the level of individual systems, however, standard ablation testing can also be impractical. As the SCADS example illustrates, metacognitive experience processing may be so integral to a system's functioning that disabling it produces a non-functional rather than merely degraded baseline, and evaluations not specifically designed to target the technique cannot easily be repurposed for this (cf. Sect. 6.4). Finally, even within the scope of this review, CMAs use metacognitive experiences for heterogeneous purposes (cf. Sect. 6.1). In the absence of a shared objective, it is unclear what a common benchmark or evaluation metric should measure, or whether a single metric could be meaningful across the full range of reviewed systems.

Consequently, we characterise the initial evidence for the effectiveness of metacognitive experiences in CMAs as cautiously promising.

Overall, our observations are in line with the general impression that the field is fragmented (Brown 1987; Cox and Raja 2011a; Caro Piñeres et al. 2014), and inadequately evaluated (Kotseruba and Tsotsos 2020). In the Introduction, we cite Gordon et al. (2011), who argue that research focuses on algorithms rather than representation, leaving models of mental representations under-researched. Our qualitative and quantitative data—specifically the limited algorithmic detail in relevant reports—suggest that the problem lies more in focusing on conceptualising full-featured architectures. While this is important, more detailed information on developed components would ensure reproducibility and comparability.

## 7.2 Limitations

This review has some limitations that must be considered when interpreting results.

### 7.2.1 Limitations of the evidence

As data availability and detail vary considerably across items and CMAs (cf. Sect. 3.2), sections may be imbalanced, since we can only report data we were able to collect. While reporting on items with limited data is necessarily restricted to a few CMAs, sections with

abundant detailed data will be more representative. Furthermore, we cannot guarantee the collected data's reliability, although we avoided recording incorrect information[14].

Most relevant CMAs rely on symbolic or hybrid data (Sect. 5.1). D20 (Daglarli 2020) is the only identified sub-symbolic-only CMAs with detailed information for just 11 of our 20 items. This could indicate under-representation of sub-symbolic CMAs—whose cognitive capacities grow from training neural networks rather than relying on predefined knowledge bases and algorithms—in our synthesis. However, evidence suggests this imbalance may reflect the current research landscape accurately: Kotseruba and Tsotsos (2020) identified no exclusively sub-symbolic[15] architecture as metacognitive. Notably, their criteria for a "metacognitive" CCAs were more generous than our requirements for episodic metacognitive experiences, yet no emergent systems qualified. We see two plausible explanations: first, emergent CCAs form a relatively small minority of CCAs overall, at least among those analysed by Kotseruba and Tsotsos. Second, emergent CCAs tend to be less transparent, as understanding how their sub-symbolic processes arrive at specific outcomes remains notoriously difficult even with full access to the underlying networks (Kotseruba and Tsotsos 2020, p. 24). This raises non-trivial problems for metacognition: if the underlying processes are poorly understood, how can systems be designed to reflect them? How could we verify whether a network truly leverages self-information metacognitively if even its object-level reasoning remains a black box? Since these findings derive primarily from data collected in 2018 (Kotseruba and Tsotsos 2020) and our synthesis spans publications only through 2023, very recent work beyond these timeframes could introduce new developments in emergent CMAs. Nevertheless, within the current body of evidence, emergent approaches incorporating metacognitive functionality, including metacognitive experiences, remain exceptionally rare.

### 7.2.2 Limitations of the scoping review process

Our literature search relied on specific search terms ("metacognition", "monitoring", "tracing") from an initial literature exploration. While the query was designed to be inclusive, it may have missed CMAs that use alternative terminology as observed in Sect. 4.2. A future iteration based on that terminology could address this issue.

Our eligibility criteria emphasised explicit references to cognitive architectures and metacognition. This focus may favour well-established or more academically oriented frameworks, while overlooking newer or industry-based approaches not positioned under the cognitive architecture banner. Furthermore, our Definition 3 of metacognition is strict and excludes systems that observe object-level actions rather than object-level cognition. For instance, MCl3 (M'balé and Josyula 2013) traces only "primitive" actions that the host agent performs in the environment, not its internal events, and was therefore excluded. The related approaches discussed in Sect.2.1.3 make the boundaries of this restriction explicit and indicate adjacent research areas that may warrant dedicated future surveys.

Each search record was evaluated by one screener (Sect. 2.4) except in ambiguous cases, and data items were collected by only one screener for relevant reports except for five

[14] An example is item (A6) and Thórisson and Helgason's (2012, p. 10) claim that ACT-R is "a predecessor to Soar"—inaccurate given that Soar was described by Laird and Newell in 1983, while ACT-R was first presented by Anderson in 1993.

[15] Recall from Sect. 5.1 that Kotseruba and Tsotsos (2020) use the term *emergent* while we use *sub-symbolic*.

CMAs, i.e., we mostly forwent duplicate screening. While this approach kept the screening process feasible within the project's time constraints, it may introduce subjectivity and inconsistency in categorising records. While a post-hoc reliability check (cf. Sect. 2.4) suggests near-perfect agreement regarding the record screening, we did not compute inter-rater reliability coefficients for the data charting step, because most items were qualitative or descriptive rather than categorical. Agreement on factual items was high; for interpretive items, divergence reflected the intended openness of our coding scheme and was subsequently reconciled through iterative discussion during synthesis. In this sense, what could be termed "disagreement resolution" was subsumed by the narrative synthesis process itself.

Although our data-charting framework is broad, it is primarily oriented towards conceptual, representational, and methodological aspects of metacognitive experience traces. Consequently, this review does not provide a systematic engineering-level comparison of computational trade-offs such as scalability, memory overhead, retrieval complexity, or runtime implications. We discuss such aspects where they are explicitly addressed in the relevant reports, for example Soar's use of delta encoding to reduce memory requirements (cf. Sect. 5.7), but we did not infer them by deconstructing implementations or by extending the charting framework retrospectively. An analysis of the computational trade-off between costs and benefits of metacognitive mechanisms has partly been addressed for metacognition more generally by Zilberstein (2011).

A further limitation concerns the reproducibility of the narrative synthesis. While our search strategy, eligibility criteria, screening decisions, and data-charting process are documented in detail, the synthesis itself necessarily involves qualitative interpretation by the authors. In particular, the organisation of findings, the grouping of architectures, and the emphasis placed on recurring patterns reflect author judgement and cannot be reproduced in the same mechanical sense as the search and selection procedure.

Finally, our grouping of CMAs into "families" involves interpretation that, while carefully done, could inadvertently overlook possibly meaningful architectural differences.

### 7.3 Outlook

The challenges we identified call for a more unified yet competitive research landscape. We believe that an agreed-upon terminology, open-source code, precise model specifications, and explicit algorithm details will promote exchange of ideas between cognitive architectures. First, successful components could be better adopted, and accumulated metacognitive experience could be shared, mutually providing training data for emergent machine learning approaches—both would advance the field as a whole. Second, improving reproducibility and facilitating comparisons and benchmarking will help identify the most promising research directions.

This also involves testing systems on more complex, dynamic domains that go beyond toy problems, targeting the real world as the ultimate testing ground. In experiments unrelated to metacognitive experiences, several CMAs investigated in this review (other than D20) have operated in real-world settings when embodying and controlling robots; some, such as ArmarX and CRAM, are specifically tailored for this purpose. A first quantitative comparison seems within reach. Meanwhile, research must create alternative evaluation standards: while common ablation tests can isolate the contribution of metacognitive experiences to overall performance, this is less obvious when contrasting distinct systems. We

argue that this requires test platforms targeting metacognitive skills such as self-correction and goal-driven learning, for example as part of the *RoboCup* (Asada and Kitano 1999; Wisspeintner et al. 2009).

Such standardisation and rigorous testing are critical, given that cognitive science often uses cognitive architectures to test theories of human cognition. Equally, both the AI community and society in general have much to gain from employing metacognitive experiences. Rapid change is underway, driven by deep learning breakthroughs: neural networks increasingly enter everyday life—from household robots that mop and vacuum (Lee 2021; Huang 2023), self-driving cars (Kim and Joe 2022), and virtual assistants (Chamola et al. 2023) to systems designed to replace human workers (Tschang and Almirall 2021). However, particularly in critical areas such as medicine or traffic, testing and ensuring the reliability of AI systems is essential. Memories of metacognitive experiences in AI systems not only enable introspective insights by the systems themselves but also make their processes transparent and comprehensible for users, e.g., through interfaces such as Metacat's (Sect. 6.1). Unlike symbolic or hybrid systems, where interpreting cognitive processes is straightforward (which might be why most relevant reports do not address interpretation and monitoring, implicitly taking them for granted), emergent systems are largely opaque—many qualify as black boxes—and require a strong research focus on interpretation. Although not framed as metacognition, such interpretation is what Explainable Artificial Intelligence (XAI) is largely aimed at (Gunning et al. 2019). The European Union in particular drives forward research programs and legislation promoting XAI, such as the *Horizon Europe Program* (Eurpoean Commission 2024) and the *AI Act* (Pavlidis 2024).

An example of such a methodology applied to the LLM context is illustrated by the case study by Lindsay et al. (2025) that we used as an example in the introduction, where the LLM output is matched against an *attribution graph* of concept activations. Other XAI techniques, e.g., saliency maps or attention scores, may highlight which factors of an input were important in deciding a solution. It is important, however, to acknowledge remaining difficulties with such XAI methods. Current explanation methods have been observed to be incomplete (Seth and Sankarapu 2025; Rudin 2019), unreliable (Abusitta et al. 2024; Longo et al. 2024; Budhkar et al. 2025), or even detached from reality (Rudin 2019; Longo et al. 2024). In particular, the question that is relevant to our purpose—how exactly a model arrives at a solution—cannot be answered reliably by current XAI methods (Weber et al. 2024).

But even if XAI methods were to deliver complete decompositional transparency of a target black box, they typically constitute black boxes themselves. As we cannot be sure of their own faithfulness, the opacity problem is merely relocated (Rudin 2019). For instance, attribution graphs are unintelligible for roughly three quarters of tested prompts, and even in the intelligible cases there is no guarantee of completeness, i.e., that all relevant computations are reflected (Lindsay et al. 2025). An early evaluation also "raises questions about [their] reliability" (Ameisen et al. 2025). In general, the faithfulness of XAI methods is a topic of active research. Ergo, while the work by Lindsay et al. showed a mismatch between the attribution graph and Claude's self-explanation—indicating that the latter is not faithful, i.e., that "self-explanation" is not a reliable method to access the workings of the LLM—this leaves open the possibility that the attribution graph itself may also be unfaithful.

Instead of trying to monitor and interpret sub-symbolic systems such as LLMs post hoc, investigating whether necessary metacognitive capabilities may already be *emergent*

(see, e.g., Berti et al. 2025; Schaeffer et al. 2023) has therefore become fruitful. However, because the field is new and the systems involved are complex and semantically opaque, no consensus exists about whether, or in what sense, LLMs possess metacognitive abilities. A central difficulty is methodological: it remains unclear what should count as evidence for metacognition in general, and for metacognitive experiences in particular. Some hints may be obtained from the psychological literature that established think-alouds as accurate and reliable indicators of human thinking (Ericsson and Simon 1993). What is particularly important in such approaches is the possibility to use the existence of a putative entity (e.g., metacognitive skill) to explain a diverse set of observations. If the entity exists, there should be many ways to access and triangulate it, and these different ways should show a certain coherence such that it would be possible to predict what the entity would do if accessed one way, from its response to another way of access. As the current state of research remains conflicted, such a triangulation of metacognition in LLMs cannot be determined with sufficient certainty. For example, some work has claimed evidence of metacognition in LLMs, such as GPT-4 appearing able to judge the skill level needed to solve mathematical problems (Didolkar et al. 2024), while other results dispute metacognition in LLMs (Comsa and Shanahan 2025; Griot et al. 2025; Lindsay et al. 2025).

An early observation has been that LLMs performance appears to improve when prompted to "think step by step" (Kojima et al. 2022; Wei et al. 2022), a technique known as Chain-of-Thought (CoT) prompting, with one commenter characterizing this as working memory being an external organ for these systems (Newman 2023). Recent approaches automate CoT prompting (Zhang et al. 2023) or suggest generating CoT during training by manipulating training data (Gandhi et al. 2024; Lehnert et al. 2024) so users need not explicitly include special instructions in prompts. Industry has quickly adopted such ideas: at the time of writing, various state-of-the-art LLMs-based chatbots such as ChatGPT o3-pro, DeepSeek DeepThink (R1), and Gemini 2.0 Flash Thinking automatically generate an additional output akin to think-aloud protocols.

Recent studies, however, raise doubts regarding the effectiveness of CoT in achieving reliable reasoning in foundation models (Stechly et al. 2024; Valmeekam et al. 2024). Importantly, we argue that CoT itself should not be equated with metacognitive experiences in the sense of Definition 3. One might interpret CoT as a foundation model thinking about and steering its own thinking process. Yet Definition 3 requires some faithful modelling relation.[16] between the trace and the system's cognitive processes. Current evidence gives reason to be cautious on precisely this point. The questionable faithfulness of the generated CoT remains a key issue: has the LLM actually adhered to it, or simply hallucinated a CoT that sounds sensible (Zhao et al. 2025)? Unfortunately, evidence—beyond the Claude example (Lindsay et al. 2025) from our Introduction—is accumulating that this is often the case (Lanham et al. 2023; Turpin et al. 2023; Paul et al. 2024; Arcuschin et al. 2025). Whether such seemingly "think-aloud" CoTs should be interpreted as traces of reasoning has also been questioned in expert commentary: since they are often not valid traces of an algorithm, Kambhampati (2024, 2025); Stetchly et al. (2025) argues that they are better understood less as (meta)cognitive functions than as prompt augmentations for query-answering systems. *Faithful* CoT generation now attracts attention from the research community but remains an unsolved problem.

[16] This is because Sanz et al. (2007), whose terminology Definition 1 to 3 explicitly build upon, do so; in their terms, the relation should even be *isomorphic*.

Analyses of modern LLM-based agents suggest that such systems increasingly move beyond plain end-to-end prompt-response pipelines and instead comprise elaborate "harnesses" (i.e., collections of specialized tools and data flow orchestrations) around large-scale neural networks. Claude, for instance, has been reported—following the public discussion surrounding its source-code leak—to combine repeated calls to a foundation model in iterative self-control loops with specialised tool use, external memory, and telemetry-like monitoring infrastructure (Mohmmad ali pour & Fazel 2026) . At this architectural level, such systems resemble long-standing principles from CCA and CMA research such as cognitive cycles, dedicated subsystems, episodic storage, and trace-based control. In our context, self-correction or reflection loops that repeatedly critique and revise an answer may implement a useful metacognitive strategy at the system level, but should not automatically be interpreted as metacognitive experiences in the sense of Definition 3. Depending on how the system processes and reuses them, intermediate textual states may simply remain part of the object-level problem-solving process, offering additional task context. However, it is also possible that these textual states emergently encode partial self-models: either explicitly or opaquely encoded through specific syntactic patterns or word choices. Given our lack of reliable methods for comparing even explicit descriptions of a system's processing with its actual processing (see above), the latter possibility complicates proving or disproving this hypothesis, even with access to all relevant data.

In summary, it remains unclear whether the techniques examined in this review can be incorporated into foundation models or whether parallel capabilities may emerge on their own with increasing complexity, and we have left surveying this active field for future work. If performance were the only concern, it would not matter whether CoTs and self-correcting loops are actually metacognitive in nature or merely allow LLMs to better simulate the benefits of metacognition. However, when explainability and transparency are the focus, we must consider that LLM-presented trains of thought may in fact be hallucinations that, instead of being truly transparent, may misleadingly give the impression of human-like thinking. A key open research question is therefore how structured metacognitive traces—as studied in this review—could be aligned with internal states of opaque models such as neural networks. Progress on this question would help connect the CMA tradition with current LLM research, and contribute to the broader challenge of hybrid AI: understanding how symbolic and sub-symbolic structures can be mapped onto one another.

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**Author contributions** R.N. conceived and designed the study, developed the methodology, and coordinated all phases of the systematic review. R.N. performed the majority of the literature screening and data extraction (as Screener A), conducted the complete synthesis and analysis, designed and implemented the accompanying website, conducted the search for related work, and wrote the manuscript draft (approximately 95%). M.P. contributed to refining the study scope and eligibility criteria, served as Screener B, contributed to the sections describing the purpose and algorithmic aspects of the reviewed systems, and critically reviewed and proofread the manuscript. A.J. implemented the interactive data visualisations on the companion website. K.J. participated as Screener C and contributed to early discussions on the study's scope. K.J. also provided critical proofreading of the final manuscript. D.B. acted as Screener D and was involved in shaping the study scope. S.J. contributed as Screener E and to the terminology section, compiling input data. R.P. provided advisory input throughout the project. J.B., M.B., and R.M. are the principal investigators (PIs) of the overarching project that funded this research. They contributed to high-level conceptual oversight and provided management and institutional support. J.B. also proofread the manuscript.

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**Data availability** Inspired by Kotseruba and Tsotsos' (2020) overview, we visualise much of the data collected in this review and offer interactive versions of these figures on a companion website (https://dml.uni-bremen.de/ease/review) that allow, e.g., to filter the presented data by the CMAs. The collected raw data, filled out PRISMA-ScR checklist, and certain analysis protocols are available in a GitHub repository (https://github.com/mrnolte/cma_review_data)

## Declarations

**Conflict of interest** Nine authors (R.N., M.P., A.J., D.B., S.J., R.P., J.B., M.B., and R.M.) are current or former principal investigators or employees of the Collaborative Research Center 1320 "EASE" at the University of Bremen (cf. the Funding), which develops and evaluates the CRAM cognitive architecture. This review includes one of our authored CRAM reports (Nolte et al. 2023), resulting in a professional and reputational interest. We have no other conflict of interest to declare that are relevant to the content of this article.

**Study information** This review was not prospectively registered, and no formal protocol was prepared. This was a deliberate decision to retain methodological flexibility: fixing a rigid protocol in advance would have constrained the explorative synthesis. Although largely narrative in nature, key analytical procedures (e.g., the later addition of many visualisations) only emerged as suitable after charting of the qualitative data.

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## References

Abusitta A, Li MQ, Fung BC (2024) Survey on explainable AI: techniques, challenges and open issues. Expert Syst Appl 255:124710. https://doi.org/10.1016/j.eswa.2024.124710

Ameisen E, Lindsey J, Pearce A, Gurnee W, Turner NL, Chen B, Citro C, Abrahams D, Carter S, Hosmer B, Marcus J, Sklar M, Templeton A, Bricken T, McDougall C, Cunningham H, Henighan T, Jermyn A, Jones A, Persic A, Qi Z, Ben Thompson T, Zimmerman S, Rivoire K, Conerly T, Olah C, Batson J (2025) Circuit tracing: Revealing computational graphs in language models. Transformer Circ Thread

Anderson JR (1993) Rules of the Mind. Lawrence Erlbaum Associates Inc

Anderson JR (2007) How can the human mind occur in the physical universe? Oxford University Press, New York

Anderson ML, Gomaa W, Grant J, Perlis D (2008) Active logic semantics for a single agent in a static world. Artif Intell 172(8):1045–1063. https://doi.org/10.1016/j.artint.2007.11.005

Anderson ML, Oates T, eds (2005) Proceedings of the 2005 AAAI spring symposium on metacognition in computation. Palo Alto: AAAI Press. https://aaai.org/proceeding/spring-2005-04/

Anderson ML, Oates T (2007) A review of recent research in metareasoning and metalearning. AI Mag 28(1):7–16. https://doi.org/10.1609/aimag.v28i1.2025

Apperly I (2011) Mindreaders: the cognitive basis of "Theory of Mind," 1st edn. Psychology Press, Hove

Arcos JL, Mülâyim O, Leake DB (2011) Using introspective reasoning to improve CBR system performance, In: Cox and Raja (eds), Chapter 11, pp 167–182. https://doi.org/10.7551/mitpress/9780262014809.003.0011

Arcuschin I, Janiak J, Krzyzanowski R, Rajamanoharan S, Nanda N, Conmy A (2025) Chain-of-thought reasoning in the wild is not always faithful. In: Proceedings of the workshop on reasoning and planning for large language models at the 13th International Conference on Learning Representations (ICLR 2025). https://openreview.net/forum?id=L8094Whth0

Asada M, Kitano H (1999) The RoboCup challenge. Robot Auton Syst 29(1):3–12. https://doi.org/10.1016/S0921-8890(99)00033-0

Atkinson R, Shiffrin R (1968) Human memory: a proposed system and its control processes. Psychology of learning and motivation, vol 2. Academic Press, pp 89–195. https://doi.org/10.1016/S0079-7421(08)60422-3

Bartsch-Spörl B, Lenz M, Hübner A (1999) Case-based reasoning—survey and future directions. In: Puppe F (ed) Proceedings of the 5th Biannual German Conference on Knowledge-Based Systems (XPS 1999), vol 1570. Lecture Notes in Artificial Intelligence. Springer, Würzburg, pp 67–89

Beetz M, Beßler D, Haidu A, Pomarlan M, Bozcuoglu AK, Bartels G (2018) KnowRob2.0—a 2nd generation knowledge processing framework for cognition-enabled robotic agents. Proceedings of the IEEE International Conference on Robotics and Automation (ICRA 2018). IEEE Computer Society, Brisbane, pp 512–519

Beetz M, Mösenlechner L, Tenorth M (2010) CRAM—a cognitive robot abstract machine for everyday manipulation in human environments. Proceedings of the IEEE/RSJ international conference on Intelligent Robots and Systems (IROS 2010). IEEE Computer Society, Taipei, pp 1012–1017

Berkeley ISN, Rice C (2012) Machine mentality? In: Müller VC (ed) Philosophy and theory of artificial intelligence. Springer, Berlin, Heidelberg, Germany, pp 1–15. https://doi.org/10.1007/978-3-642-31674-6_1

Berti L, Giorgi F, Kasneci G (2025) Emergent abilities in large language models: A survey. abs/2503.05788.https://doi.org/10.48550/ARXIV.2503.05788

Beßler D, Porzel R, Pomarlan M, Vyas A, Höffner S, Beetz M, Malaka R, Bateman J (2021) Foundations of the socio-physical model of activities (SOMA) for autonomous robotic agents. In: Neuhaus F, Broderic B (eds) Proceedings of the 12th international conference on Formal Ontology in Information Systems (FOIS 2021), Frontiers in Artificial Intelligence and Applications (FAIA), vol 344. IOS Press, Bozen-Bolzano, pp 159–174

Birlo M, Tapus A (2011) The crucial role of robot self-awareness in HRI. In: Billard A, Kahn P, Adams JA, Trafton JG (eds) Proceedings of the 6th International Conference on Human-Robot Interaction (HRI 2011). Association for Computing Machinery, New York, pp 115–116

Block N (2002) Concepts of consciousness. In: Chalmers DJ (ed) Philosophy of mind: classical and contemporary readings. Oxford University Press, pp 206–218

Bostock M, Ogievetsky V, Heer J (2011) $D^3$ data-driven documents. IEEE Trans Visual Comput Graphics 17(12):2301–2309. https://doi.org/10.1109/TVCG.2011.185

Brody J, Cox MT, Perlis D (2013) The processual self as cognitive unifier. In: Proceedings of the 2013 Meeting of the International Association for Computing and Philosophy (IACAP 2013), College Park, Maryland, USA.International Association of Computing and Philosophy (2013). ,2013_proceedings/ https://www.iacap.org/conferences/iacap2013/iacap_

Brown AL (1987) Metacognition. In: Weinert FE, Kluwe RH (eds) executive control, self-regulation, and other more mysterious mechanisms. In Metacognition, motivation and understanding. Lawrence Erlbaum Associates, Hillsdale, pp 65–116

Budhkar A, Song Q, Su J, Zhang X (2025) Demystifying the black box: a survey on explainable artificial intelligence (XAI) in bioinformatics. Comput Struct Biotechnol J 27:346–359. https://doi.org/10.1016/j.csbj.2024.12.027

Burstein M, Sridharan M, McDonald D, eds (2021) Proceedings of the 9th annual conference on advances in Cognitive Systems (ACS 2021), Virtual event. Cognitive Systems Foundation

Canbaloğlu G, Treur J (2023) Modeling mental models: their use, adaptation and control. In: Canbaloglu et al. (2023), pp 51–70. ISBN: 978-3-031-28737-4

Canbaloglu G, Treur J, Wiewiora A (eds) (2023) Computational Modeling of Multilevel Organisational Learning and Its Control Using Self-modeling Network Models, Studies in Systems. Decision and Control. Springer, Cham

Carbonell JG (1985) Derivational analogy: A theory of reconstructive problem solving and expertise acquisition. Technical report, Carnegie Mellon University

Carbonell JG, Etzioni O, Gil Y, Joseph R, Knoblock C, Minton S, Veloso M (1991) Prodigy: an integrated architecture for planning and learning. SIGART Bull 2(4):51–55. https://doi.org/10.1145/122344.122353

Caro Piñeres MF (2015) Metamodel for personalized adaptation of pedagogical strategies using metacognition in intelligent tutoring systems. Ph.D. thesis, Universidad Nacional de Colombia Sede Medellín

Caro Piñeres MF, Builes J (2013) Analysis of models and metacognitive architectures in intelligent systems. DYNA 80(180):50–59
Caro Piñeres MF, Josyula DP, Jiménez-Builes JA (2013) Metamemory for information retrieval from long-term memory in artificial cognitive systems. In: Proceedings of the Workshop on Metacognition in Situated Agents at the 2nd Annual Conference on Advances in Cognitive Systems (ACS 2013), Baltimore, Maryland, USA.Cognitive Systems Foundation, pp 17–33
Caro Piñeres MF, Josyula DP, Cox MT, Jiménez-Builes JA (2014) Design and validation of a metamodel for metacognition support in artificial intelligent systems. Biol Inspired Cogn Archit 9:82–104. https://doi.org/10.1016/j.bica.2014.07.002
Caro Piñeres MF, Josyula DP, Gómez Salgado AA, Kennedy CM (2018) Introduction to the CARINA metacognitive architecture, pp 530–540
Caro Piñeres MF, Josyula DP, Madera-Doval DP, Kennedy CM, Gómez Salgado AA (2019a) The CARINA metacognitive architecture. Int J Cogn Inf Natural Intell 13(4):71–90. https://doi.org/10.4018/IJCINI.2019100104
Caro Piñeres MF, Madera-Doval DP, Cox MT, Sun R, Josyula DP, Kennedy CM, Ackerman R (2019b) The double metareasoning cycle in the CARINA metacognitive architecture. Proceedings of the 7th annual conference on Advances in Cognitive Systems (ACS 2019), Advances in Cognitive Systems. Cognitive Systems Foundation, Cambridge, pp 518–536
Castelfranchi C, Falcone R (2019) Self-awareness implied in human and robot intentional action. In Chella A, Gamez D, Lincoln P, Manzotti R, Pfautz J (eds) Proceedings of the symposium toward conscious AI systems (TOCAIS 2019). CEUR Workshop Proceedings, vol 2287. Stanford. CEUR-ws.org. https://ceur-ws.org/Vol-2287/
Chamola V, Hassija V, Sulthana AR, Ghosh D, Dhingra D, Sikdar B (2023) A review of trustworthy and explainable artificial intelligence (XAI). IEEE Access 11:78994–79015. https://doi.org/10.1109/ACCESS.2023.3294569
Choi Y, Zhou D, Das D, Chen D, Artzi Y, Fan A, Rush A, Oh A, Choudhury M, Yatskar M, eds (2024) Proceedings of the 1st Conference on Language Modeling (COLM 2024)
Chong HQ, Tan AH, Ng GW (2009) Integrated cognitive architectures: a survey. Artif Intell Rev 28:103–130. https://doi.org/10.1007/s10462-009-9094-9
Cognitive Systems Foundation (2013) Proceedings of workshop on metacognition in situated agents at the 2nd annual conference on Advances in Cognitive Systems (ACS 2013). Cognitive Systems Foundation, Baltimore
Comsa IM, Shanahan M (2025) Does it make sense to speak of introspection in large language models? *CoRR* abs/2506.05068. https://doi.org/10.48550/ARXIV.2506.05068. arXiv:2506.05068
Cox MT (1994) Machines that forget: Learning from retrieval failure of mis-indexed explanations. In Ram A, Eiselt K (eds) Proceedings of the 16th annual conference of the Cognitive Science Society, Atlanta. Routledge
Cox MT (1995) Representing mental events (or the lack thereof). In Cox MT, Freed M (eds) Proceedings of the 1995 AAAI spring symposium on representing mental states and mechanisms, Palo Alto, pp 22–30. AAAI Press. https://cdn.aaai.org/Symposia/Spring/1995/SS-95-05/SS95-05-004.pdf
Cox MT (1996) An empirical study of computational introspection: evaluating multistrategy learning in the meta-aqua system. In Michalski RS, Wnek J (eds) Proceedings of the 3rd international conference on Multistrategy Learning (MSL 1996), Harpers Ferry, pp 131–142. The AAAI Press. https://cdn.aaai.org/MSL/1996/MSL96-014.pdf
Cox MT (1997a) An explicit representation of reasoning failures. In Leake and Plaza (eds)Proceedings of the 2nd International Conference on Case-Based Reasoning Research and Development (ICCBR 1997), Lecture Notes in Computer Science, vol 1266, Providence, Rhode Island, USA. Springer, pp 211–222. ISBN: 978-3-540-69238-6
Cox MT (1997b) Loose coupling of failure explanation and repair: using learning goals to sequence learning methods. In Leake and Plaza (eds)Proceedings of the 2nd International Conference on Case-Based Reasoning Research and Development (ICCBR 1997), Lecture Notes in Computer Science, vol 1266, Providence, Rhode Island, USA. Springer, pp 425–434. ISBN: 978-3-540-69238-6
Cox MT (2005a) Metacognition in computation: A selected history. In: Anderson and Oates (eds) Proceedings of the 2005 AAAI Spring Symposium on Metacognition in Computation, Palo Alto, California, USA. AAAI Press, pp 1–17. https://cdn.aaai.org/Symposia/Spring/2005/SS-05-04/SS05-04-002.pdf
Cox MT (2005b) Metacognition in computation: a selected research review. Artif Intell 169(2):104–141. https://doi.org/10.1016/j.artint.2005.10.009
Cox MT (2007) Perpetual self-aware cognitive agents. AI Mag 28(1):32–45. https://doi.org/10.1609/aimag.v28i1.2027
Cox MT (2011) Metareasoning, monitoring, and self-explanation, Cox and Raja (eds), Chapter 9, pp 131–149. https://doi.org/10.7551/mitpress/9780262014809.003.0009

Cox MT (2013) Question-based problem recognition and goal-driven autonomy. Proceedings of the 2nd annual conference on Advances in Cognitive Systems (ACS 2013). Presented as a poster. Cognitive Systems Foundation, Baltimore, pp 29–45

Cox MT, Alavi, Z, Dannenhauer D, Eyorokon V, Muñoz-Avila H, Perlis D (2016) MIDCA: A metacognitive, integrated dual-cycle architecture for self-regulated autonomy. In Schuurmans D, Wellman MP (eds) Proceedings of the 30th AAAI Conference on Artificial Intelligence (AAAI 2016). AAAI Press, Phoenix, pp 3712–3718. https://aaai.org/proceeding/01-aaai-2016-aaai-30/

Cox MT, Maynord M, Oates T, Paisner M, Perlis D (2013) The integration of cognitive and metacognitive processes with data-driven and knowledge-rich structures. In: Proceedings of the 2013 Meeting of the International Association for Computing and Philosophy (IACAP 2013), College Park, Maryland, USA. International Association of Computing and Philosophy. ,2013_proceedings/ https://www.iacap.org/conferences/iacap2013/iacap_

Cox MT, Mohammad Z, Kondrakunta S, Gogineni VR, Dannenhauer D, Larue O (2021) Computational metacognition. In: Burstein et al (eds)Proceedings of the 9th Annual Conference on Advances in Cognitive Systems (ACS 2021), virtual event. Cognitive Systems Foundation, pp 1–20

Cox MT, Oates T, Perlis D (2011) Toward an integrated metacognitive architecture. In: Langley P (ed) Proceedings of the 2011 AAAI Fall Symposium on Advances in Cognitive Systems, Arlington, Virginia, USA. AAAI Press,pp 74–81. ISBN: 978-1-57735-545-8. https://aaai.org/proceeding/01-fall-2011/

Cox MT, Raja A (2011a) Metareasoning: an introduction, Cox and Raja (eds), Chapter 1, pp 3–14. https://doi.org/10.7551/mitpress/9780262014809.003.0001

Cox MT, Raja A (eds) (2011b) Metareasoning: thinking about thinking. MIT Press, Cambridge

Cox MT, Ram A (1991) Using introspective reasoning to select learning strategies. In Michalski RS, Tecuci G (eds) Proceedings of the 1st international conference on Multistrategy Learning (MSL 1991), Harpers Ferry, pp 217–230. ISBN: 1-55860-305-0. https://apps.dtic.mil/sti/tr/pdf/ADA282947.pdf

Cox MT, Ram A (1992) Multistrategy learning with introspective meta-explanations. In Sleeman D, Edwards P (eds) Proceedings of the 9th international workshop on machine learning (ML 1992). Morgan Kaufmann Publishers, San Mateo, pp 123–128

Cox MT, Ram A (1994) Interacting learning-goals: Treating learning as a planning task. In J. P. Haton, M. Keane, and M. Manago (Eds.), Proceedings of the 2nd European Workshop on Advances in Case-Based Reasoning (EWCBR 1994). Lecture Notes in Computer Science (LNCS), vol 984. Springer, Chantilly, pp 66–74

Cox MT, Ram A (1999) Introspective multistrategy learning: on the construction of learning strategies. Artif Intell 112(1):1–55. https://doi.org/10.1016/S0004-3702(99)00047-8

Crowder, JA, Carbone JN, Friess SA (2014) Artificial cognition architectures. Springer. ISBN: 978-1-4614-8072-3

Daglarli E (2020) Computational modeling of prefrontal cortex for meta-cognition of a humanoid robot. IEEE Access 8:98491–98507. https://doi.org/10.1109/ACCESS.2020.2998396

Dannenhauer D, Cox MT, Gupta S, Paisner M, Perlis D (2014) Toward meta-level control of autonomous agents. In Samsonovich AV, Robertson P (eds) Proceedings of the 5th annual international conference on Biologically Inspired Cognitive Architectures (BICA 2014) Procedia computer science, vol 41. Elsevier, Cambridge, pp 226–232. https://www.sciencedirect.com/science/journal/18770509/41

Dannenhauer D, Cox MT, Muñoz-Avila H (2018) Declarative metacognitive expectations for high-level cognition. Adv Cogn Syst 6:231–250

Dawidpuljak L (2021) Language restrictions in systematic reviews should not be imposed in the search strategy but in the eligibility criteria if necessary pieper. J Clin Epidemiol 132:146–147. https://doi.org/10.1016/j.jclinepi.2020.12.027

Dehaene S, Lau H, Kouider S (2017) What is consciousness, and could machines have it? Science 358(6362):486–492. https://doi.org/10.1126/science.aan8871

Didolkar AR, Goyal A, Ke NR, Guo S, Valko M, Lillicrap TP, Rezende DJ, Bengio Y, Mozer MC, Arora S (2024) Metacognitive capabilities of LLMs: an exploration in mathematical problem solving. In Proceedings of the workshop on AI for Math @ ICML 2024. https://openreview.net/forum?id=0MsI3bSmmD

DOAJ (2014) Some journals say they are indexed in DOAJ but they are not. Blog post. https://blog.doaj.org/2014/08/28/some-journals-say-they-are-in-doaj-when-they-are-not/

Epstein SL Petrovic S (2011) Learning expertise with bounded rationality and self-awareness, Cox and Raja (eds), Chapter 4, pp 43–57. https://doi.org/10.7551/mitpress/9780262014809.003.0004

Ericsson KA, Simon HA (1993) Protocol analysis: verbal reports as data, vol 04. MIT Press

Eurpoean Commission (2024) Horizon Europe work programme 2023–2025. https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/wp-call/2023-2024/wp-7-digital-industry-and-space_horizon-2023-2024_en.pdf

Flavell JH (1979) Metacognition and cognitive monitoring: a new area of cognitive-developmental inquiry. Am Psychol 34:906–911. https://doi.org/10.1037/0003-066x.34.10.906

Fletcher L, Carruthers P (2012) Metacognition and reasoning. Philosophical Trans R Soc B 367(1594):1366–1378. https://doi.org/10.1098/rstb.2011.0413
Flórez MA, Gómez Salgado AA, Caro Piñeres MF (2018) Formal representation of introspective reasoning trace of a cognitive function in CARINA. Wang et al. (ed), pp 620–628
Foltyn L, Tozicka J, Rollo M, Pechoucek M, Jisl P (2006) Reflective-cognitive architecture: From abstract concept to self-adapting agent. In Marík V, Gruver WA, Pechoucek M, Preucil L (eds) Proceedings of the IEEE workshop on distributed intelligent systems: collective intelligence and its applications (DIS 2006). IEEE Computer Society, Prague, pp 326–334
Frey BJ, Dueck D (2007) Clustering by passing messages between data points. Science 315(5814):972–976. https://doi.org/10.1126/science.1136800
Friedman SE, Forbus KD, Sherin B (2011) Constructing and revising commonsense science explanations: a metareasoning approach. In: Langley (ed)Proceedings of the 2011 AAAI Fall Symposium on Advances in Cognitive Systems, Arlington, Virginia, USA. AAAI Press, pp 121–128. ISBN: 978-1-57735-545-8. https://aaai.org/proceeding/01-fall-2011/
Gandhi K, Lee D, Grand G, Liu M, Cheng W, Sharma A, Goodman ND (2024) Stream of search(SoS): learning to search in language. In: Choi Y, Zhou D, Das D, Chen D, Artzi Y, Fan A, Rush A, Oh A, Choudhury M, Yatskar M (eds) Proceedings ofthe 1st Conference on Language Modeling (COLM 2024). https://openreview.net/forum?id=2cop2jmQVL
Gholamian S, Ward PAS (2021) A comprehensive survey of logging in software: from logging statements automation to log mining and analysis. arXiv Preprint. https://doi.org/10.48550/arXiv.2110.12489
Goel AK, Jones J (2011) Metareasoning for self-adaptaion in intelligent agents, Cox and Raja (eds), Chapter 10, pp 151–165. https://doi.org/10.7551/mitpress/9780262014809.003.0010
Goldberg MD (2022) Time-situated metacognitive agency and other aspects of commonsense reasoning, Ph.D. thesis. College Park
Gordon AS, Hobbs JR, Cox MT (2011) Anthropomorphic self-models for metareasoning agents, Cox and Raja (eds), Chapter 19, pp 295–305. https://doi.org/10.7551/mitpress/9780262014809.003.0019
Griot M, Hemptinne C, Vanderdonckt J, Yuksel D (2025) Large language models lack essential metacognition for reliable medical reasoning. Nat Commun. https://doi.org/10.1038/s41467-024-55628-6
Grüninger M, Fox MS (1995) Methodology for the design and evaluation of ontologies. Proceedings of the workshop on basic ontological issues in knowledge sharing @ IJCAI 95
Gunning D, Stefik M, Choi J, Miller T, Stumpf S, Yang GZ (2019) XAI—explainable artificial intelligence. Sci Robot 4:37. https://doi.org/10.1126/scirobotics.aay7120
Gupta N, Bedi P, Jindal V (2023) A survey on intrusion detection and prevention systems. SN Comput Sci 4:439. https://doi.org/10.1007/s42979-023-01926-7
Gusenbauer M, Haddaway NR (2020) Which academic search systems are suitable for systematic reviews or meta-analyses? Evaluating retrieval qualities of Google Scholar, Pubmed, and 26 other resources. Res Synth Methods 11(2):181–217. https://doi.org/10.1002/jrsm.1378
Helgason HP (2013) General attention mechanism for artificial intelligence systems. Ph.D. thesis, Reykjavík University
Hernández Corbato C (2013) Model-based Self-awareness patterns for autonomy. Ph.D. thesis, Universidad Politécnica de Madrid
Herrmann JW (2023) Metareasoning for robots: adapting in dynamic and uncertain environments, 1st edn. Synthesis Lectures on Computer Science. Springer International Publishing, Cham
Huang Q (2023) Towards indoor suctionable object classification and recycling: developing a lightweight AI model for robot vacuum cleaners. Appl Sci 13:18. https://doi.org/10.3390/app131810031
Hudlicka E (2005) Modeling interactions between metacognition and emotion in a cognitive architecture. In:Anderson and Oates (eds) Proceedings of the 2005 AAAI Spring Symposium on Metacognition in Computation, Palo Alto, California, USA. AAAI Press, pp 55–61. ISBN: 1-57735-230-0. https://aaai.org/proceeding/spring-2005-04/
Hugging Face I (2021) multi-QA-MPNET-base-cos-v1. https://huggingface.co/sentence-transformers/multi-qa-mpnet-base-cos-v1. Accessed on 30 Sept 2010
International Association of Computing and Philosophy (2013) Proceedings of the 2013 meeting of the International Association for Computing and Philosophy (IACAP 2013). International Association of Computing and Philosophy, College Park. https://www.iacap.org/conferences/iacap2013/iacap_2013_proceedings/
Jackson PC (2014) Toward human-level artificial intelligence—representation and computation of meaning in natural language. Ph.D. thesis, Tilburg University
Jackson PC (2019) Toward metascience via human-level AI with metacognition, pp 527–534. https://doi.org/10.1016/j.procs.2020.02.214
Janes A, Xiaozhou L, Lenarduzzi V (2023) Open tracing tools: overview and critical comparison. J Syst Softw. https://doi.org/10.1016/j.jss.2023.111793

Jang B, Kim M, Harerimana G, Kim JW (2019) Q-learning algorithms: a comprehensive classification and applications. IEEE Access 7:133653–133667. https://doi.org/10.1109/ACCESS.2019.2941229
Kalish MQ, Samsonovich AV, Coletti MA, De Jong KA (2010) Assessing the role of metacognition in GMU BICA. In Samsonovich et al. (eds)Proceedings of the 1st Annual Meeting of the Biologically Inspired Cognitive Architectures Society (BICA 2010), Frontiers in Artificial Intelligence and Applications, vol 221, Washington, District of Columbia, USA. IOS Press, pp 72–77. ISBN: 978-1-60750-660-7
Kambhampati S (2024) Think "prompt augmentations" rather than chains and thoughts! Social media post. https://x.com/rao2z/status/1865985285034553478
Kambhampati S (2025) No, virginia, those rorschach musings by lrms may not be "reasoning traces" in any meaningful way. Social media post. https://x.com/rao2z/status/1891260345165263128
Kelley TD, Thomson R, Milton JR (2018) Standard model of mind: episodic memory. In Samsonovich and Lebiere (eds)Proceedings of the 9th Annual International Conference on Biologically Inspired Cognitive Architectures (BICA 2018), Procedia Computer Science, vol 145, Prague, Czech Republic. Elsevier, pp 717–723
Kennedy CM (2003) Distributed reflective architectures for anomaly detection and autonomous recovery. Ph.D. thesis, University of Birmingham
Kennedy CM (2010) Decentralised metacognition in context-aware autonomic systems: Some key challenges. In Raja A, Josyula DP (eds) Proceedings of the 4th AAAI workshop on metacognition for robust social systems (AAAIWS 2010). AAAI Press, Menlo Park, pp 34–41. https://aaai.org/proceeding/ws10-04/
Kennedy CM (2011) Distributed metamanagement for self-protection and self-explanation, Cox and Raja (eds), Chapter 15, pp 233–247. https://doi.org/10.7551/mitpress/9780262014809.003.0015
Kennedy CM (2012) Towards a theory of mind for ethical software agents. In Guarini M, Bello P (eds) Proceedings of the symposium on moral cognition & theory of mind held at the AISB-IACAP World Congress 2012. AISB, Birmingham, pp 37–40. ISBN: 978-1-908187-22-2. https://frontiersinai.com/turingfiles/July/15.pdf
Kennedy CM (2018) Computational modelling of metacognition in emotion regulation. In 8th workshop on emotion and computing held at the German conference on AI (KI 2018). Presented, but not included in the workshop proceedings
Kennedy CM (2021) Modelling behaviour change using cognitive agent simulations. *arXiv CoRR preprint*. https://doi.org/10.48550/arXiv.2110.08645
Kennedy CM, Sloman A (2002) Acquiring a self-model to enable autonomous recovery from faults and intrusions. J Intell Syst 12(1):1–40. https://doi.org/10.1515/JISYS.2002.12.1.1
Kennedy CM, Sloman A (2003) Autonomous recovery from hostile code insertion using distributed reflection. Cogn Syst Res 4(2):89–117
Kim HS, Joe I (2022) An XAI method for convolutional neural networks in self-driving cars. PLoS ONE 17(8):1–17. https://doi.org/10.1371/journal.pone.0267282
Kojima T, Gu SS, Reid M, Matsuo Y, Iwasawa Y (2022) Large language models are zero-shot reasoners. In: Koyejo et al (eds.)Proceedings of the 36th Annual Conference on Advances in Neural Information Processing Systems (NeurIPS 2022). , pp 22199–22213
Kotseruba I, Tsotsos JK (2018) A review of 40 years of cognitive architecture research: core cognitive abilities and practical applications. *arXiv CoRR preprint*. https://doi.org/10.48550/arXiv.1610.08602
Kotseruba I, Tsotsos JK (2020) 40 years of cognitive architectures: core cognitive abilities and practical applications. Artif Intell Rev 53(1):17–94. https://doi.org/10.1007/s10462-018-9646-y
Koyejo S, Mohamed S, Agarwal A, Belgrave D, Cho K, Oh A, eds (2022) Proceedings of the 36th annual conference on advances in neural information processing systems (NeurIPS 2022). Curran Associates Inc
Kralik JD, Lee JH, Rosenbloom PS, Jackson PC, Epstein SL, Romero OJ, Sanz R, Larue O, Schmidtke HR, Lee SW, McGreggor K (2018) Metacognition for a common model of cognition. In Samsonovich and Lebiere (eds.)Proceedings of the 9th Annual International Conference on Biologically Inspired Cognitive Architectures (BICA 2018), Procedia Computer Science, vol 145, Prague, Czech Republic. Elsevier, pp 730–739
Krulwich B (1991) Determining what to learn in a multi-component planning system. In Hammon KJ, Centner D (eds) Proceedings of the 13th annual conference of the cognitive science society (CogSci 1991). Lawrence Erlbaum Associates, Chicago, , pp 102–107
Laird JE (2012) The SOAR cognitive architecture. The MIT Press, London
Laird JE (2019) The SOAR cognitive architecture, Reprint. The MIT Press, London
Laird JE (2021) Analysis and comparison of ACT-R and soar.In: Burstein et al (eds)Proceedings of the 9th Annual Conference on Advances in Cognitive Systems (ACS 2021), virtual event. Cognitive Systems Foundation

Laird JE, Lebiere C, Rosenbloom PS (2017) A standard model of the mind: toward a common computational framework across artificial intelligence, cognitive science, neuroscience, and robotics. AI Mag 38(4):13–26. https://doi.org/10.1609/aimag.v38i4.2744
Laird JE, Newell A (1983) A universal weak method: summary of results, vol 2. Morgan Kaufmann Publishers Inc, pp 771–773
Landis JR, Koch GG (1977) The measurement of observer agreement for categorical data. Biometrics 33(1):159–174. https://doi.org/10.2307/2529310
Langley P, ed (2011) Proceedings of the 2011 AAAI fall symposium on advances in cognitive systems. AAAI Press, Arlington. ISBN: 978-1-57735-545-8. https://aaai.org/proceeding/01-fall-2011/
Langley P, Laird JE, Rogers S (2009) Cognitive architectures: research issues and challenges. Cogn Syst Res 10(2):141–160. https://doi.org/10.1016/j.cogsys.2006.07.004
Lanham T, Chen A, Radhakrishnan A, Steiner B, Denison C, Hernandez D, Li D, Durmus E, Hubinger E, Kernion J, Lukosiute K, Nguyen K, Cheng N, Joseph N, Schiefer N, Rausch O, Larson R, McCandlish S, Kundu S,Kadavath S, Yang S, Henighan T, Maxwell T, Telleen-Lawton T, Hume T, Hatfield-Dodds Z, Kaplan J, Brauner J, Bowman SR, Perez E (2023) Measuring faithfulness in chain-of-thought reasoning. CoRR abs/2307.13702. https://doi.org/10.48550/ARXIV.2307.13702
Leake DB, Plaza E, eds. (1997) Proceedings of the 2nd international conference on case-based reasoning research and development (ICCBR 1997), vol 1266. Lecture Notes in Computer Science (LNCS). Springer, Providence
Lee I (2021) Service robots: a systematic literature review. Electronics 10:21. https://doi.org/10.3390/electronics10212658
Lehnert L, Sukhbaatar S,Su D, Zheng Q, McVay P, Rabbat M, Tian Y (2024) Beyond A*: better planning with transformers via search dynamics bootstrapping. In: Choi Y, Zhou D,Das D, Chen D, Artzi Y, Fan A, Rush A, Oh A, Choudhury M, Yatskar M (eds) Proceedings of the 1st Conference on Language Modeling (COLM 2024). https://openreview.net/forum?id=SGoVIC0u0f
Lehrer A (1995) Prefixes in English word formation. Folia Linguistica 29(1–2):133–148. https://doi.org/10.1515/flin.1995.29.1-2.133
Lindsay J, Gurnee W, Ameisen E, Chen B, Pearce A, Turner NL, Citro C, Abrahams D, Carter S, Hosmer B, Marcus J, Sklar M, Templeton A, Bricken T, McDougall C, Cunningham H, Henighan T, Jermyn A, Jones A, Persic A, Qi Z, Thompson TB, Zimmerman S, Rivoire K, Conerly T, Olah C, Batson J (2025) On the biology of a large language model. Online
Longo L, Brcic M, Cabitza F, Choi J, Confalonieri R, Ser JD, Guidotti R, Hayashi Y, Herrera F, Holzinger A, Jiang R, Khosravi H, Lecue F, Malgieri G, Páez A, Samek W, Schneider J, Speith T, Stumpf S (2024) Explainable artificial intelligence (XAI) 2.0: a manifesto of open challenges and interdisciplinary research directions. Inf Fusion 106:102301. https://doi.org/10.1016/j.inffus.2024.102301
Madera-Doval DP, Caro Piñeres MF, Gómez Salgado AA, Cardozo-Soto AM, Jiménez-Builes JA (2018) Design of metacognitive expectations of cognitive functions through ontological representations. Revista DYNA 85(206):194–201. https://doi.org/10.15446/dyna.v85n206.71687
Marshall JB (2002) Metacat: a self-watching cognitive architecture for analogy-making. In: Gray WD, Schunn CD (eds) Proceedings of the 24th annual conference of the cognitive science society. Routledge, New York, pp 631–637
Marshall JB (2006) A self-watching model of analogy-making and perception. J Exp Theor Artif Intell 18(3):267–307. https://doi.org/10.1080/09528130600758626
Mascarenhas S, Guimarães M, Prada R, Santos PA, Ja D, Paiva A (2022) Fatima toolkit: toward an accessible tool for the development of socio-emotional agents. ACM Trans Interact Intell Syst. https://doi.org/10.1145/3510822
M'Balé KM (2017) Behavior oriented intelligence. Ph.D. thesis, Bowie State University
M'Balé KM, Josyula DP (2013) Integrating metacognition into artificial agents. In Lebiere C, Rosenbloom PS (eds) Proceedings of the 2013 AAAI symposium on integrated cognition. Arlington, pp 55–62. https://cdn.aaai.org/ocs/7626/7626-32590-1-PB.pdf
M'Balé, KM, Josyula DP (2014) Handling seasonality using metacognition. In Lounis H, Josyula DP (eds) Proceedings of the 6th international conference on advanced cognitive technologies and applications (COGNITIVE 2014). IARIA, Venice, pp 205–210. ISBN: 978-1-61208-340-7. https://www.thinkmind.org/index.php?view=instance&instance=COGNITIVE+2014
McCarthy J (1995) Making robots conscious of their mental states. In Cox MT, Freed M (eds) Proceedings of the AAAI spring symposium on representing mental states & mechanisms (AAAI-SS 1995). AAAI Press, Palo Alto, pp 89–95
McDermott D (2007) Artificial intelligence and consciousness. Cambridge Handbooks in Psychology. Cambridge University Press, Chapter 6, pp 117–150
Michaelian K (2016) Mental time travel: episodic memory and our knowledge of the personal past. The MIT Press, London

Miller M, Perlis D (1993) Presentations and this and that: logic in action. Lawrence Erlbaum Associates, pp 83–88

Minsky M (1975) A framework for representing knowledge. In: Winston PH (ed) The psychology of computer vision. McGraw-Hill

Mohammad Z, Cox MT, Molineaux M (2020) Rebel agents that adapt to goal expectation failures. In Dannenhauer Z, Roberts M, Vaquero T (eds) Proceedings of the integrated execution/goal reasoning workshop (IntEx/GR 2020) held at the 13th international conference on automated planning and scheduling (ICAPS 2020), Virtual event

Moorman KM (1997) A functional theory of creative reading: process, knowledge, and evaluation. Ph.D. thesis, Georgia Institute of Technology

Morgan B (2010) Funk2: a distributed processing language for reflective tracing of a large critic-selector cognitive architecture. In Kellenberger P (ed) Proceedings of the 4th IEEE international conference on self-adaptive and self-organizing systems workshops (SASOW 2010). IEEE Computer Society, Budapest, pp 269–274

Morgan B (2013) A substrate for accountable layered systems. Ph.D. thesis, Massachusetts Institute of Technology

Nelson TO, Narens L (1990) Metamemory: a theoretical framework and new findings. Psychol Learn Motiv 26:125–173. https://doi.org/10.1016/S0079-7421(08)60053-5

Newman S (2023) GPT-4 doesn't figure things out, it already knows them. Blog post. https://amistrongeryet.substack.com/p/gpt-4-capabilities

Newton N (2001) Emergence and the uniqueness of consciousness. J Conscious Stud 8(9–10):47–59

Nolte R, Pomarlan M, Beßler D, Porzel R, Malaka R, Bateman J (2023) Towards an ontology for robot introspection and metacognition. In Aussenac-Gilles N, Hahmann T, Galton A, Hedblom MM (eds) Proceedings of the 13th international conference on formal ontology in information systems (FOIS 2023), vol 377. Frontiers in Artificial Intelligence and Applications (FAIA). IOS Press, Sherbrooke, pp 318–333

Ortony A, Clore GL, Collins A (1988) The cognitive structure of emotions. Cambridge University Press, Cambridge

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372. https://doi.org/10.1136/bmj.n71

Paul D, West R, Bosselut A, Faltings B (2024) Making reasoning matter: measuring and improving faithfulness of chain-of-thought reasoning. Proceedings of the 2024 conference in empirical methods in natural language processing (EMNLP 2024)

Pavlidis G (2024) Unlocking the black box: analysing the EU artificial intelligence act's framework for explainability in AI. Law Innov Technol 16(1):293–308. https://doi.org/10.1080/17579961.2024.2313795

Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, Blondel M, Prettenhofer P, Weiss R, Dubourg V, Vanderplas J, Passos A, Cournapeau D, Brucher M, Perrot M, Duchesnay E (2011) Scikit-learn: machine learning in python. J Mach Learn Res 12:2825–2830

Peller-Konrad F, Kartmann R, Dreher CRG, Meixner A, Reister F, Grotz M, Asfour T (2023) A memory system of a robot cognitive architecture and its implementation in armarx. Robot Auton Syst. https://doi.org/10.1016/j.robot.2023.104415

Perlis D, Cox MT, Maynord M, McNany E, Paisner M, Shivashankar V, Hand E, Shamwell J, Oates T, Du T, Josyula DP, Caro Piñeres MF (2013) A broad vision for intelligent behavior: perpetual real-world cognitive agents. In: Proceedings of the Workshop on Metacognition in Situated Agents at the 2nd Annual Conference on Advances in Cognitive Systems (ACS 2013), Baltimore, Maryland, USA. Cognitive Systems Foundation, pp 1–16

Peters MA (2022) Towards characterizing the canonical computations generating phenomenal experience. Neurosci Biobehav Rev 142:104903. https://doi.org/10.1016/j.neubiorev.2022.104903

Prezenski S, Brechmann A, Wolff S, Russwinkel N (2017) A cognitive modeling approach to strategy formation in dynamic decision making. Front Psychol. https://doi.org/10.3389/fpsyg.2017.01335

Proust J (2007) Metacognition and metarepresentation: is a self-directed theory of mind a precondition for metacognition? Synthese 159:271–295. https://doi.org/10.1007/s11229-007-9208-3

Ram A (1989) Decision models: a theory of volitional explanation. Ph.D. thesis, Yale University

Ram A, Cox MT (1995) Introspective reasoning using meta-explanations for multistrategy learning. The MIT Press, London, pp 211–239. https://doi.org/10.7551/mitpress/3314.003.0012

Recht S, Jovanovic L, Mamassian P, Balsdon T (2022) Confidence at the limits of human nested cognition. Neurosci Consciousness 2022(1):1–11. https://doi.org/10.1093/nc/niac014
Reimers N, Gurevych, I (2019) Sentence-BERT: sentence embeddings using Siamese BERT-networks. In Inui K, Jiang J, Ng V, Wan X (eds) Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP 2019). Association for Computational Linguistics, pp 3982–3992
Reitter D, Lebiere C (2012) Social cognition: memory decay and adaptive information filtering for robust information maintenance. In Hoffmann J, Selman B (eds) Proceedings of the 26th AAAI conference on artificial intelligence (AAAI 2012). AAAI Press, Toronto, pp 242–248. https://aaai.org/proceeding/01-aaai-2012/
Rhodius A (2009) Argonautica. Loeb Classical Library. Harvard University Press, Cambridge
Rudin C (2019) Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat Mach Intell 1(5):206–215. https://doi.org/10.1038/s42256-019-0048-x
Samsonovich AV (2006) Biologically inspired cognitive architecture for socially competent agents. In Upal MA, Sun R (eds) Proceedings of the AAAI workshop on cognitive modeling and agent-based social simulation. AAAI Press, Menlo Park, pp 36–48. https://aaai.org/proceeding/ws06-02/
Samsonovich AV (2009) The constructor metacognitive architecture. In Samsonovich AV (ed) Proceedings of the 2009 AAAI fall symposium on biologically inspired cognitive architectures (BICA 2009), vol FS-09-01. AAAI Press, Arlington, pp 124–134
Samsonovich AV (2010) Toward a unified catalog of implemented cognitive architectures. In Samsonovich et al. (eds)Proceedings of the 1st Annual Meeting of the Biologically Inspired Cognitive Architectures Society (BICA 2010), Frontiers in Artificial Intelligence and Applications, vol 221, Washington, District of Columbia, USA. IOS Press, pp 195–244. ISBN: 978-1-60750-660-7
Samsonovich AV (2013a) Emotional biologically inspired cognitive architecture. In Proceedings of the 4th annual international conference on biologically inspired cognitive architectures (BICA 2013), vol 6. Biologically Inspired Cognitive Architectures (BICA). Elsevier, Kiev, pp 109–125
Samsonovich AV (2013) How to develop fluid intelligence via metacognitive self-organization. In: Proceedings of the Workshop on Metacognition in Situated Agents at the 2nd Annual Conference on Advances in Cognitive Systems (ACS 2013), Baltimore, Maryland, USA.Cognitive Systems Foundation, pp 99–110
Samsonovich AV, De Jong KA (2002) General purpose meta-cognitive systems: from philosophical ideas to a computational framework. Shtuchnyy intelekt 4:67–73
Samsonovich AV, De Jong KA (2003) Meta-cognitive architecture for team agents. In Alterman R, Kirsch D (eds) Proceedings of the 25th annual conference of the cognitive science society (CogSci 2003). Lawrence Erlbaum Associates, Boston, pp 1029–1034
Samsonovich AV, De Jong KA (2005) Designing a self-aware neuromorphic hybrid. In Thórisson KR, Vilhjalmsson H, Marsella S (eds) Proceedings of the AAAI workshop on modular construction of human-like intelligence. AAAI Press, Pittsburgh, pp 71–78
Samsonovich AV, De Jong KA, Kitsantas A (2009) The mental state formalism of GMU-BICA. Int J Mach Consciousness 1(1):111–130. https://doi.org/10.1142/s1793843009000116
Samsonovich AV, Johannsdottir KR, Chella A, Goertzel B, eds (2010) Proceedings of the 1st annual meeting of the biologically inspired cognitive architectures society (BICA 2010), vol 221. Frontiers in Artificial Intelligence and Applications (FAIA)IOS Press, Washington, DC. ISBN: 978-1-60750-660-7
Samsonovich AV, Lebiere C, eds (2018) Proceedings of the 9th annual international conference on biologically inspired cognitive architectures (BICA 2018), vol 145. Procedia Computer Science. Elsevier, Prague
Samsonovich AV, Nadel L (2005) Fundamental principles and mechanisms of the conscious self. Cortex 41(5):669–689. https://doi.org/10.1016/S0010-9452(08)70284-3
Sant'Anna A, Michaelian K, Andonovski N (2024) Autonoesis and episodicity: perspectives from philosophy of memory. WIREs Cognit Sci 15(1):e1665. https://doi.org/10.1002/wcs.1665
Sanz R, López I, Rodríguez M, Corbato CH (2007) Principles for consciousness in integrated cognitive control. Neural Netw 20(9):938–946. https://doi.org/10.1016/j.neunet.2007.09.012
Schaeffer R, Miranda B, Koyejo S (2023) Are emergent abilities of large language models a mirage? In Oh A, Naumann T, Globerson A, Saenko K, Hardt M, Levine S (eds) Proceedings of the 37th annual conference on neural information processing systems advances in neural information processing systems (NeurIPS 2023), vol 36. Curran Associates Inc, pp 55565–55581
Schank RC (1986) Explanation patterns: understanding mechanically and creatively. Psychology Press

Schmill MD, Josyula DP, Anderson ML, Wilson S, Oates T, Perlis D, Wright D, Fults S (2007) Ontologies for reasoning about failures in AI systems. In Proceedings of the workshop on metareasoning in agent based systems held at the 6th international conference on Autonomous Agents and Multiagent Systems (AAMAS 2007), Honolulu, p 15

Schneider C, Barker A, Dobson S (2015) A survey of self-healing systems frameworks. Softw Practice Exp 45(10):1375–1398. https://doi.org/10.1002/spe.2250

Schwartz BL (2006) Tip-of-the-tongue states as metacognition. Metacogn Learn 1(2):149–158. https://doi.org/10.1007/s11409-006-9583-z

Schwartz BL, Cleary AM (2016) Tip-of-the-tongue states, déjà vu experiences, and other odd metacognitive experiences, In Dunlosky J, Tauber SUK (eds) The Oxford handbook of metamemory. Oxford University Press. ISBN: 978-0-19-933674-6. https://doi.org/10.1093/oxfordhb/9780199336746.013.5

Schwarz N (2010) Meaning in context: metacognitive experiences. In: Barrett LF, Mesquita B, Smith ER (eds) The mind in context. The Guilford Press, pp 105–125

Scott AE, Neumann D, Niess J, Woźniak PW (2023) Do you mind? User perceptions of machine consciousness. In Schmidt A, Väänänen K, Goyal T, Kristensson PO, Peters A, Mueller S, Williamson JR, Wilson ML (eds) Proceedings of the conference on human factors in computing systems (CHI 2023). Association for Computing Machinery (ACM), New York

Seth P, Sankarapu VK (2025) Bridging the gap in xai-why reliable metrics matter for explainability and compliance. CoRR abs/2502.04695. https://doi.org/10.48550/ARXIV.2502.04695

Shapiro SC, Bona JP (2010) The glair cognitive architecture. Int J Mach Consciousness 02:307–332. https://doi.org/10.1142/S1793843010000515

Shrager J, Siegler RS (1998) SCADS: a model of children's strategy choices and strategy discoveries. Psychol Sci 9(5):405–410

Sloman A, Chrisley RL (2005) More things than are dreamt of in your biology: information-processing in biologically inspired robots. Cogn Syst Res 6(2):145–174. https://doi.org/10.1016/j.cogsys.2004.06.004

Stechly K, Valmeekam K, Kambhampati S (2024) Chain of thoughtlessness? An analysis of cot in planning. In Globerson A, Mackey L, Belgrave D, Fan A, Paquet U, Tomczak J, Zhang C (eds) Proceedings of the 38th annual conference on neural information processing systems (NeurIPS 2024). Curran Associates, Inc, pp 29106–29141. https://openreview.net/forum?id=kPBEAZU5Nm

Stetchly K, Valmeekam K, Gundawar A, Palod V, Kambhampati S (2025) Beyond semantics: the unreasonable effectiveness of reasonless intermediate tokens. arXiv:2505.13775 arXiv preprint

Stevens CA, de Weerd H, Cnossen F, Taatgen NA (2016) A metacognitive agent for training negotiation skills. In Reitter D, Ritter FE (eds) Proceedings of the 14th International Conference on Cognitive Modeling (ICCM 2016), University Park, Pennsylvania. ISBN: 978-0-9985082-0-7. https://acs.ist.psu.edu/iccm2016/proceedings/

Stroulia E (1994) Failure-driven learningas model-based self-redesign. Ph.D. thesis, Georgia Institute of Technology

Sturm A, Shehory O (2014) The evolution of MAS tools. In: Shehory O, Sturm A (eds) Agent-oriented software engineering. Springer, Berlin, Heidelberg, Germany, pp 275–288. https://doi.org/10.1007/978-3-642-54432-3_14

Sun R (2016) Anatomy of the mind: exploring psychological mechanisms and processes with the clarion cognitive architecture. Oxford University Press

Sun R (2018) Why is a computational framework for motivational and metacognitive control needed? J Exp Theor Artif Intell 30(1):13–37. https://doi.org/10.1080/0952813X.2017.1413141

Sun R, Allen J, Werbin E (2022) Modeling emotion contagion within a computational cognitive architecture. J Cogn Cult 22(1–2):60–89. https://doi.org/10.1163/15685373-12340125

Sun R, Zhang X, Mathews R (2005) Modeling meta-cognition in a cognitive architecture. In Bara BG, Barsalou L, Bucciarelli M (eds) Proceedings of the 27th annual conference of the cognitive science society (CogSci 2005). Psychology Press, Stresa, pp 2104–2109

Sun R, Zhang X, Mathews R (2006) Modeling meta-cognition in a cognitive architecture. Cogn Syst Res 7(4):327–338. https://doi.org/10.1016/j.cogsys.2005.09.001

Thórisson KR, Helgason HP (2012) Cognitive architectures and autonomy: a comparative review. J Artif Gen Intell 3(2):1–30. https://doi.org/10.2478/v10229-011-0015-3

Tiku N (2022) The Google engineer who thinks the company's AI has come to life. The Washington Post

...Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, Moher D, Peters MDJ, Horsley T, Weeks L, Hempel S, Akl EA, Chang C, McGowan J, Stewart L, Hartling L, Aldcroft A, Wilson MG, Garritty C, Lewin S, Godfrey CM, Macdonald MT, Langlois EV, Soares-Weiser K, Moriarty J, Clifford T, Tunçalp O, Straus SE (2018) Prisma extension for scoping reviews (PRISMA-ScR): checklist and explanation. Ann Intern Med 169(7):467–473. https://doi.org/10.7326/M18-0850

Tschang FT, Almirall E (2021) Artificial intelligence as augmenting automation: implications for employment. Acad Manag Perspect 35(4):642–659. https://doi.org/10.5465/amp.2019.0062

Tulving E (1972) Episodic and semantic memory. In Tulving E, Donaldson W (eds) Organization of memory, Chapter 10. Academic Press, pp 381–402

Tulving E (1985) Elements of episodic memory. Oxford Psychology Series. Oxford University Press

Turpin M, Michael J, Perez E, Bowman S (2023) Language models don't always say what they think: unfaithful explanations in chain-of-thought prompting. In Oh A, Naumann T, Globerson A, Saenko K, Hardt M, Levine S (eds) Proceedings of the 37th annual conference on advances in neural information processing systems (NeurIPS 2023). Curran Associates Inc, pp 74952–74965

Valmeekam K, Stechly K, Kambhampati S (2024) LLMs still can't plan; can LRMs? A preliminary evaluation of openAI's o1 on PlanBench. In Globerson A, Mackey L, Belgrave D, Fan A, Paquet U, Tomczak J, Zhang C (eds) Proceedings of the workshop on open-world agents at the 38th annual conference on neural information processing systems (NeurIPS 2024). Curran Associates Inc. https://openreview.net/forum?id=Gcr1Lx4Koz

van Ments L, Treur J (2021) Reflections on dynamics, adaptation and control: a cognitive architecture for mental models. Cogn Syst Res 70:1–9. https://doi.org/10.1016/j.cogsys.2021.06.004

Vinokurov Y, Lebiere C, Hard S, O'Reilly R (2011) A metacognitive classifier using a hybrid ACT-R/Leabra architecture, pp 50–55

Wang P, Li X, Hammer P (2018a) Self in NARS, an AGI system. Front Robot AI. https://doi.org/10.3389/frobt.2018.00020

Wang Y, Kwong S, Feldman J, Howard N, Sheu PC, Widrow B, eds (2018b) Proceedings of the 17th international conference on cognitive informatics & cognitive computing (ICCI*CC 2018). IEEE Computer Society, Berkeley

Weber RO, Johs AJ, Goel P, Silva JM (2024) XAI is in trouble. AI Mag 45(3):300–316. https://doi.org/10.1002/aaai.12184

Wei J, Wang X, Schuurmans D, Bosma M, ichter b, Xia F, Chi E, Le QV, Zhou D (2022) Chain-of-thought prompting elicits reasoning in large language models. In: Koyejo et al. (eds)Proceedings of the 36th Annual Conference on Advances in Neural Information Processing Systems (NeurIPS 2022)., pp 24824–24837

Wisspeintner T, Tvd Z, Iocchi L, Schiffer S (2009) RoboCup@Home: scientific competition and benchmarking for domestic service robots. Interact Stud 10(3):392–426. https://doi.org/10.1075/is.10.3.06wis

York W, Swan J (2012) Taking Turing seriously (but not literally). In Müller VC, Ayesh A (eds) Proceedings of symposium on revisiting turing and his test held at the AISB-IACAP World Congress 2012. The Society for the Study of Artificial Intelligence and Simulation of Behaviour, Birmingham, pp 54–59. ISBN: 978-1-908187-15-4

Zhang Z, Zhang A, Li M, Smola A (2023) Automatic chain of thought prompting in large language models. In Proceedings of the 11th International Conference on Learning Representations (ICLR 2023). https://openreview.net/forum?id=5NTt8GFjUHkr

Zhao C, Tan Z, Ma P, Li D, Jiang B, Wang Y, Yang Y, Liu H (2025) Is chain-of-thought reasoning of LLMs a mirage? A data distribution lens

Zheng Y, Recht S, Rahnev D (2023) Common computations for metacognition and meta-metacognition. Neurosci Consciousness 2023(1):1–8. https://doi.org/10.1093/nc/niad023

Zilberstein S (2011) Metareasoning and bounded rationality, Cox and Raja (eds), Chapter 3, pp 27–40. https://doi.org/10.7551/mitpress/9780262014809.003.0003

Mohammad Ali Pour, Fazel (2026) The Claude Code Leak: A Complete Technical & Security Investigation 2026-03-31 SSRN preprintt https://doi.org/10.2139/ssrn.6504920

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